Weighted configurations and the nonexistence of arithmetic ball quotient compactifications

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Abstract

We prove that, for every $n \geq 9$, no GIT quotient of n ordered points on $\mathbb{P}^1$ with positive rational weights smaller than one and of total weight two is isomorphic to the Baily–Borel compactification of an arithmetic complex ball quotient. The weights may lie on stability walls, and the lattice may have torsion; the quotient map may be branched along divisors. No compatibility with a prescribed period map is assumed. We study homomorphisms from the tangent bundle pulled back to complete curves into line bundles of negative degree and compare the resulting constraints with normal bundles of the images of balls fixed by finite reflection groups in the arithmetic lattice. We describe these homomorphisms through coincident marked points and prove vanishing statements for their kernels. The comparison, together with integrality of Chern classes and the invariant theory of finite reflection groups, gives the nonexistence theorem without a Fano assumption on the GIT quotient.

Use of AI. The authors used OpenAI Codex as a discussion partner for mathematical arguments, to assist with revising proofs and their presentation, and to proofread the manuscript. The authors take responsibility for the mathematics, computations, and references.

1 Introduction

Let $n \geq 9$, and choose rational weights

$$0 < \mu_i < 1, \qquad \sum_{i=1}^{n} \mu_i = 2. \tag{1.1}$$

For $I \subset \{1, \dots, n\}$, write $\mu(I) = \sum_{i \in I} \mu_i$. After clearing denominators, let

$$P_n = (\mathbb{P}^1)^n, \qquad X_\mu = P_n /\!/_{\mathcal{O}(m\mu_1, \dots, m\mu_n)} \mathrm{SL}(2, \mathbb{C}).$$

The labels are retained, and the quotient is independent of the integer $m > 0$ used to clear denominators. A configuration is stable precisely when $\mu(I) < 1$ for every set I of labels whose points coincide. We use $\mathbb{B}^d$ for the complex ball of dimension d, and the superscript BB for the Baily–Borel compactification.

Theorem 1.1. For every $n \geq 9$ and every weight vector satisfying (1.1), there is no arithmetic lattice $\Gamma \subset \mathrm{PU}(1, n-3)$ with an algebraic isomorphism

$$X_\mu \simeq (\mathbb{B}^{n-3}/\Gamma)^{\mathrm{BB}}.$$

We impose no compatibility with label permutations or with a specified period map. Failure of an integrality condition for a Deligne–Mostow period map [10] alone would not prove the theorem. The weights may lie on stability walls. The lattice may have torsion and need not

be a normal or congruence subgroup of a prescribed group; the quotient map may be branched along divisors. No Fano hypothesis is imposed on X_μ.

The lower bound $n \geq 9$ is sharp for a theorem that applies to every weight vector satisfying (1.1): for eight equal weights, X_μ is the Baily–Borel compactification of an arithmetic ball quotient [10, 16]. This does not classify all weights with fewer than nine labels. The theorem also includes chambers whose quotient is projective space. For example, choose a rational number $0 < \varepsilon < 1/(n-2)$ and put

$$\mu = \left(1 - \varepsilon, \left(\frac{1+\varepsilon}{n-1}\right)^{n-1}\right).$$

The heavy point cannot meet any light point, every proper subset of the light points may coincide, and all $n-1$ light points cannot coincide. Place the heavy point at infinity. The affine coordinates of the light points, modulo translations and nonzero scalar multiplication, give

$$X_\mu \simeq \mathbb{P}\big(\mathbb{C}^{n-1}/\mathbb{C}\mathbf{1}\big) \simeq \mathbb{P}^{n-3}, \qquad \mathbf{1} = (1, \ldots, 1).$$

The unstable affine configurations are those in which all light points coincide, and the residual group is the affine group of the line. Thus, for $d \geq 6$, the theorem implies that $\mathbb{P}^d$ is not isomorphic to the Baily–Borel compactification of an arithmetic ball quotient.

The case of twelve equal weights answers Question 2.3 of Hulek–Kondö–Maeda [15]. The companion paper [24] proves the nonexistence theorem for seven equal weights by a comparison involving all fixed components. It cites the propositions on tangent bundles and arithmetic groups proved here. The proof of Theorem 1.1 does not use that result.

The universal family of projective lines with labeled sections determines homomorphisms from the tangent bundle to line bundles of negative degree and controls their kernels. The finite reflection groups in the original arithmetic lattice fix balls whose normal directions are described by polynomial invariants, with finite characters and central translations at the cusps that descend together. We compare these structures on complete curves obtained from quotients of fixed balls by their full normalizers. Three reductions shorten the proof. The degrees of the relative tangent line bundles along the labeled sections first give decompositions of effective curve classes and the initial bounds for degrees of curves without an external theorem on generators of the Mori cone. Next, an irreducible factor of a finite stabilizer has basic invariants of equal degree and rank at most 2 if it is the full generic stabilizer of its fixed ball and every full interior stabilizer along that ball is the product of this factor and a group acting trivially on the normal space, and then the Baily–Borel compactification of the quotient of its fixed ball by the full normalizer is a quotient with one or two labels merged. Strong induction from $n = 9, 10$ excludes every such factor for $n \geq 11$. Finally, after the stabilizers are of type A, write $N = n - 2$ and

$$b_N = \prod_{p \mid N} p^{\lceil 2v_p(N)/3 \rceil}, \qquad C_N = \frac{2N^2}{b_N}. \tag{1.2}$$

Throughout, L is the rational class of the automorphic line bundle of the original lattice or its specified pullback. Let $W = X_\nu$ be the smooth GIT quotient obtained by perturbing μ to weights ν in a nearby chamber, with the small projective morphism $\pi : W \to X_\mu$ of Lemma 2.1. Using integrality of Chern classes and complete curves obtained by moving marked points, we show that $K_W + C_N L$ is nef even when W is isomorphic to projective space. For $N \geq 11$,

$$C_N < \frac{N^2 - 4N + 1}{2},$$

contradicting the lower bound of Proposition 3.11 for curves obtained from disks fixed by reflection groups of type A.

The proof proceeds through the following steps. The first row gives the common geometric input; the remaining rows indicate where the arguments depend on the dimension.

Range	Main steps
All n	Propositions 2.2–2.4 control homomorphisms from the tangent bundle to line bundles of negative degree and their kernels; Proposition 3.1 compares fixed components, retaining the full normalizer, inertia characters, and cusp widths.
$n = 9$	Theorems 4.6 and 4.14 treat the initial case; the D_5 argument uses successive kernels, and the G_{33} argument uses the theta calculation in Lemma A.1.
$n = 10$	Theorem 5.5 gives the second induction case by proving nefness of K_W plus smaller multiples of L and treating the exceptional stabilizers.
$n = 11, 12$	Propositions 6.2 and 6.3 combine the induction with exceptional invariant calculations and the examination of every case in the finite ranges of Appendix C.
$n \geq 13$	Proposition 6.4 reduces the remaining stabilizers to type A and applies the strict inequality for C_N stated after (1.2).

After the induction and the stabilizer reductions, the argument of Proposition 6.4 applies uniformly for $n \geq 13$. The numerical inequality itself already holds for $N \geq 9$; the separate treatments of $n = 11, 12$ establish the required geometric hypotheses, including the exclusion of the remaining exceptional stabilizers. The appendices retain the local algebra, the arguments for fixed curves in projective space, and the finite ranges used in the proof. The notation $H = -K_W$ does not assert ampleness; in a calculation on projective space its hyperplane class is identified separately.

2 Curves in GIT quotients, arithmetic groups, and cusps

Put $d = n - 3$, $N = d + 1 = n - 2$, and $\mu(I) = \sum_{i \in I} \mu_i$. Throughout this section $n \geq 6$. For a weight vector ν, put $\nu(I) = \sum_{i \in I} \nu_i$, and call I short or long if $\nu(I) < 1$ or $\nu(I) > 1$, respectively. A configuration is ν-stable if $\nu(I) < 1$ for every set I of labels whose points coincide. Stable configurations have at least three distinct points, so the action of $\mathrm{PGL}(2, \mathbb{C})$ is free. A pair $\{i, j\}$ is allowable if $\nu_i + \nu_j < 1$. At a wall $\mu(I) = 1$, the vertex is genuine if $|I|, |I^c| \geq 3$.

2.1 The family with labeled sections and quotients of the tangent bundle onto line bundles of negative degree

Lemma 2.1. There is a smooth GIT quotient $W = X_\nu$, obtained by perturbing μ to weights ν in a nearby chamber, and a small projective morphism $\pi : W \to X_\mu$ such that K_W is π-nef. The original stable quotient X_μ^{s} is an open subset of W. At a genuine wall vertex the local model is

$$\mathrm{Cone}\big(\mathbb{P}^{|I|-2} \times \mathbb{P}^{|I^c|-2}, \mathcal{O}(1,1)\big).$$

The resolution retains the smaller factor $\mathbb{P}^u$, where $u = \min(|I|, |I^c|) - 2$, and

$$\mathrm{codim}_W \mathrm{Exc}(\pi) \geq \lceil (n-2)/2 \rceil.$$

Proof. After fixing one mark at each of $0, \infty$, the transverse torus weights are $(1^{|I|-1}, -1^{|I^c|-1})$. Products of opposite coordinates generate its invariant ring, with the two by two minors as relations. Keeping the smaller factor gives normal bundle $\mathcal{O}(-1)^{\oplus(v+1)}$, $v = \max(|I|, |I^c|) - 2$, and degree against $-K_W$ $u - v \leq 0$. All these choices are made simultaneously by

$$\nu = (1 + \epsilon)\mu - \epsilon(2/n, \ldots, 2/n) + \epsilon^2 \eta, \qquad \sum_i \eta_i = 0,$$

with η generic and $\epsilon > 0$ sufficiently small. Every I with $\mu(I) = 1$ and $|I| < n/2$ becomes long; when $|I| = n/2$, η decides which of I and I^c becomes long. The exceptional codimension is $d - u$. In particular, it is at least four for $n \geq 9$. □

From the principal $\mathrm{PGL}(2, \mathbb{C})$-bundle of stable configurations, we obtain the associated bundle $\mathcal{P} \to W$ with fiber $\mathbb{P}^1$ and labeled sections σ_i. For a nonconstant map $f : C \to W$ from a smooth projective connected curve, write the pullback of $\mathcal{P}$ as $\mathbb{P}_C(V)$, using the convention of quotients of dimension one. A section supplies a divisor of relative degree one, so such a vector bundle V of rank 2 exists. Write $V \twoheadrightarrow L_i$ for the marked quotients and put $M_i = L_i^2(\det V)^{-1}$, $m_i = \deg M_i$, and $a = -K_W \cdot f_* C$. The infinitesimal action and the brackets of distinct sections give

$$\begin{aligned} 0 \longrightarrow \mathrm{End}_0(V) \longrightarrow \bigoplus_i M_i \longrightarrow f^* T_W \longrightarrow 0, \\ a = \sum_i m_i, \qquad m_i + m_j = 2 \deg D_{ij} \geq 0 \quad (\sigma_i \neq \sigma_j). \end{aligned} \tag{2.1}$$

Here D_{ij} is the effective zero divisor of the bracket. If a set I of sections coincides, the smooth locus Z_I^{s} where they coincide satisfies

$$N_{Z_I^{\mathrm{s}}/W} = M_I \otimes_{\mathbb{C}} (\mathbb{C}^I / \mathbb{C}\mathbf{1}). \tag{2.2}$$

Proposition 2.2. Suppose Q is a line bundle of negative degree and $\mathrm{Hom}(f^* T_W, Q) \neq 0$. Every homomorphism $f^* T_W \to Q$ factors through the stalkwise quotient

$$D = M^{\oplus(\ell-1)}, \qquad \psi : f^* T_W \twoheadrightarrow D, \qquad \psi \, df = 0.$$

The $\ell \geq 2$ labels associated with this quotient form a set I of identical sections of weight $\nu(I) < 1$, and $M = M_I$ has degree $-e < 0$. For $j \notin I$, write D_j for the divisor of the bracket of σ_j with this common section, and put $\delta = \sum_{j \notin I} \deg D_j$. Then

$$M_j = M^{-1}(2D_j), \quad a = (n - 2\ell)e + 2\delta, \quad \mathrm{Hom}(f^* T_W, Q) = H^0(C, QM^{-1})^{\oplus(\ell-1)}. \tag{2.3}$$

In particular, $\deg Q = -b < 0$ implies $e \geq b$. If this Hom space has dimension one, then $\ell = 2$ and $a \geq (n-4)b$.

Proof. By (2.1), two line bundles M_i of negative degree cannot belong to distinct sections. Thus all line bundles M_i of negative degree arise from one section, and its brackets give the first two formulas. For $j \notin I$, all lines M_j have positive degree and map trivially to Q. Evaluation of $\mathrm{End}_0(V)$ onto the common tangent line is surjective at every stalk, and its image in the I components is exactly the diagonal. Quotienting those components by the diagonal proves the assertion and the Hom formula. If there is only one such component it is removed entirely, so $\ell \geq 2$. The derivatives of identical sections coincide, proving $\psi \, df = 0$. □

Proposition 2.3. Put $F = f^* T_{Z_I^{\mathrm{s}}}$, let $T_f \subset F$ be the saturated image of df, and put $\mathcal{K} = F/T_f$. Then

$$f^* T_W = F \oplus D, \qquad N_f^{\mathrm{sat}} = \mathcal{K} \oplus D, \qquad \bigoplus_{j \notin I} M_j \twoheadrightarrow F \twoheadrightarrow \mathcal{K}.$$

For every line bundle Q and every integer $q \geq 1$,

$$\deg Q < qe \implies \mathrm{Hom}(\textstyle\bigwedge^q F, Q) = \mathrm{Hom}(\bigwedge^q \mathcal{K}, Q) = 0.$$

For $C = \mathbb{P}^1$, all line summands of F have degree at least e. If a line bundle of degree less than e is a direct summand of $f^* T_W$, or of N_f^{sat}, then it is isomorphic to M. If f is immersed, N_f^{sat} is the normal bundle of f.

Proof. Split the I components of (2.1) into their diagonal and the subbundle with sum zero. The infinitesimal action has zero projection to the subbundle with sum zero, so $f^*T_W = F \oplus D$ and $N_f^{\mathrm{sat}} = \mathcal{K} \oplus D$. Stalkwise evaluation onto the diagonal removes that diagonal and gives the surjection $\bigoplus_{j\notin I} M_j \twoheadrightarrow F$. The exterior powers of $\bigoplus_{j\notin I} M_j$ have summands of degrees at least qe, proving the vanishing. On $\mathbb{P}^1$, this also says that $F(-e)$ is globally generated. For a line summand of degree less than e, its retraction vanishes on F, or on $\mathcal{K}$, and factors through D. The resulting idempotent of D, after tensoring by M^{-1}, is a constant matrix; its image of rank 1 is trivial. □

Proposition 2.4. Let $\mathcal{Q}$ be a vector bundle of rank q on C with $\deg\det\mathcal{Q} < 0$, admitting a surjection from f^*T_W or N_f^{sat}. Then the set I of coincident sections and the bundles F, $\mathcal{K}$, and D in Proposition 2.3 exist. If the restriction to F, respectively $\mathcal{K}$, has generic rank k, then

$$\deg\det\mathcal{Q} \geq (2k-q)e \geq -qe. \tag{2.4}$$

If $\deg\det\mathcal{Q} < -(q-2)e$, the surjection factors through D and $\ell \geq q+1$. If $\deg\det\mathcal{Q} = -qe$, then $\mathcal{Q} = M^{\oplus q}$. If f takes values in the stable locus of the original equal weight quotient with $n = 7, 8$ and $q = 2$, we have $\ell = 3$, $\mathcal{Q} = M^{\oplus 2}$, and $e = a/\varepsilon_n$, where $\varepsilon_7 = 1$, $\varepsilon_8 = 2$. On that stable locus for $n = 6$, no such quotient of rank 2 exists.

Proof. The inequality $\deg\det\mathcal{Q} < 0$ forces at least two coincident sections with negative degrees m_i: stalkwise evaluation removes a single such section. Nonzero minors show that the determinant of the saturated image of F, or $\mathcal{K}$, has degree at least ke; the quotient generated by D contributes at least $-(q-k)e$. This proves (2.4) and the strict case. Equality makes $M^{-1}\mathcal{Q}$ globally generated with determinant of degree zero, hence trivial. For $q = 2$, negativity gives $k = 0$ and $\ell \geq 3$. Stability for the original equal weights forces $\ell = 3$ and $D_j = 0$ for $n = 7, 8$, so $D \twoheadrightarrow \mathcal{Q}$ is an isomorphism and (2.3) gives e; $n = 6$ forbids $\ell \geq 3$. □

2.2 Effective curve classes without a Mori theorem for moduli spaces

For the equal weight quotients X_n, with $\mu_i = 2/n$, put $s_n = \lfloor (n-1)/2 \rfloor$ and $\varepsilon_n = n - 2s_n$.

Proposition 2.5. The smooth stable quotient X_n^{s} is simply connected, and the descent H of $\mathcal{O}_{P_n}(2,\dots,2)$ is ample and equals $-K$. For odd n, we have $X_n = X_n^{\mathrm{s}}$ and $\mathrm{Pic}(X_n) = \{b \in \mathbb{Z}^n : \sum_i b_i \in 2\mathbb{Z}\}$, $H = (2,\dots,2)$. For even $n = 2h$, the descent J of $\mathcal{O}(1,\dots,1)$ is very ample, $\mathrm{Pic}(X_n) = \mathbb{Z}J$, and $H = 2J = -K_{X_n}$. The smooth locus equals the stable locus; each of the $\frac{1}{2}\binom{n}{h}$ singularities is the cone over $\mathbb{P}^{h-2} \times \mathbb{P}^{h-2}$ with polarization $\mathcal{O}(1,1)$. Its link is simply connected, and its two small resolutions are crepant. For curves $f : C \to X_n^{\mathrm{s}}$, Proposition 2.2 gives $\ell \leq s_n$, $a \geq \varepsilon_n e$, and vanishing whenever $\deg Q < -a/\varepsilon_n$.

Proof. General position and the homotopy sequence of the principal bundle give simple connectedness. The character $(-1)^{\sum b_i}$ gives odd descent. Even descent requires $\sum_{i\in I} b_i = \sum_{i\notin I} b_i$ at every closed orbit, forcing constant multidegree; perfect matching brackets give very ampleness [13, Theorem 2.3]. The tangent sequence gives $H = -K$. Lemma 2.1, the primitive circle class $(1,1)$, and equal factor dimensions give the cones, links, and crepancy. Stability and (2.3) give the curve bounds. □

Proposition 2.6. Let $n = 2h+1$, and let E be the line bundle of multidegree b on X_n. Set $q = \frac{1}{2}\sum_i b_i$ and $d_I = q - \sum_{i\in I} b_i$ for $|I| = h$. Then $Z_I = \mathbb{P}^{h-1}$, $M_I = \mathcal{O}(-1)$, $H|_{Z_I} = \mathcal{O}(1)$, $E|_{Z_I} = \mathcal{O}(d_I)$. The conditions $d_I \geq 0$ for every I, global generation of E, and nefness of E are equivalent. If ℓ_I is a line in Z_I, then $\overline{\mathrm{NE}}(X_n) = \sum_I \mathbb{R}_{\geq 0}\ell_I$. Every ample integral E has $E - H$ globally generated.

Proof. The affine quotient at infinity gives $\mathbb{P}((\mathbb{C}^{h+1}/\mathbb{C}\mathbf{1})^\vee)$, represented by $\mathcal{O}\oplus\mathcal{O}(1)$, proving the restrictions. Partitioning the other labels gives $b_j = d_I + d_{I'} \geq 0$. List b_i copies of label i consecutively in a circle of length $2q$, and pair opposite positions. The inequalities prevent coincident pairs, so bracket products generate E; $q = 0$ is immediate. Pass to real classes by clearing denominators and limits. Since $H\ell_I = 1$, cone duality gives the Mori cone; ampleness gives $d_I \geq 1$. □

Proposition 2.7. For equal weights, if $|I| = \ell$ and $n > 2(\ell+1)$, there is a complete embedded $\mathbb{P}^1 \subset Z_I^{\mathrm{s}}$ on which M_I is trivial. Thus no finite surjection onto Z_I^{s} can make M_I^{-1} ample.

Proof. Moving one outside label with the others distinct and fixed gives an embedded stable $\mathbb{P}^1$, since $\ell + 1 < n/2$. Three fixed labels trivialize the common tangent line, whose pullback to every complete curve above it still has degree zero. □

Proposition 2.8. The cone $\overline{\mathrm{NE}}(W)$ is generated by coordinate curves γ_i, obtained by moving a single mark for which every pair containing that mark is allowable, and by lines β_J in the closed loci of positive dimension

$$Z_J = \mathbb{P}^{n-|J|-2}, \qquad \nu(J) < 1, \qquad \nu(J) + \nu_j > 1 \quad (j \notin J).$$

For these curves,

$$-K_W \cdot \gamma_i = 2, \qquad -K_W \cdot \beta_J = n - 2|J|, \qquad N_{Z_J/W} = \mathcal{O}(-1)^{\oplus(|J|-1)}.$$

More precisely, if all $m_i \geq 0$, then

$$[f_*C] = \tfrac{1}{2}\sum_i m_i[\gamma_i].$$

If I is the set of labels i with $m_i < 0$, possibly of size one, there is a complete rational curve β_I on which the degrees m_i are -1 on I and 1 elsewhere, and

$$[f_*C] = e[\beta_I] + \sum_{j\notin I} \deg D_j[\gamma_j], \qquad [\beta_I] = [\beta_J] + \sum_{j\in J\setminus I}[\gamma_j] \tag{2.5}$$

for every maximal short set $J \supset I$. Every coordinate curve with positive coefficient in these formulas exists.

Proof. The M_i generate $\mathrm{Pic}(W)_{\mathbb{Q}}$. Pull a rational section of a line bundle B to the principal bundle and close its invariant divisor in $(\mathbb{P}^1)^n$. Its bundle is $\mathcal{O}(b_1,\dots,b_n)$; its linearization agrees with the standard $\mathrm{SL}(2,\mathbb{C})$-linearization because $\mathrm{SL}(2,\mathbb{C})$ has no characters. Squaring and descending gives $B^2 = \bigotimes_i M_i^{b_i}$. Thus the degrees m_i determine numerical curve classes.

If $m_i > 0$ and all degrees are nonnegative, each distinct section has a bracket of positive degree with i, so the corresponding pair actually collides and must be allowable. Identical sections also form an allowable pair by stability. Moving i, keeping the other marks general, therefore gives γ_i, with degrees $2\delta_{ij}$. This proves the first formula.

In the negative case all outside pairs are allowable, by their positive bracket degrees or stability. On $\mathbb{P}^1$, take the quotient of $\mathcal{O}\oplus\mathcal{O}(1)$ onto $\mathcal{O}$ for I, and the quotients onto $\mathcal{O}(1)$ given by $(s_j, 1)$ for the other labels, with s_j general linear forms. The common section is disjoint from the others; every other collision is an allowable pair. This constructs β_I. If $\deg D_j > 0$, the original collision of j with I is stable, and all pairs containing j are allowable; hence γ_j exists. Comparing the degrees m_i proves the first equality in (2.5).

For J maximal short, put its common section at infinity. Every proper subset of the outside labels is short, since $\nu(J) + \nu_j > 1$. The affine coordinates, modulo translations and scalars, give $Z_J = \mathbb{P}^{n-|J|-2}$ and $M_J = \mathcal{O}(-1)$, whence the normal bundle and degree against $-K_W$. In

(2.5) the outside labels of J are at least three: their total weight exceeds one, but every pair outside I is allowable. Thus this Z_J has positive dimension and β_J exists. All pairs containing $j \in J \setminus I$ are allowable, either inside J or outside I. Comparison of the degrees m_i proves the second equality. Every curve class is now in the cone on the indicated finite set, which proves the assertion after closure. In particular no line in a locus of dimension zero where sections coincide has been included. □

Corollary 2.9. If $W \neq \mathbb{P}^d$, then $K_W + (n-4)\pi^* A_X$ is nef for every ample line bundle A_X on X_μ.

Proof. Only negative extremal rays require checking. They are noncontracted because K_W is π-nef, so $\pi^* A_X$ has positive integral degree on their indicated generators. A maximal short set J with $|J| = 1$ would give $Z_J = W = \mathbb{P}^d$; otherwise their degrees against $-K_W$ are at most $n-4$. □

Lemma 2.10. Let $A = \pi^* A_X$ for an ample line bundle A_X on X_μ, and suppose the general point of $f(C)$ is in X_μ^{s}. Put $b = \deg f^* A$. Then $a \leq (n-2)b$; if all degrees m_i are nonnegative, $a \leq 2b$. A nonzero map $f^* T_W \to Q$ with $\deg Q = q \leq 0$ implies $a \leq (n-4)b$. More generally, if such a map satisfies

$$a - (n-4)q > 2b,$$

then at least two coincident sections have negative degrees m_i, $q < e$, and $e + \delta \leq b$. The splitting of Proposition 2.3 applies, and all maps to lines of degree less than e vanish on F.

Proof. The perturbation in Lemma 2.1 does not make an original pair of weight at least one allowable. All required γ_i in the preceding proof therefore lie in X_μ^{s}. Also $\mu(I) < 1$, and the constructed β_I has only collisions of allowable pairs of labels outside I and never meets its common section. Thus A has positive integral degree on each curve appearing in the first decomposition of (2.5). The decompositions give $a \leq 2b$ or $e + \delta \leq b$, respectively, and hence $a \leq (n-2)b$. For $q \leq 0$, nonnegative degrees m_i already give $a \leq 2b \leq (n-4)b$. Otherwise, for $j \notin I$, the lines M_j have positive degree and map trivially to Q; diagonal evaluation removes a single negative component, so $\ell \geq 2$ and $a = (n-2\ell)e + 2\delta \leq (n-4)b$. For the last assertion, nonnegative degrees m_i give no map when $q < 0$, and $a - (n-4)q \leq 2b$ when $q \geq 0$. If a negative degree exists but $q \geq e$, then $a \leq (n-2)e + 2\delta \leq (n-4)q + 2b$, again a contradiction. Thus $q < e$, the columns for the labels $j \notin I$ vanish, and diagonal evaluation gives $\ell \geq 2$. Proposition 2.3 concludes. □

2.3 The original arithmetic group and its automorphic line bundle

Suppose $X_\mu \simeq (\mathbb{B}^d/\Gamma)^{\mathrm{BB}}$ with Γ arithmetic. We keep this group throughout. The ball parametrizes negative lines, and $\mathcal{L}$ denotes the automorphic line bundle obtained from the tautological line bundle, with rational class L. The orbifold canonical class is NL. A normalized reflection acts identically on its mirror.

Proposition 2.11. The original Γ, all finite point stabilizers, and all full cusp stabilizers are generated by complex reflections. There is a Hermitian space (V, h) of dimension N over a CM extension K/F, of signature $(1, d)$ at the distinguished embedding and definite at all other real embeddings of F. The normalized reflections generate a group $\widehat{\Gamma} \subset \mathrm{U}(h, K)$ preserving one full $\mathcal{O}_K$-lattice and mapping onto Γ, with kernel a finite group $C_0 \subset \mu(K)$ of scalars. For $c = |C_0|$, $\mathcal{L}^c$ is an ample line bundle on X_μ. In the nonuniform case $F = \mathbb{Q}$ and $c \mid w_K = |\mu(K)| \in \{2, 4, 6\}$.

Proof. The genuine wall link is simply connected: it is the circle bundle of primitive Chern class $(1, 1)$ over a product of two projective spaces of positive dimension. Its second Betti number is one, but a finite linear quotient link is a rational homology sphere. Thus a genuine

wall vertex cannot be an interior ball quotient point. The rational smooth projective variety W is simply connected, by weak factorization [1]. Removing the inverse images of the cusps, of codimension at least two, preserves its fundamental group. Brown–Higgins [9, Propositions 1.8 and 2.8(iii)] now makes Γ generated by stabilizers. Interior smoothness and Chevalley–Shephard–Todd [11, Theorem 1.3] show that they are generated by reflections. The local cusp argument in Lemma 2.16, which does not use arithmeticity, gives the same assertion for the full cusp stabilizers.

Choose a normal torsion-free subgroup of Γ of index b_0. Reflection orders divide b_0, so determinants of normalized lifts lie in μ_{b_0}, and the scalars among them lie in μ_{Nb_0}. Their generated group is consequently a discrete finite central extension of the original arithmetic lattice. Apply [29, Corollary 4.4] to this group generated by reflections; its subgroup of elements of determinant one need not itself be generated by reflections. This gives the stated Hermitian arithmetic datum.

For completeness, the entire projective group is rational over F, not just a finite-index subgroup. Put $G = \mathrm{PGU}(h)$, $\mathfrak{g} = \mathrm{Lie}(G)$, and choose an arithmetic $\Delta \subset G(F)$ commensurable with Γ. For $\gamma \in \Gamma$, suitable finite-index subgroups $\Delta_1, \Delta_2 \subset \Delta$ satisfy $\gamma\Delta_1\gamma^{-1} = \Delta_2$. Borel density [6] and absolute irreducibility of the adjoint representation give $\operatorname{span}_F \mathrm{Ad}(\Delta_i) = \mathrm{End}_F(\mathfrak{g})$. Skolem–Noether therefore writes $\mathrm{Ad}(\gamma) = rB$, with $B \in \mathrm{GL}_F(\mathfrak{g})$ and r real at the distinguished place. Preservation of the bracket gives $B[X,Y] = r[BX,BY]$; a nonzero component gives $r \in F$. Thus $\gamma \in G(F)$. Hilbert 90 for the center $\mathrm{Res}_{K/F}\mathbb{G}_m$ of $\mathrm{GU}(h)$ supplies a K-rational similitude representative. For a reflection its mirror eigenvalue has multiplicity d; because $2d > N$, that eigenvalue lies in K. Dividing by it gives a normalized unitary reflection over K. The normalized lifts preserve a full lattice: take a finite-index normal integral subgroup and sum the finitely many translates of its invariant lattice. This uses the same lattice for the full group.

Interior stabilizers act on the tautological fiber through C_0, so $\mathcal{L}^c$ descends to the interior of the coarse quotient. At a cusp let ℓ generate its rational isotropic line. Every normalized cusp reflection fixes ℓ. With h linear in its second variable,

$$s([v]) = v/h(\ell, v)$$

is a nonzero tautological frame whose c-th power is invariant under the full cusp stabilizer. On a small cusp neighborhood U, glue $\mathcal{O}_U$ to the interior bundle by $1 \mapsto s^c$ on the punctured neighborhood. This first constructs a holomorphic line bundle M on X_μ. A sufficiently divisible ample Baily–Borel bundle [4], killing its finite determinant character, agrees with M^r on the interior for some $r > 0$. The isomorphism and its inverse extend across the finitely many cusps by normality, since $d \geq 2$. They remain inverse; GAGA and the ampleness of that Baily–Borel bundle show that M is algebraic and ample. Finally, a nonzero isotropic K-vector remains isotropic at every embedding. In the nonuniform case this excludes all other real places of F, so $F = \mathbb{Q}$. □

2.4 Fixed balls of finite reflection groups

Lemma 2.12. Let T be a finite reflection group in Γ, let $V_0 \subset V$ be its common fixed space, and put $E = V_0^{\perp}$, $s = \dim E$, $m = d - s \geq 1$. Assume $\mathbb{B}(V_0) \neq \varnothing$ and T is its full generic stabilizer. The effective image Π of its full normalizer is arithmetic and gives a finite map

$$j : (\mathbb{B}(V_0)/\Pi)^{\mathrm{BB}} \longrightarrow X_\mu$$

birational onto its image. The normalizer has finite image on E, after retaining C_0. If every full interior stabilizer along this ball is $T \times J_x$, with J_x acting trivially on E, the restriction of j to $\mathbb{B}(V_0)/\Pi$ is immersed and its normal bundle has components indexed by the degrees q of the basic invariants of T. A component of multiplicity u_q has finite projective monodromy and first Chern class $-qu_q j^*L$. The same assertion applies over the open subset X_μ^{s} of W.

Proof. The spaces V_0, E are defined over K. Its matrices on E preserve the same $\mathcal{O}_K$-lattice and are bounded at every Archimedean embedding. Minkowski discreteness therefore makes that image finite. The algebraic stabilizer of their orthogonal decomposition has a compact factor on E; its arithmetic subgroup projects to an arithmetic lattice on the ball by Borel–Harish-Chandra [7]. The effective full normalizer is now a discrete overgroup of this lattice and hence has finite index over it. After a finite-index passage for this construction, a rational flag of hyperplanes and the local extension in [21, Section 5] give the extension of j across the cusps; the flag lies in $E^{\perp}$, and all its normals are positive. When $F \neq \mathbb{Q}$ there are no cusps and properness is immediate. The extended map is proper because its source is projective; its interior fibers are discrete by discreteness and the local finite point groups, hence compact and finite. There are finitely many cusp preimages, so the extended map is finite. A generic identification carries the full generic stabilizer T to itself and therefore belongs to the full normalizer, proving birationality. When every full interior stabilizer along this ball is $T \times J_x$, with J_x acting trivially on E, the local map of coarse quotients is $y \mapsto (0, y)$ in invariant coordinates for T. The linear indecomposable invariants of degree q transform by finite matrices times the inverse q-th automorphic factor. Decomposable invariants have zero normal differential. This gives the stated normal bundles, including their finite characters. □

Lemma 2.13. If Z is the GIT quotient of $m \geq 5$ labeled points on $\mathbb{P}^1$ for some weights and the relative tangent line bundle M_* along the section labeled $*$ extends to an antiample $\mathbb{Q}$-Cartier divisor class on Z, then

$$Z = \mathbb{P}^{m-3}, \qquad M_* = \mathcal{O}(-1).$$

Proof. First assume that the weights lie on no stability wall. If $\mu_* + \mu_j < 1$, keep all marks except j at distinct fixed points and move j over $\mathbb{P}^1$. At each point where the marks of a pair $\{j, k\}$ that is not allowable collide, blow up that point and contract the original fiber. The complementary marks then coincide with weight $2 - \mu_j - \mu_k < 1$, so the completed family is stable. The centers avoid $*$; each contraction increases the self-intersection of the section labeled $*$ by one. Thus M_* has nonnegative degree on a complete nonconstant curve, a contradiction. All pairs with $*$ are therefore long. Putting $*$ at infinity and dividing affine coordinates by translations and scalars gives $\mathbb{P}^{m-3}$, with $M_* = \mathcal{O}(-1)$. At a genuine wall vertex the local class group is $\mathrm{Pic}(\mathbb{P}^{|I|-2} \times \mathbb{P}^{|I^c|-2})/\mathbb{Z}[\mathcal{O}(1,1)] \simeq \mathbb{Z}$. The class of M_* is its non-torsion generator, with sign determined by the side containing $*$. Hence M_* does not extend to a $\mathbb{Q}$-Cartier divisor class across that vertex. At a wall with a factor of dimension zero the quotient for the chamber retaining that factor is locally isomorphic to the quotient for the weights on the wall. A perturbation of the weights as in Lemma 2.1 makes these choices consistently, and the argument above for weights on no stability wall applies. □

Proposition 2.14. In Lemma 2.12, suppose that every full interior stabilizer along $\mathbb{B}(V_0)$ is $T \times J_x$, with J_x acting trivially on E, that T is irreducible on E, and that $s \leq d - 2$. The image of j is the closed locus $Z_I = \mathbb{P}^{d-s}$ where the sections in a set I with $|I| = s + 1$ coincide. More precisely, $(\mathbb{B}(V_0)/\Pi)^{\mathrm{BB}}$ is isomorphic to the GIT quotient of the original merged weights

$$\mu^I = (\mu(I), (\mu_j)_{j \notin I}), \qquad 0 < \mu(I) < 1, \qquad \sum \mu_i^I = 2,$$

on $n - s$ labels, including its stability walls. The automorphic line bundle of $\mathbb{B}(V_0)/\Pi$ is the restriction of the original $\mathcal{L}$. All basic invariants of T have the same degree δ, and M_I extends to the antiample $\mathbb{Q}$-Cartier divisor class $-\delta L|_{Z_I}$. In particular $s \leq 2$. For a cocompact arithmetic lattice, the case $s = d - 1 \geq 3$, with the other hypotheses unchanged, is also impossible. For equal weights, such a group T is also impossible when $n > 2(s + 2)$.

Proof. In $(\mathbb{B}(V_0)/\Pi)^{\mathrm{BB}}$ choose a general complete intersection curve avoiding the finitely many cusps and wall vertices, and not contained in any proper inverse image of a locus where sections

coincide. After a finite cover its normal bundle is a sum of line bundles with degrees $-q_\alpha t$, $t > 0$. Proposition 2.2 factors its stalkwise normal quotient through $M_I^{\oplus(|I|-1)}$, so $|I| - 1 \geq s$. The curve selection implies that the entire image lies in Z_I; dimension gives the reverse inequality. The selected curve lies in X_μ^{s}, so $\mu(I) < 1$. Thus $|I|-1 = s$, and exactness of $\mathrm{SL}(2,\mathbb{C})$-invariants identifies the closed diagonal image with the normal GIT quotient of the original merged weights μ^I. Finite birationality then identifies $(\mathbb{B}(V_0)/\Pi)^{\mathrm{BB}}$ with this quotient. Negative lines in V_0 are the same lines in V, so their automorphic line bundles agree. The normal bundle is $M_I^{\oplus s}$. Its semistability on the curve forces all the q_α to coincide. The reflection classification [11, Table 1] has no irreducible example with basic invariants of equal degree in rank at least 3.

On the stable interior, the determinant transition differs from the automorphic factor of degree $-s\delta$ by a finite character. For an integer $r > 0$ killing that character and all scalar denominators, there is an isomorphism

$$M_I^{sr} \simeq (\mathcal{L}|_{Z_I})^{-s\delta r}.$$

Since $\dim Z_I \geq 2$, this extends M_I to a $\mathbb{Q}$-Cartier divisor class across the finitely many points of Z_I outside the stable interior; its inverse is ample. Lemma 2.13 gives the projective space assertion. In the compact case of rank $d-1$, use the whole curve $\mathbb{B}(V_0)/\Pi$, which has no boundary, for the same quotient onto a line bundle of negative degree and semistability argument. The resulting contradiction for basic invariants of equal degree occurs before any use of Lemma 2.13; that lemma requires at least five merged labels. For equal weights, Proposition 2.7 excludes the antiample extension when $n > 2(s+2)$. □

Lemma 2.15. In the nonuniform case, a finite reflection group of rank $s \leq d-2$ embeds, with the same reflection representation and reflection orders, in the quotient of a full cusp stabilizer of the original Γ by its intersection with the Heisenberg group.

Proof. Its fixed Hermitian space over the imaginary quadratic field has dimension at least three. The underlying indefinite rational quadratic form has at least six variables, so Meyer [28, Chapter IV, Section 3.2] supplies a fixed rational isotropic vector. The Heisenberg lattice is torsion-free, hence the finite group injects into the quotient by the Heisenberg lattice, with unchanged nonzero roots and eigenvalues. This argument does not apply in rank $d-1$. □

2.5 Central translations in full cusp stabilizers and cusps whose germs are cones over products of projective spaces

The local arguments of this subsection apply to any nonuniform ball lattice. Use Siegel coordinates $(z,w) \in \mathbb{C}^r \times \mathbb{C}$, where $r = d-1$, and let Z be the center of the Heisenberg group, acting by $w \mapsto w + c$. For a full cusp stabilizer P, let N_P be its intersection with the Heisenberg group, $F_P = P/N_P$, and

$$P \cap Z = N_P \cap Z = c_*\mathbb{Z}, \qquad q_* = \exp(2\pi i w/c_*).$$

The lattice Λ_P is the image of N_P in the group of translations of $\mathbb{C}^r$. The affine action on $A_P = \mathbb{C}^r/\Lambda_P$ is faithful and has no nonidentity pure translations; no common fixed point is assumed. For $\lambda \in \Lambda_P$, write a lift in N_P as $(z,w) \mapsto (z+\lambda, w + a_\lambda(z))$. The translation multipliers $\exp(2\pi i a_\lambda(z)/c_*)$ define a negative line bundle $\mathcal{V}_*$, with dual ample bundle M_* and its induced F_P-linearization. The invariant metric is $|q_*|^2 \exp(4\pi\|z\|^2/c_*)$, of negative curvature. The cusp contracts the zero section of $\mathcal{V}_*/F_P$, with ring

$$R_P = \bigoplus_{k\geq 0} H^0(A_P, M_*^k)^{F_P}. \tag{2.6}$$

It is enough to use a high Veronese subring to construct the contraction; generation in degree one is not assumed. This is the germ at the cusp of the minimal compactification [23, Sections 2.1–2.3]; in the arithmetic case, it is the germ of the Baily–Borel compactification [21, Section 4.3]. For a neat subgroup $P_0 \subset P$ of finite index, let $f_0 : A_0 \to A_P$ be the induced covering of abelian varieties. Write $P_0 \cap Z = c_0\mathbb{Z}$, put $q_0 = \exp(2\pi i w/c_0)$, and let M_0 be the dual of the negative line bundle for P_0. If $c_0 = e_0 c_*$, then $q_* = q_0^{e_0}$ and $f_0^* M_* = M_0^{e_0}$. Both the constant multipliers and the remaining finite action are retained in these identities.

Lemma 2.16. If the punctured cusp germ is smooth and simply connected, P is generated by complex reflections and F_P is a reflection group. At a smooth cusp, the linear action of F_P on $\mathbb{C}^r$ is irreducible. A cusp with germ $\operatorname{Cone}(\mathbb{P}^a \times \mathbb{P}^b, \mathcal{O}(1,1))$, $a, b \geq 1$, has exactly two irreducible factors in the linear action of F_P on $\mathbb{C}^r$, of ranks a, b. An irreducible factor, with the lift of its action to M_*, is strongly crystallographic and its ring of invariant sections is polynomial.

Proof. Apply Brown–Higgins to a contractible deep horoball and then Chevalley–Shephard–Todd at its smooth points. The first Betti number of the link excludes nonzero vectors of $\mathbb{C}^r$ fixed by F_P. Write the orthogonal irreducible factors as F_j of ranks r_j. Their invariant cohomology is $\mathbb{C}[\omega_j]_j/(\omega_j^{r_j+1})_j$, $\deg \omega_j = 2$, by [14, Theorem 1.2]. The Gysin sequence for the negative bundle gives $b_{2i} = h_i - h_{i-1}$, where $\sum h_i t^i = \prod_j (1 + t + \cdots + t^{r_j})$, in the first half of the link dimensions. Comparison with a link that is a sphere or with the link of the Segre cone gives, respectively, one factor or two factors of ranks a, b. Rains [26, Proposition 2.6] applies to the lift of the action to M_*, making the action strongly crystallographic; [26, Theorem 2.7] gives the polynomial ring. Evaluation and averaging on free orbits in A_P give invariant sections in every sufficiently large degree. Thus the degrees of the generators of the ring of invariant sections have greatest common divisor one in the grading by powers of M_*. If a group μ_g of scalars acting only on the fibers of $\mathcal{V}_*$ is first removed, the correct new coordinate is $q_{\text{new}} = q_{\text{old}}^g$, with period $c_{\text{new}} = c_{\text{old}}/g$, and the degrees of the generators divided by g. □

Lemma 2.17. Let $(F_P)_x$ be the full stabilizer of $x \in A_P$, and let χ_x be its character on $(\mathcal{V}_*)_x$. Its kernel T lifts to the full finite stabilizer of a nonzero point over x in a sufficiently deep horoball. If T fixes no nonzero vector of $\mathbb{C}^r$, the ball fixed by T is a disk ending at this cusp. For the disk modulo its full normalizer, the local coordinate of the coarse quotient at this end is

$$Q = q_*^{\kappa_x}, \qquad \kappa_x = |\operatorname{im} \chi_x|.$$

Proof. Choose a lift of a nonzero fiber point. Unique correction by the freely acting Heisenberg lattice lifts exactly $\ker \chi_x$. Its fixed Hermitian plane gives the disk when the fixed space of T in $\mathbb{C}^r$ is zero. The full central translations retain the period c_*, and the residual stabilizer acts by χ_x, giving the coordinate Q. □

Lemma 2.18. Let $\mathbb{C}^*$ act on $\mathbb{C}^{p+q}$, $p, q \geq 2$, with positive weights u_i and negative weights $-v_j$. Assume that, after any one weight is deleted, the remaining weights have greatest common divisor one. If the affine quotient is smooth off the vertex and has embedding dimension pq at the vertex, all weights equal one.

Proof. At an orbit on which only X_i and Y_j are nonzero, the transverse stabilizer is $\mu_{\gcd(u_i, v_j)}$. The assumption on greatest common divisors excludes ineffective elements and pseudoreflections, so smoothness forces $\gcd(u_i, v_j) = 1$. The pq monomials $X_i^{v_j} Y_j^{u_i}$ are therefore indecomposable. If $u_i > 1$, take $j \neq k$ and $1 \leq e < u_i$ with $v_j + e v_k \equiv 0 \pmod{u_i}$. The invariant $X_i^{(v_j + e v_k)/u_i} Y_j Y_k^e$ has an indecomposable factor divisible by none of the monomials $X_i^{v_j} Y_j^{u_i}$, contradicting the embedding dimension. Exchange signs to conclude. □

Theorem 2.19. For a cusp with germ $\mathrm{Cone}(\mathbb{P}^a \times \mathbb{P}^b, \mathcal{O}(1,1))$ as in Lemma 2.16, there are product decompositions $A_P = A_1 \times A_2$ and $F_P = F_1 \times F_2$, with $\dim A_1 = a$ and $\dim A_2 = b$. Let P_j be the subgroup generated by all reflections whose roots lie in the j-th factor. Then $P_j \cap Z = P \cap Z$. With degrees measured by c_*, its ring of invariant sections is polynomial on r_j+1 generators of degree one, where $r_1 = a, r_2 = b$. The contraction is that of $\mathcal{O}_{\mathbb{P}^a \times \mathbb{P}^b}(-1,-1)$. For $r_j \geq 3$, F_j is S_{r_j+1} or $G(m_j, 1, r_j)$, with $m_j \in \{2,3,4,6\}$. The full F_P occurs as a finite stabilizer arbitrarily deep in the cusp.

Proof. A reflection's displacement and central multiplier belong to its root factor, and orthogonal reflections commute. For $j = 1, 2$, put $N_j = P_j \cap N_P$, and let Λ_j be its image in the group of translations of $\mathbb{C}^r$. Since P is generated by reflections, $P = P_1 P_2$, $N_P = N_1 N_2$, and $\Lambda_P = \Lambda_1 \oplus \Lambda_2$. This gives the product decomposition $A_P = A_1 \times A_2$. Write $P_j \cap Z = c_j \mathbb{Z}$ with $c_j = d_j c_*$, so $\gcd(d_1, d_2) = 1$. For the j-th factor, let w_j be the coordinate along the center in its Siegel coordinates, put $q_j = \exp(2\pi i w_j / c_j)$, and let M_j be the dual of the negative line bundle on A_j defined by the translation multipliers of P_j. Put $Z_* = P \cap Z = c_* \mathbb{Z}$. Rains gives $R_1 = \mathbb{C}[x_0, \ldots, x_a]$ and $R_2 = \mathbb{C}[y_0, \ldots, y_b]$, with degrees α_i, β_j of greatest common divisor one. Including constants and characters in the multipliers gives

$$q_* = q_1^{d_1} q_2^{d_2}, \qquad M_* = M_1^{d_1} \boxtimes M_2^{d_2}, \qquad R_P = \bigoplus_{k \geq 0} (R_1)_{d_1 k} \otimes (R_2)_{d_2 k}.$$

Thus R_P is the torus invariant ring of weights $d_2\alpha_i, -d_1\beta_j$.

The residual $\mu_{d_j} = P_j Z_* / P_j$ acts freely off the vertex of the j-th factor cone. Otherwise combine a fixed factor point with a free orbit in the other factor. Its finite stabilizer lies in $P_j Z_*$, but smoothness of the punctured germ makes it generated by reflections in P_j, a contradiction. On the coordinate axis $x_i \neq 0$, the residual group μ_{d_1} has stabilizer of order $\gcd(d_1, \alpha_i)$, since its action is $x_i \mapsto \zeta^{\alpha_i} x_i$. Freeness therefore gives $\gcd(d_1, \alpha_i) = 1$ for every i, and similarly $\gcd(d_2, \beta_j) = 1$ for every j. After deleting the positive weight $d_2\alpha_i$, the remaining weights have greatest common divisor

$$\gcd\left(d_2 \gcd_{k \neq i} \alpha_k, d_1 \gcd_j \beta_j\right) = \gcd\left(d_2 \gcd_{k \neq i} \alpha_k, d_1\right) = 1.$$

Deleting a negative weight gives the same conclusion with the factors exchanged. This argument includes $a = 1$ or $b = 1$, because each factor still has a generator after one weight is deleted. The analytic germ comparison gives smoothness away from the vertex and embedding dimension $(a+1)(b+1)$. Lemma 2.18 forces $d_1 = d_2 = \alpha_i = \beta_j = 1$. Consequently $A_j / F_j = \mathbb{P}^{r_j}$, the characters of all point stabilizers on the fibers of the negative bundles are trivial, and the negative bundles descend to $\mathcal{O}(-1)$ and $\mathcal{O}(-1,-1)$. The faithful affine actions have no pure translations. The classification of smooth quotients [3, Theorem 1.1(3)] gives a common fixed point, since the generators have fixed points; [2, Theorem 1.1] gives the indicated irreducible groups. The stabilizer of their product acts trivially on the fiber of $\mathcal{V}_*$. Lemma 2.17, followed by central imaginary translation, realizes the full group arbitrarily deep. □

3 Fixed curves

The results of this section apply for $n \geq 6$, subject to the hypotheses of each statement. Put $d = n - 3$, $N = d + 1$, $H = -K_W$, $a = H \cdot f_* C$, and $t = L \cdot f_* C > 0$. Here $W = X_\nu$ is the smooth GIT quotient of Lemma 2.1, obtained by perturbing μ to weights ν in a nearby chamber, and $f : C \to W$ comes from the quotient $\mathbb{B}(V_0)/\Pi$ of Lemma 2.12 in the original lattice. Let $g = g(C)$ be the genus of C. We do not assume $a > 0$, immersion, or separate descent of an automorphic factor and a finite character. All degrees of invariant sections at cusps below are measured by the positive generator c_* of $P \cap Z$ for the full cusp stabilizer P.

3.1 The determinant of the differential and one row of its inverse

Proposition 3.1. Let X be the Baily–Borel compactification of an arithmetic ball quotient of dimension d, and let K be the full generic stabilizer of a ball of dimension $m = d - k \geq 1$ fixed by a finite subgroup. Assume that K is generated by reflections and has rank k. Let $f : Y \to X$ be the finite morphism from the compactified quotient of this ball by the full effective normalizer of K in the original lattice. Assume that K_X is $\mathbb{Q}$-Cartier near $f(Y)$, and that the general point of every interior divisor of Y maps to the smooth locus of X. Put $H = -K_X$, and write D_G for the sum of the degrees of the basic invariants of a reflection group G. There is an effective $\mathbb{Q}$-divisor on Y with class

$$f^*H + (D_K + m + 1)f^*L.$$

Its interior support consists of the divisors with a larger full stabilizer. For $m > 1$, let Σ_K be the sum, as reduced cycles, of the images in X of all fixed balls of dimension m whose full generic stabilizer acts on the normal space by a complex reflection representation isomorphic to that of K, and define Σ_J in the same way for the groups J of rank $k+1$. Let $n_{K,J}$ count the lines ℓ in the normal representation of J whose full pointwise stabilizer J_ℓ acts on the directions transverse to ℓ by a representation isomorphic to that of K. Then

$$(H + (D_K + m + 1)L)\Sigma_K = \sum_{\operatorname{rank} J = k+1} (D_J - D_K - 1)\frac{|K| n_{K,J}}{|J|}\Sigma_J.$$

There is no cusp divisor term when $m > 1$.

For $m = 1$, write $Y = C$, let $S \subset C$ be the cusp set, and assume that X is smooth near $f(C)$, with polynomial rings of invariant sections at its ends. Put $a = \deg f^*H$, $t = \deg f^*L$, and $e_y = |N_{J_y}(K_y)|/|K_y|$, where J_y is the full interior stabilizer and K_y the pointwise stabilizer of the corresponding local axis. At an end c, let $b_{c,1}, \ldots, b_{c,d}$ be the degrees of the generators of the ring of invariant sections, measured by the positive generator c_* of $P \cap Z$ for the full cusp stabilizer P, and put $A_c = \sum_i b_{c,i}$. Let κ_c be the order of the character of the full stabilizer on the fiber of $\mathcal{V}_*$, so that the local coordinate Q of the coarse curve at c and the coordinate $q = q_*$ for the full cusp stabilizer P at $f(c)$ satisfy $Q = q^{\kappa_c}$. Then

$$a + (D_K + 2)t = \sum_{y \in C \setminus S,\ J_y \neq K_y} \frac{D_{J_y} - D_K - 1}{e_y} + \sum_{c \in S} \frac{A_c}{\kappa_c}. \tag{3.1}$$

No immersion or separate descent of the line bundle for a basic invariant is required. If H is nef, every component with $m > 1$ contains an interior stratum of rank $k + 1$.

Proof. Lemma 2.12 gives Y and the finite representations on invariant normal directions. Indecomposable invariants of degree e transform by $\mathcal{L}^{-e}$ with their finite representation; terms in the square of the invariant ideal disappear from the normal differential. Repeated degrees are kept in one vector bundle. The determinant of the tangent bundle of the fixed ball has automorphic factor $\mathcal{L}^{-(m+1)}$ and a finite character. Thus the determinant of the domain of the differential of the quotient map is $\mathcal{L}^{-(D_K+m+1)}$ with a finite character. Take a sufficiently divisible power N to remove this character, make NK_X Cartier, and make $\mathcal{L}^N$ descend to a Baily–Borel line bundle. The determinant then gives a nonzero section of the line bundle

$$f^*\mathcal{O}_X(-NK_X) \otimes (f^*\mathcal{L}^N)^{\otimes(D_K+m+1)}.$$

These finite representations are removed on a normal neat cover; their automorphic factors are retained.

At an interior divisor with full group J, discard the common fixed directions and write u for the remaining axis coordinate, η_i for the basic invariants of K, and F_j for those of J. Homogeneity and generic invertibility give

$$\det \frac{\partial(F_1,\dots,F_{k+1})}{\partial(u,\eta_1,\dots,\eta_k)}(u,0) = c_0 u^{D_J - D_K - 1}, \qquad c_0 \neq 0.$$

The group $N_J(K)/K$ acting on the axis has order $e = |N_J(K)|/|K|$, so the divisor coefficient after division by N is $(D_J - D_K - 1)/e$. Changes of invariant coordinates multiply the determinant by a unit, and the tangent column is included. The section is regular in codimension one away from cusps and extends across the remaining interior subset by normality of Y. When $m > 1$, the cusp set also has codimension at least two, so the same extension applies. In a toroidal cover, a boundary divisor maps to a point and has zero pushforward as a divisor on Y.

For $m = 1$, extend in the logarithmic frame containing $q\,\partial/\partial q$ at each end. The frame of Proposition 2.11 on the punctured neighborhood of the cusp, whose c-th power is invariant under the full cusp stabilizer, introduces no power of q_*. At an end c, let x_c be the point of A_P determined by the curve f. Write the invariant theta functions $\theta_{c,j}$ in local invariant coordinates $\boldsymbol{\eta}$, centered at x_c, as $\bar\theta_j(\boldsymbol{\eta})$. Thus $\theta_{c,j}(x_c) = \bar\theta_j(0)$; we omit the index c when one end is under consideration. At an end the contraction and its constant differential are

$$U_j = q^{b_{c,j}}\bar\theta_j(\boldsymbol{\eta}), \qquad J_c = [(b_{c,j}\bar\theta_j(0))_j, D_{\boldsymbol{\eta}}\bar\theta(0)]. \tag{3.2}$$

The differential is $\operatorname{diag}(q^{b_{c,j}})J_c$, and J_c is invertible by smoothness of the punctured quotient with full stabilizer K. If $Q = (q')^r$ on the neat cover, its determinant has order rA_c/κ_c, giving A_c/κ_c after descent. This proves effectivity and the curve formula. For $m > 1$, summing over the orbits of local axes gives $\sum 1/e = |K|n_{K,J}/|J|$; pushing the divisors forward proves the formula for $(H + (D_K + m + 1)L)\Sigma_K$. Finally, $(f^*H + (D_K + m + 1)f^*L)(f^*L)^{m-1} > 0$ if H is nef, so the effective divisor is nonzero and contains the asserted interior stratum. □

Lemma 3.2. For $r \geq 5$, the linear normalizer of $G(m,p,r)$, $m \geq 2$, $p \mid m$, lies in $\mathbb{C}^*G(m,1,r)$ and preserves the line spanned by each elementary symmetric invariant and the line spanned by $P_0 = (x_1 \cdots x_r)^{m/p}$.

Proof. The bounds $m^k k!(r-k)!/p$, $k > 0$, and $r!$, $k = 0$, for point stabilizers are logarithmically convex. The strict inequality $m^{r-1} > pr$ identifies the coordinate axes by maximal stabilizers. A normalizer permutes these axes; the reflections of $G(m,p,r)$ exchanging two coordinates up to factors in μ_m then force all diagonal ratios into μ_m. □

Lemma 3.3. If $u \mapsto \zeta_e u$ acts on a line by ζ_e^a, $0 \leq a < e$, the line bundle of its invariant sections on the coarse curve has degree equal to its orbifold degree minus a/e.

Proof. The invariant frame is u^a times a function of u^e. □

For $r = d - 1 \geq 3$, $m \geq 2$, and $p \mid m$, put $T = G(m,p,r)$ and

$$f_j = e_j(x_1^m,\dots,x_r^m)\ (1 \leq j < r), \quad P_0 = (x_1 \cdots x_r)^{m/p}, \quad D = \frac{mr(r-1)}{2} + \frac{rm}{p}.$$

Set $k = rm$ if $p = 1$, and $k = (r-1)m$ otherwise.

Proposition 3.4. Suppose every larger interior group on a disk fixed by T is $G(m,p,r+1)$, with the prescribed coordinate axis, or $T \times \mu_e$, with μ_e trivial on the normal space. Suppose every end is a smooth cusp whose full stabilizer H_c in F_P of the point $x_c \in A_P$ has $\ker\chi_c = T$, $|\operatorname{im}\chi_c| = \kappa_c$. Suppose the full normalizer preserves the degree k indecomposable direction and the stabilizers of points of the curve in its effective image fix that direction. Let D_R be the

reduced divisor of the points with stabilizer $G(m,p,r+1)$, put $R=\deg D_R$, and let s count the ends. Put $\sigma=\sum_{\text{product}}(1-e^{-1})$, where e is the effective inertia order at the corresponding point with stabilizer $T\times\mu_e$. Put

$$B_c=\frac{\sum_i b_{c,i}}{\kappa_c},\quad \nu_c=\left\lfloor\frac{\max_i b_{c,i}}{\kappa_c}\right\rfloor,\quad \beta=r+\frac{1}{p}-\frac{1}{m}.$$

There is a line bundle E_k of degree $-kt$, and

$$2g-2+s+(1-m^{-1})R+\sigma=2t, \tag{3.3}$$

$$a=\sum_c B_c+\beta R+\sigma-(D+2)t, \tag{3.4}$$

$$0\neq f^*T_W\longrightarrow E_k(D_R+\sum_c \nu_c[c]),\qquad h=-kt+R+\sum_c\nu_c. \tag{3.5}$$

Proof. Let j_γ be the factor of automorphy of the original automorphic line bundle $\mathcal{L}$ for the chosen lift of γ, and let $\chi_q(\gamma)$ be the factor from the finite action on the coefficient line of an invariant of polynomial degree q. On the line bundle for this invariant, the transition is $j_\gamma^{-q}\chi_q(\gamma)$. Scalar lifts cancel between the two factors, so their product is effective. A neat cover of degree v, of any genus, computes degree $-qvt$. This proves the assertion for E_k; Riemann–Hurwitz for the disk gives (3.3). Put $\delta=m/p$. The local map at a point with stabilizer $G(m,p,r+1)$ is

$$g_1=u^m+f_1,\quad g_j=f_j+u^m f_{j-1}\ (2\le j<r),\quad g_r=P_0^p+u^m f_{r-1},\quad Z=u^\delta P_0. \tag{3.6}$$

Its Jacobian determinant along the axis is $mu^{rm+\delta-1}$, so Proposition 3.1 gives β. The residual action sends P_0 to $\zeta_m^{-\delta}P_0$, so passing to the coarse curve changes the degree of the line bundle for P_0 by $-(1-1/p)R$. For $p>1$, write $P_0=u^{m-\delta}\psi(u^m)$. After elimination of the tangent column, the block of the elementary symmetric invariants has diagonal $v=u^m$ and superdiagonal 1. For $p=1$, the same follows from $dg_1=dv+df_1$. One row of the inverse differential has a simple pole: $\dot P_0=v^{-1}\dot Z$, or $\dot f_{r-1}=v^{-1}\dot g_r$. At a point with stabilizer $T\times\mu_e$, the inverse differential has no pole in the normal directions. At a cusp invert the differential in (3.2). The rows of the inverse for the directions of A_P have pole at most $q^{-\max b_{c,i}}$. For the row of the inverse differential being used, the finite inertia group at the cusp acts trivially on its target invariant line, so the entries of this row are functions of $Q=q^{\kappa_c}$, and the pole bound $\max_i b_{c,i}$ in q becomes ν_c in Q. Equation (3.1) now gives (3.4). □

Whenever a unique degree m line descends, mt is integral. Repeated degrees justify only the greatest common divisor of degrees of lines actually shown to descend.

3.2 Elimination of coefficients and injections into sums of line bundles

Lemma 3.5. Let $Q,E_1,\dots,E_q$ be line bundles on C, and consider homomorphisms $f^*T_W\to Q$ with zero coefficient on the tangent direction of C. Suppose their coefficients $b_i\in H^0(C,QE_i^{-1})$ with respect to the line bundles E_i determine them injectively. At a reduced divisor D, suppose the conditions for their coefficients to extend across D come from the upper bidiagonal differential with first row $(a(u),1,0,\dots)$, $a(u)\not\equiv 0$, and remaining diagonal u and superdiagonal 1. At each end c, suppose the differential is $\operatorname{diag}(u^{\nu_i(c)})J(u)$. Assume the exponents are nondecreasing and every leading block at a strict increase is invertible at zero. If

$$\deg QE_i^{-1}(-D-\sum_c \nu_i(c)[c])<0\quad(1\le i<q),$$

and $\deg QE_q^{-1}(-D-\sum_c\nu_q(c)[c])=0$, the space of these homomorphisms has dimension at most one.

Proof. The vanishing of the coefficient on the tangent direction kills the coefficient of the first row at D. Once the preceding coefficients b_i vanish, the next is divisible by u. The invertible leading blocks at ends give the analogous successive divisibility. Negative degree kills the first $q-1$ coefficients, and degree zero bounds the last by one. □

Lemma 3.6. Suppose $\bigoplus_{i=1}^{r} \mathcal{O}_{\mathbb{P}^1}(-ij) \to V$, $j \geq 1$, is invertible away from R points and has the bidiagonal matrix just described in its individual invariant frames. Then $V = \bigoplus_{i=1}^{r} \mathcal{O}(R-ij)$. For $p > 1$, the block of the elementary symmetric invariants has length $r-1$, and the separate direction of P_0 has degree $(R-rj)/p$.

Proof. A map $V \to \mathcal{O}(q)$ has coefficients $b_i \in H^0(\mathcal{O}(q+ij))$ satisfying $b_1 = vc_1$, $b_i = c_{i-1} + vc_i$. Induction kills the coefficients with $q + ij < R$; thereafter evaluation at R points is surjective. Thus the space of maps $V \to \mathcal{O}(q)$ has dimension $\sum_i \max(q+ij+1-R, 0)$, which determines the splitting. In the case of the line bundle corresponding to P_0, add the contribution R of its elementary modification at the R points to the change $-(1-1/p)R$ on passing to the coarse curve and to its degree $-rj/p$ before these changes. □

Proposition 3.7. Suppose $C = \mathbb{P}^1$ satisfies the hypotheses of Proposition 3.4 with every invariant direction preserved. Put $j = mt$, $\rho = \operatorname{lcm}_c \kappa_c$, and $B = \sum_c \max_i b_{c,i}/\kappa_c$. When at most two widths are nontrivial, use the cyclic degree ρ cover $\varrho : C' \to C$ branched at two values containing those ends and avoiding the points with a larger stabilizer. Use another end as the second value if possible; let s_0 count auxiliary branch values, and put $B_T = B + s_0(1-1/\rho)$. For $\rho = 1$ use the identity cover and $s_0 = 0$. Let $(d_i) = (m, 2m, \dots, rm)$ for $p = 1$, or $(m, \dots, (r-1)m, rm/p)$ otherwise. Then $C' = \mathbb{P}^1$, and the degrees of the line bundles in the following injections are integers:

$$(f\varrho)^* T_W \hookrightarrow \mathcal{O}(\rho B_T + 2) \oplus \bigoplus_i \mathcal{O}(\rho(B_T - d_i t)) \qquad (R = 0), \tag{3.7}$$

$$\varrho^* N_f^{\mathrm{sat}} \hookrightarrow \begin{cases} \bigoplus_{i=1}^{r} \mathcal{O}(\rho(B + R - ij)), & p = 1, \\ \bigoplus_{i=1}^{r-1} \mathcal{O}(\rho(B + R - ij)) \oplus \mathcal{O}(\rho(B + (R - rj)/p)), & p > 1. \end{cases} \tag{3.8}$$

For $R = 0$, the comparison using the normal bundle allows arbitrarily many nontrivial widths and any finite cover on which they become trivial; its genus need not be zero. The critical divisor of the original map has degree

$$c_f = \sum_c \left(\min_{i : \theta_{c,i}(x_c) \neq 0} b_{c,i}/\kappa_c - 1 \right).$$

If the quotient is constructed from ℓ coincident labels of common degree $-e < 0$, sort the degrees of the target line bundles, divided by ρ, as $v_1 \geq \cdots \geq v_q$. A necessary condition is

$$\#\{i : v_i \geq e\} \geq k_0, \qquad A \leq \sum_{i=1}^{k_0} v_i, \tag{3.9}$$

where $(k_0, A) = (n - \ell - 2, a + (\ell - 1)e)$ for the comparison using the tangent bundle, and $(k_0, A) = (n - \ell - 3, a - 2 - c_f + (\ell - 1)e)$ for the comparison using the normal bundle. Failure at the least allowed e excludes every larger e with all other data unchanged.

Proof. By the Riemann–Hurwitz formula, the cover branched at two points is rational. At an end of ramification index e_c, write $q = x^{e_c/\kappa_c}$. Saturating the inertia gives the degrees $-\rho ij$ of the line bundles for the elementary symmetric invariants and the degree $-\rho rj/p - \rho(1-1/p)R$ of the line bundle corresponding to P_0. Extend through the points with stabilizer $G(m, p, r+1)$ by (3.6); Lemma 3.6 gives the summands of the normal bundle. For $R = 0$, the local charts given

by the full normalizer glue to an extension of $\bigoplus_i \mathcal{O}_{C'}(-\rho d_i t)$ by $T_{C'}$ on the cover C', which splits by $H^1(\mathcal{O}(2+\rho d_i t)) = 0$. At the cusps, the inverse of the differential has poles of total degree ρB; an auxiliary branch value adds only a pole of order $\rho - 1$ in the tangent direction. This proves both injections. When we use a cover of arbitrary genus for the comparison using the normal bundle, no interior modification is needed and the line bundles for the basic invariants already form a direct sum. A cover on which all widths become trivial is the normalization of $y^\rho = \prod_c (Q - a_c)^{\rho/\kappa_c}$; extra branching adds no pole in the directions of A_P. Saturating the image of the differential commutes with finite pullback, and differentiating the image of the curve under the map in (3.2) gives c_f. The kernel in Proposition 2.3, or its quotient by the saturated image T_f of df, has rank k_0, degree A, and is generated by line bundles of degrees at least e. Maps to target line bundles of degree below e vanish. A nonzero maximal minor proves the degree inequality in (3.9). As e increases, the two necessary conditions in (3.9) become more restrictive. □

3.3 Two geometric inequalities

Lemma 3.8. At a fixed point $x \in A_P$ the cusp width is $\kappa = \gcd\{b_i : \theta_i(x) \neq 0\}$. If $b_i \in \{1, 3\}$ and $\kappa = 1$, the curve f is immersed at the end.

Proof. The fiber stabilizer of this fixed point x is defined by $\lambda^{b_i} = 1$ for all nonzero coordinates. A width of one forces a nonzero section of degree one, giving a nonzero derivative in $U_i(Q)$. Properness extends the curve to W, and $d\pi\, df_W = df_X$ preserves this immersion. □

Lemma 3.9. A k-dimensional family of distinct complete embedded rational curves in the smooth coarse ball quotient has at least $k + 3$ branch values on a general member, provided it avoids the strata of codimension at least two defined by stabilizers and meets the branch divisor transversely. In particular, a coordinate curve γ_i on W satisfies

$$L \cdot \gamma_i \geq \frac{d-2}{2(d+1)} = \frac{n-5}{2(n-2)}. \tag{3.10}$$

Proof. On a normal neat cover the pullback families are smooth proper after restricting parameters; choose connected components of fixed degree after finite base change. Each cyclic inertia acts freely on sheets, so every component retains all branch values. With b branch values, finite permutation monodromy tuples leave at most $b - 3$ parameters. The cases $b \leq 2$ would make these components rational, impossible in a ball quotient. Let X' be the smooth ball quotient associated with the normal neat cover. For one such component D, fixed as an abstract curve, and every nonconstant map $v : D \to X'$, negative bisectional curvature and the integrated Bochner formula give $H^0(D, v^* T_{X'}) = 0$ [19]. Hence $k \leq b - 3$. The formula for the canonical class, evaluated on γ, gives $(d+1)L\gamma = -2 + \sum(1 - e_x^{-1}) \geq -2 + b/2$, where e_x is the local ramification index of the induced neat covering of the curve at the branch value x. For an existing coordinate curve γ_i, all pairs containing i are allowable. The open set where the other marks are distinct is therefore the universal smooth projective line bundle over $M_{0,n-1}$. Its fibers are the complete embedded coordinate curves and its base has dimension $d - 1$. The chamber perturbation never allows a pair of original weight at least one, so the family lies in X_μ^{s}. The finite cusp set has proper image in the base. A stabilizer stratum of codimension at least two has base image of dimension at most $d - 2$, so a general fiber avoids it. A branch divisor with nondominant base image is also avoided; otherwise its projection is generically smooth in characteristic zero, making a general fiber transverse. The family thus satisfies the hypotheses already proved, and $H \cdot \gamma_i = 2$ gives the bound. □

Lemma 3.10. Let $Z = \mathbb{P}^k$, $k \geq 2$, be closed in the Baily–Borel compactification of an arithmetic ball quotient, with $Z^\circ = Z \setminus S$ in its smooth interior for a finite set S containing

the boundary points. Suppose $N_{Z^\circ} = \mathcal{O}(-1)^{\oplus c}$ and the class L of the original lattice restricts to $\lambda\mathcal{O}(1)$. Retain the finite maps j of Lemma 2.12, defined by the full normalizers, and the pullbacks j^*L. Let the full stabilizer of a general point of the ball over Z° have rank $s \le c$. If $s = c$, the Baily–Borel compactification of the quotient of its fixed ball by the full normalizer is Z. If $s < c$, then $\lambda \ge 1$; equality makes the inverse image of Z° locally totally geodesic at its smooth points in the fixed ball of that stabilizer.

Proof. For $s = c$, the quotient of the fixed ball by the full normalizer maps finitely to Z with a unique generic preimage over Z°, so its compactification is finite birational to normal Z, hence isomorphic to it. For $q = c - s > 0$, pass to a neat cover Y_0 of the quotient $\mathbb{B}(V_0)/\Pi$ and resolve the graph of the Gauss map of the inverse image of Z°. Denote this resolution by $\widetilde{Z}$, let $\rho : \widetilde{Z} \to Y_0$ be the resulting map, and put $T = T_{Y_0}$. The saturation of the image of $d\rho$ gives a subbundle $\mathcal{S} \subset \rho^*T$ of rank k. The quotient $\mathcal{Q}$ has rank q, maps generically injectively to the pullback of $\mathcal{O}(-1)^c$, and $\mathcal{Q} \otimes \rho^*\mathcal{L}$ is semipositive. The curvature of the ball metric in a direction perpendicular to $\mathcal{S}$ is $-\|d\rho(v)\|^2\|\xi\|^2$; the curvature of the quotient bundle $\mathcal{Q}$ adds the nonnegative square of the second fundamental form [19, Section 1.6]. On a complete curve in $\widetilde{Z}$ over a line in Z avoiding S, with degree $v > 0$ against the pullback of $\mathcal{O}(1)$, this gives

$$0 \le \deg\det(\mathcal{Q} \otimes \rho^*\mathcal{L}) \le q(\lambda - 1)v.$$

Thus $\lambda \ge 1$. At equality the integral of the nonnegative curvature term is zero, so the second fundamental form vanishes along every such curve. Such curves over general lines cover all tangent directions at the points where ρ is an immersion, proving that the inverse image of Z° is locally totally geodesic at its smooth points. This argument uses the quotient of the fixed ball by the full normalizer; it does not lift a splitting of the tangent bundle of the coarse quotient through ramification. □

The conclusion at equality is supplementary and is not needed for Theorem 1.1.

Proposition 3.11. Let the curve $f : C \to W$ have full generic stabilizer A_r, $r \ge 2$, larger groups only A_{r+1} and $A_r \times \mu_e$, and ends at which the full groups F_P have rings of invariant sections generated in degree one. Put $q = r + 1$, let R count A_{r+1} points, let s count the ends, put $\sigma = \sum(1 - e^{-1})$ over the points with stabilizer $A_r \times \mu_e$, where e is the effective inertia order there, and put $h = -qt + R + s$. If $h < 0$, the space of homomorphisms to the line bundle associated with the invariant of greatest degree has dimension one. In all cases

$$a \ge \frac{r^2 - 3}{2}t + \sigma. \tag{3.11}$$

Proof. The lines E_j, $2 \le j \le q$, associated with the polynomial invariants on the normal space have degrees $-jt$, with the products of automorphic factors and finite characters as transition functions. At A_{r+1}, expand

$$(z + qu)\left((z - u)^q + \sum_{j=2}^{q} f_j(z - u)^{q-j}\right).$$

The differential has diagonal entry $-q(q+1)u$ in the tangent direction, diagonal entries $(q+1)u$ in the normal directions, and superdiagonal one. Let D_R be the reduced divisor of A_{r+1} points, and let S be the reduced divisor of ends. The line bundle associated with the invariant of greatest degree is $Q = E_q(D_R + S)$. After division by the zeros forced by the differential, its coefficient on the tangent direction of C has degree $-(q-2)t - \sigma < 0$. Lemma 3.5 kills the coefficients for all $j < q$, whose degrees after this division are $-(q-j)t$; the last is constant. Let $N = N_f^{\mathrm{sat}}$, and put $E = E_{q-1} \oplus E_q$. The inverse normal differential followed by projection onto these two directions defines a morphism $\psi : N \to E(2D_R + S)$. Define $\mathcal{Q} = \operatorname{im}\psi$,

without taking its saturation in the target. This image is a vector bundle of rank 2 on C, and $N \to \mathcal{Q}$ is surjective. At an A_{r+1} point, put $\alpha = (q+1)u$; the stalk is generated, in the frames of E_{q-1}, E_q, by $(\alpha^{-1}, 0)$ and $(-\alpha^{-2}, \alpha^{-1})$. At a point with stabilizer $A_r \times \mu_e$, the normal differential is invertible and the stalk is E. At a cusp with parameter v, the invertibility of J_c in (3.2), after removing the radial direction, gives the stalk $v^{-1}E$. Projection onto the second coordinate is consequently surjective onto $Q = E_q(D_R + S)$ at every point, and its kernel is $E_{q-1}(D_R + S)$. Thus

$$0 \longrightarrow E_{q-1}(D_R + S) \longrightarrow \mathcal{Q} \longrightarrow Q \longrightarrow 0, \qquad \deg\det \mathcal{Q} = -(2q-1)t + 2R + 2s = 2h + t.$$

These equalities of stalks exclude any additional quotient of finite length. If both $h < 0$ and $2h + t < 0$, Proposition 2.4 forces at least three coincident labels. Proposition 2.2 then shows that the Hom space has dimension at least two, a contradiction. Thus $h < 0$ implies $2h+t \geq 0$. Substituting the definition of h into equation (3.1) gives

$$qh = a - \frac{(q-2)(q+1)}{2}t - \sigma.$$

Substitute $h \geq -t/2$; for $h \geq 0$ this identity is stronger. □

3.4 Integral classes and local flags

Proposition 3.12. On a smooth coarse quotient of dimension d, let c_i^{orb} denote the average of tangent Chern classes on a neat cover. Then $c_i^{\mathrm{orb}} = (-1)^i \binom{d+1}{i} L^i$. If all stabilizers in codimension at most i have orders dividing M_i, then $M_i c_i^{\mathrm{orb}}$ is integral. It extends over a removed subset of codimension at least $i+1$. If all reflections have order two,

$$\mathcal{T} = 2NL = 2K_W + B \in \mathrm{Pic}(W)$$

for the reduced branch divisor B. For products of type A, the denominators $M_2 = 12$, $M_3 = 24$ suffice. On a closed $\mathbb{P}^k$, $k \geq 3$, write $\mathcal{T} = b\mathcal{O}(1)$, $b > 0$. Then

$$\frac{3(N-1)b^2}{2N},\ \frac{(N-1)(N-2)b^3}{2N^2} \in \mathbb{Z}, \qquad N^2 \mid b^3, \qquad L \cdot \mathrm{line} \geq \frac{1}{2N^{1/3}}. \tag{3.12}$$

Proof. The following identities do not require a lower bound on the number of labels. Let U be a smooth coarse ball quotient of dimension d, let $p : Y \to U$ be a finite neat cover of degree v, and let $L = c_1(\mathcal{L})$. The Euler sequence, whose middle term is a flat bundle twisted by a line bundle, gives $c(T_Y) = (1 - p^*L)^{d+1}$ in rational cohomology. If G_x is the full stabilizer at $x \in U$, proper pushforward of constructible functions and the normalization of Chern–Schwartz–MacPherson classes give

$$c_{\mathrm{SM}}\big(x \longmapsto |G_x|^{-1}\big) = v^{-1} p_*\big(c(T_Y) \cap [Y]\big), \qquad c_i^{\mathrm{orb}} = (-1)^i \binom{d+1}{i} L^i. \tag{3.13}$$

Here the orbifold classes are defined by the average in (3.13), and the cohomology equality uses the duality of the smooth U. We use the natural transformation C_* from constructible functions to Chern–Schwartz–MacPherson classes underlying [25]. For a fixed ball, the identity from the Euler sequence holds on its neat cover, with its own dimension and the restricted pullback of the class L of the original lattice. Set the constructible function to zero on strata of codimension greater than i. They do not change the dimension $d - i$ component, and multiplying by M_i gives an integral function and integral Chern class. For a removed support of codimension c, local cohomology is Borel–Moore homology in degree $2d - q$, so vanishes for $q < 2c$; this gives the stated extension. The formula for the canonical class when all reflections have order two gives $\mathcal{T}$; the small morphism π of Lemma 2.1 has no exceptional divisor. The low rank type A group orders give 12 and 24. Substitution into $12c_2^{\mathrm{orb}}$ and $24c_3^{\mathrm{orb}}$ gives the two fractions. For odd $p \mid N$, the second fraction in (3.12) forces $3v_p(b) \geq 2v_p(N)$. For $v_2(N) \geq 2$, use $v_2(N-2) = 1$; for $v_2(N) = 1$, the first fraction in (3.12) forces b even. □

Proposition 3.13. For $R = A_r, D_r, E_6, E_7$, use $A_\tau = Q(R) \otimes (\mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z}))$ with the polarization associated with the root lattice and with the equivariant structure for the Weyl group that is trivial at the origin. Choose basic homogeneous polynomial invariants $p_1, \dots, p_r$ of the Weyl group, put $d_i = \deg p_i$, and let m_i be the degrees of the corresponding generators of the ring of invariant sections, with distinguished generator ϕ_0 of degree one. Then $\phi_0(0) \neq 0$, and the first derivatives of invariant sections of degree at most j, divided by the corresponding powers of ϕ_0, project isomorphically onto $\langle dp_i : m_i \leq j \rangle$. The statement holds at all common fixed points at which the Weyl group acts trivially on the fiber of the line bundle of the polarization, by lifts of translations to the theta group, retains the components along the differentials of basic invariants of higher degree, and respects $z \mapsto -z$. In type D, the direction of P_0 is distinguished from the other invariant directions by its sign character.

Proposition 3.14. At the inclusions in the following table, use increasing degrees k_i of the polynomial invariants on the normal space, the coordinate $v = u^e$ on the coarse quotient of the axis, and the frames p_i/u^{ϵ_i} of the normal lines over the punctured axis. Coordinate changes in the target alone make the normal differential upper triangular with the indicated successive orders:

$P \subset G$	e	(ϵ_i)	orders
$D_5 \subset E_6$	1	0	$(3, 2, 3, 3, 4)$
$E_6 \subset E_7$	2	$k_i \bmod 2$	$(2, 2, 2, 2, 3, 3)$
$E_7 \subset E_8$	2	$k_i \bmod 2$	$(3, 3, 3, 4, 4, 5, 6)$
$G_{33} \subset G_{34}$	6	$(-2, 0, -2, 0, 0)$	$(1, 2, 2, 3, 4)$.

The last row of the inverse differential has pole order at most the last integer in the table; on its kernel, the preceding row has pole order at most the preceding integer.

The degrees k_i in the four rows are, respectively, $(2, 4, 5, 6, 8)$, $(2, 5, 6, 8, 9, 12)$, $(2, 6, 8, 10, 12, 14, 18)$, and $(4, 6, 10, 12, 18)$. The frames and successive orders in the table refer to these same directions, rather than to new coordinates. Lemma 4.3 uses the first row; Theorem 5.5 uses the second, Proposition 6.2 the third, and Lemmas 4.2 and 4.9 the fourth. At the ends, Proposition 3.13 identifies the same directions with the successive projections of the first derivatives of invariant sections; Lemma A.1 supplies the exponents $(1, 3, 3, 4, 6)$ of the powers of q_* used when dividing the coefficients for the fourth row. Proofs of both statements, including the complete flags of ideals in the local algebras that justify elimination without changing these directions, are in Appendix A.

3.5 Fixed curves without ends and with generic stabilizer $G(m, p, r)$

Lemma 3.15. If $n = r + 4$, $r \geq 5$, and X_μ is the Baily–Borel compactification of an arithmetic ball quotient, then no curve obtained from a disk fixed by a finite reflection group, without ends and with full generic stabilizer $G(m, p, r)$, has only the points with a larger stabilizer allowed in Proposition 3.4. The ambient lattice may be compact or nonuniform.

Proof. The image of this compact curve lies in the smooth interior of the coarse ball quotient, where the small projective morphism $\pi : W \to X_\mu$ is an isomorphism by Lemma 2.1 and Proposition 2.11. Thus all calculations take place on W, even if the ambient quotient has cusps elsewhere. Put $j = mt \in \mathbb{Z}_{>0}$, $A = r(r-1)/2 + r/p$, $\eta = r - 1 + 1/p$, and $k_0 = k/m$. Normalizer preservation and Proposition 3.4 give

$$(m-1)R + m\sigma = 2j + 2m(1-g), \quad a = 2 - 2g - Aj + \eta R, \quad h = R - k_0 j.$$

For $g \geq 1$, $R \leq 2j/(m-1)$, hence $h < 0$. For any map to the target associated with the invariant of greatest degree, the coefficient on the tangent direction of the curve has negative degree $R/m - (k_0 - 2/m)j - \sigma$. After division by the zeros forced by the differential, the

successive coefficients for the elementary symmetric invariants have degrees $-(k_0 - i)j < 0$. For $p > 1$, the coefficient for P_0 has, after this division, degree at most

$$(1 - 1/p)R - (r - 1 - r/p)j \leq ((r + 2)/p - r + 1)j < 0.$$

Thus the evaluation map from the Hom space has rank at most 1. Proposition 2.2 gives exactly two coincident labels for the quotient D, so $a = re + 2\delta > 0$. But

$$a \leq (2 + 2m\eta/(m - 1))(1 - g) - (A - 2\eta/(m - 1))j < 0,$$

since $A - 2\eta/(m-1) \geq (r-1)(r-4)/2 + (r-2)/p > 0$. Consequently $C = \mathbb{P}^1$. The coordinate $g_1 = v$ at the points with stabilizer $G(m, p, r + 1)$ and the invertible normal differential at points with stabilizer $G(m, p, r) \times \mu_e$ show that the curve is immersed. The symmetric prefixes and product block are complementary at every stalk, so Lemma 3.6 gives

$$N_f = \begin{cases} \bigoplus_{i=1}^{r} \mathcal{O}(R - ij), & p = 1, \\ \bigoplus_{i=1}^{r-1} \mathcal{O}(R - ij) \oplus \mathcal{O}((R - rj)/p), & p > 1. \end{cases}$$

If summands of negative degree occur, Proposition 2.2 makes all of them $\mathcal{O}(-e)$ and every other summand of degree at least e. Two summands of negative degree associated with elementary symmetric invariants have different degrees and are impossible. For $p > 1$, one summand of negative degree associated with an elementary symmetric invariant forces a summand of the same degree associated with P_0, hence

$$j = (p - 1)e, \qquad R = [r(p - 1) - p]e.$$

For $p = 2$, the summand associated with the preceding elementary symmetric invariant has degree zero, a contradiction. For $p \geq 3$, the Riemann–Hurwitz formula for the disk gives $ce \leq 2m$, but

$$c = (m - 1)[r(p - 1) - p] - 2(p - 1), \quad c - 2m \geq (4p - 7)m - 6p + 7 \geq 4 + (p - 3)(4p - 1) > 0.$$

The remaining negative possibilities force $(r - 1)j < R < rj$. For $m \geq 3$ the bound from the Riemann–Hurwitz formula for the disk contradicts this; for $m = 2$, it leaves only $r = 5, 6$, $j = 1$, and there is no integral R in that interval. Thus N_f is nef, $R \geq rj$, and $[r(m - 1) - 2]j \leq 2m$. This forces $m = 2$, $j = 1$, $r = 5$ or 6. For $r = 6$, $R = 6$, $a = 17$, contradicting the parity of $a = \sum_i m_i$ for ten labels. For $r = 5$, the only integral pairs are $(R, a) = (5, 12), (6, 17)$, with $t = 1/2$. Nefness of N_f permits at most one negative degree m_i. With none, Proposition 2.8 decomposes the curve into coordinate curves, and (3.10) gives $t \geq a/7$. With one, $a = 7e + 2\delta$; the first pair is impossible, and the second has $e = 1, \delta = 5$, giving $t \geq 2\delta/7 = 10/7$. Both contradict $t = 1/2$. □

4 Nine points

Let $n = 9$, and retain arbitrary weights $0 < \mu_i < 1$ of sum two. Suppose that $X = X_\mu$ is the Baily–Borel compactification of an arithmetic quotient of $\mathbb{B}^6$. Use the smooth GIT quotient $W = X_\nu$ obtained by perturbing μ to weights ν in a nearby chamber, and the small projective morphism $\pi : W \to X$ of Lemma 2.1, and write $H = -K_W$. For a curve $f : C \to W$ obtained from the full effective normalizer in the original lattice, put

$$t = \deg f^* L > 0, \qquad a = \deg f^* H, \qquad g = g(C), \qquad m_i = \deg M_i.$$

All line bundles for basic invariants below denote the line bundles of their invariant sections on the coarse curve, with their finite characters. We allow curves without ends and of arbitrary genus. The proof uses Lemma 2.10 repeatedly, without proving separately for each reflection group that K_W plus a multiple of L is nef.

4.1 Two bounds for complete curves in the compact case

Lemma 4.1. Suppose that the arithmetic quotient of $\mathbb{B}^6$ is compact. For a nonconstant complete curve, if $m_i \geq 0$ for every i, then $a \leq 7t$. If exactly one degree m_i is negative, write

$$m_0 = -e, \qquad m_j = e + 2\delta_j \ (j \neq 0), \qquad \delta = \sum_{j \neq 0} \delta_j.$$

Then $a = 7e + 2\delta$ and $7t > 6e + 2\delta = a - e$.

Proof. A general coordinate curve γ_i has at least eight branch values by Lemma 3.9. The formula for the canonical class, evaluated on γ_i, gives $7L\gamma_i \geq -2 + 8/2 = 2$. Apply the decompositions of $[f_*C]$ obtained from the degrees m_i in Proposition 2.8. They give the first assertion and reduce the second to $L\beta_0 > 6/7$. Choose a maximal short set $J \ni 0$, and set $q = |J| \leq 6$. The same decomposition gives

$$[\beta_0] = [\beta_J] + \sum_{j \in J \setminus \{0\}} [\gamma_j].$$

For $q = 1$, Proposition B.2 gives $L\beta_0 \geq 1$. For $q = 2, 3$, Proposition B.4 gives, respectively,

$$L\beta_0 \geq \frac{2}{3} + \frac{2}{7} = \frac{20}{21}, \qquad L\beta_0 \geq \frac{1}{3} + \frac{4}{7} = \frac{19}{21}.$$

For $q \geq 4$, ampleness gives $L\beta_J > 0$, hence $L\beta_0 > 2(q-1)/7 \geq 6/7$. □

Lemma 4.2. If the arithmetic quotient of $\mathbb{B}^6$ is compact, it has no full G_{34} point.

Proof. Take its G_{33} disk and then its quotient by the full normalizer. Let $R > 0$ count G_{34} points and put $\sigma = \sum(1 - 1/e)$ for the points with stabilizer $G_{33} \times \mu_e$. The line bundle of degree six gives $j = 6t \in \mathbb{Z}_{>0}$. The Riemann–Hurwitz formula for the disk, equation (3.1), and the row of the inverse differential associated with the invariant of greatest degree give

$$\frac{5}{6}R + \sigma = 2t + 2 - 2g, \quad a = 2 - 2g - 50t + \frac{38}{3}R, \quad h = -18t + 4R. \tag{4.1}$$

The degrees of the five successive coefficients for a map to a line bundle of degree d are

$$d + 4t - \frac{4}{3}R, \quad d + 6t - 2R, \quad d + 10t - \frac{7}{3}R, \quad d + 12t - 3R, \quad d + 18t - 4R. \tag{4.2}$$

These are the degrees of the line bundles for the basic invariants of the normal representation, $(-4t + R/3, -6t, -10t + R/3, -12t, -18t)$, corrected by the successive zeros $(1, 2, 2, 3, 4)$ of Proposition 3.14. If $g \geq 1$, then $a, h < 0$ and the first two coefficients for $d = h$ vanish. Proposition 2.2 gives $\ell \leq 4$ and therefore $a = (9 - 2\ell)e + 2\delta > 0$, a contradiction. If $g = 0$ and $j \geq 8$, the four lower coefficients vanish and the Hom space has dimension one. The resulting quotient D of rank 1 requires $a + 5h \geq 0$, but

$$a + 5h \leq (402 - 308t)/5 < 0.$$

For $1 \leq j \leq 7$, the conditions

$$R \leq (2j + 12)/5, \quad R \equiv 2j \pmod 3, \quad \sigma = 0 \text{ or } \sigma \geq 1/2$$

leave $(j, R) = (1, 2), (2, 1), (3, 3), (4, 2), (5, 1), (6, 3), (7, 2)$. The five rows with $h < 0$ have $a < 0$ and a Hom space of dimension one, again impossible. The other rows are

$$(t, R, \sigma, a, h) = (1/6, 2, 2/3, 19, 5), \qquad (1/2, 3, 1/2, 15, 3).$$

Formula (4.2), including the coefficient on the tangent direction of the curve, excludes every quotient onto a line bundle of negative degree. Thus at most one m_i is negative. If none is negative, Lemma 4.1 contradicts $a \leq 7t$. Otherwise $a = 7e + 2\delta$ gives $(e, \delta) = (1, 6)$ or $(1, 4)$, and the weaker inequality $t > 2\delta/7$ already contradicts the values $t = 1/6$ and $t = 1/2$. □

4.2 Successive kernels for D_5

The following sequence of kernels is used in both the compact and nonuniform cases.

Lemma 4.3. Let a curve obtained from the full normalizer have generic group D_5, larger irreducible groups E_6, D_6, and other interior groups $D_5 \times \mu_e$ acting trivially on the root space. Allow ends with the minimal D_5 polarization, generators of degrees $(1^4, 2^2)$, trivial character on the fiber of $\mathcal{V}_*$, and width one. Write R, D, S for the reduced divisors of these E_6 points, D_6 points, and ends, using the same letters R, D for their degrees and $s = \deg S$. Set $E = f^*T_W$ and $T_f = \operatorname{sat}_E df(T_C)$. Then $T_f = T_C(R)$ and E/T_f has successive line quotients Q_2, Q_4, Q_5, Q_6, Q_h of degrees

$$q_2 = -2t + 3R + D + s, \qquad q_4 = -4t + 2R + D + s, \tag{4.3}$$

$$q_5 = -5t + 3R + D/2 + s, \qquad q_6 = -6t + 3R + D + 2s, \tag{4.4}$$

$$h = -8t + 4R + D + 2s. \tag{4.5}$$

In particular $E \twoheadrightarrow Q_h$, and

$$2g - 2 + s + D/2 + \sigma = 2t, \quad a = 4h + 5t + D + \sigma, \quad \deg T_f = (h + D)/4 + \sigma + s/2. \tag{4.6}$$

If $R > 0$, then $h > 0$ and E is nef.

Proof. The line bundles associated with these directions have degrees $-2t, -4t, -5t - D/2, -6t, -8t$. The term $-D/2$ comes from the frame P/u on the coarse quotient at a D_6 point, with its character included. The full normalizer preserves their flag ordered by increasing degree. Proposition 3.14 gives successive zero orders $(3, 2, 3, 3, 4)$ at E_6. At D_6, put $v = u^2$ and $P = u\xi$; after removing the tangent column, the matrix of the normal differential is

$$\begin{pmatrix} v & 1 & 0 & 0 & 0 \\ 0 & v & 0 & 1 & 0 \\ 0 & 0 & v & 0 & 0 \\ 0 & 0 & 0 & v & 1 \\ 0 & 0 & 0 & 0 & v \end{pmatrix},$$

so every quotient gains one zero. At an end the distinguished generator ϕ_0 of Proposition 3.13 is nonzero and has degree one. After its removal, the matrix is $\operatorname{diag}(q, q, q, q^2, q^2)A(q)$, where the leading 3×3 block of $A(0)$ is invertible. The least orders of its minors using the first one, two, three, four, and five columns are $1, 2, 3, 5, 7$; their differences give $(1, 1, 1, 2, 2)$. These are orders of minors from the initial columns, not merely Smith exponents after an arbitrary change of basis of the domain. The differential on the E_6 axis has one common zero, and the differential on the coarse quotients of the other axes is invertible. This proves the flag and (4.6).

We record the local obstruction that also treats $h = 0$. Use coordinates $T_2, T_5, T_6, T_8, T_9, T_{12}$. The E_6 axis is

$$T(u) = (-3u^2, 3u^5, -3u^6/2, -3u^8/4, u^9, -3u^{12}/16).$$

The image of its ambient local ring is $A = \mathbb{C}\{u^2, u^5\}$, whose missing nonnegative exponents are $1, 3$. The rows of the inverse differential for the descending saturated quotients, divided over $\mathbb{C}\{u\}$ by the common orders of the zeros and poles of their entries, are

$$\begin{aligned} C_8 &= dT_{12} + u^3 dT_9/2 + u^6 dT_6/4, \\ C_6 &= dT_9 + u\, dT_8 + u^3 dT_6 + u^4 dT_5/2 + u^7 dT_2/4, \\ C_5 &= dT_8 + u^3 dT_5/2 + u^6 dT_2/4, \\ C_4 &= dT_6 + u\, dT_5 + u^4 dT_2, \qquad C_2 = dT_5 + 5u^3 dT_2/2. \end{aligned} \tag{4.7}$$

They annihilate $T'(u)/u$ and have a unit minor in columns $T_{12}, T_9, T_8, T_6, T_5$. The flag of ideals in the local Tjurina algebra used in Proposition 3.14 identifies their successive kernels. No unit multiple of C_8 is A-valued, since its coefficient ratio is $u^3/2 \notin A$. If a regular row on the ambient quotient annihilates the tangent direction and the quadratic invariant, then the row has an extra zero in the coefficient of the invariant of degree four: compare the missing u term in its dT_5 coefficient. If the row also kills the invariant of degree four, comparison of the missing u^3 terms in dT_6, dT_5 shows that the value of the row at zero lies in $\mathbb{C}dT_{12}$.

Suppose $R > 0$ and $h \leq 0$. For a row whose target line bundle has degree $q \leq h$, the degrees of the first two coefficients are bounded by

$$q - q_2 \leq 3h/4 - 2R - 3D/4 - s/2 < 0, \qquad q - q_4 \leq (h - D)/2 \leq 0.$$

Equality in the second bound requires $q = h = D = 0$. If $h < 0$, the independent quotients associated with coincident pairs in Proposition 2.2 all have values in $\mathbb{C}dT_{12}$, hence $\ell = 2$. The quotient associated with the invariant of greatest degree factors through this pair quotient with a unit coefficient, contradicting $u^3/2 \notin A$. For $h = 0$, the same argument first excludes quotients onto line bundles of negative degree. Two negative degrees m_i would supply one; a unique negative degree m_i would make the row associated with the invariant of greatest degree zero on the columns for the other marks, whose degrees are positive, and infinitesimal projective transformations then kill its remaining column. Thus all degrees m_i are nonnegative and the row associated with the invariant of greatest degree only uses marks i with $m_i = 0$. Their distinct positions are disjoint. Three positions trivialize the family and make the relevant row constant; one position makes it a constant sum of pair differences. Both contradict the same coefficient ratio. If exactly two positions remain, each position contains at least two marks and gives an independent quotient of degree zero associated with that pair. Their first coefficients vanish by degree. The second vanish by degree when $D > 0$, and by their extra zero when $D = 0$. Both pair quotients therefore have value in $\mathbb{C}dT_{12}$, contradicting independence. Consequently $h > 0$.

Finally (4.6) gives

$$q_2 = h/4 + 2R + 3D/4 + s/2 > 0, \quad q_4 = (h + D)/2 > 0,$$

$$q_6 = 3h/4 + D/4 + s/2 > 0, \quad \deg T_f > 0, \qquad q_5 = 3h/4 + g - 1 + \sigma/2 \geq -1/4.$$

Since h, q_5 are degrees of line bundles, hence integers, $q_5 \geq 0$. The flag has quotients of nonnegative degree and remains so on every finite cover; hence E is nef. □

Proposition 4.4. In the compact case the curve of Lemma 4.3 cannot have $R > 0$.

Proof. We have $h > 0$, E nef, and at most one negative degree m_i. Put $Y = 2t - h - D - \sigma$. If all degrees m_i are nonnegative, Lemma 4.1 gives $Y \geq 3h > 0$. With one negative degree $m_i = -e$, the eight columns for the other marks generate E, so $e \leq h$ and

$$S' = 7t + e - a > 0, \qquad Y = S' + 3h - e > 0.$$

The column for each mark, written in a frame pulled back from W, has coefficients in A. For a column killed by C_8, successive use of the rows in (4.7) and the missing exponents $1, 3$ gives extra zero orders $(R, 2R, 2R, 3R)$ in Q_6, Q_5, Q_4, Q_2. The equation $C_8 v = 0$ gives $v_9(0) = 0$, then $C_6 v = 0$ gives $v_8(0) = 0$; the u^2, u^3 terms of $C_5 v = 0$ give $[v_8]_{u^2} = v_5(0) = 0$. The u^3 terms of $C_6 v = 0$ and $C_4 v = 0$ give $v_6(0) = [v_5]_{u^2} = 0$. The adjusted degrees are

$$q_6 - R = -t - \sigma/2 - Y/2, \quad q_5 - 2R = -5t/2 - \sigma/4 - Y/4,$$

$$q_4 - 2R = -2t - h - \sigma - Y, \qquad q_2 - 3R = -h - \sigma - Y.$$

All are negative, so a column for a mark i with $m_i \geq 0$ killed by the row associated with the invariant of greatest degree lies in T_f. Let p count these columns and put $\tau = \deg T_f = (h + D)/4 + \sigma$. At a general point, the kernel of their evaluation consists of projective vector fields vanishing at complementary marked positions. There is at least one such position, hence $p \leq 3$. With a unique negative degree m_i there are two distinct complementary positions, hence $p \leq 2$. Individual columns are nonzero because a nonconstant ordered configuration has at least four distinct positions.

If all degrees m_i are nonnegative, put $A_7 = 7t - a \geq 0$. For $\tau \leq h$ use $B_0 = 9h - a \geq 0$; for $\tau > h$ use $B_3 = 6h + 3\tau - a \geq 0$. But

$$5A_7 + 4B_0 = -10t - 9D - 9\sigma < 0, \qquad 2A_7 + B_3 = -t - 21h/4 - 9D/4 < 0.$$

With one negative degree m_i, $T_p = (8 - p)h + p\tau - e - a \geq 0$, but

$$S' + T_p = -3t - 3ph/4 + (p/4 - 2)D + (p - 2)\sigma < 0 \qquad (0 \leq p \leq 2).$$

Every case is impossible. □

4.3 Nefness of $K_W + 10L$ in the compact case

Lemma 4.5. Once the proper irreducible factors satisfying the condition on full interior stabilizers in Lemma 2.12 and the compact $G(m, p, 5)$ curves with $m \geq 4$ or $(m, p) = (3, 1)$ are excluded, the divisor $K_W + 10L$ is nef.

Proof. The remaining groups of full rank 6 are $A_6, B_6, D_6, G(3, 3, 6), E_6, G_{34}$. Their reflections have order two; write B for the reduced branch divisor. Then $14L = 2K_W + B$ is integral and the groups of rank 2 are A_1^2, A_2, B_2. Set

$$C_1 = H + 7L = B/2, \quad C_2 = c_2(T_W) + 6HL + 21L^2, \quad A_0 = 3C_1(C_1 + L) - 6C_2.$$

For a normal neat cover $p : Y \to W$ with group G, Burnside's fiber count gives

$$\mathbf{1}_W = |G|^{-1} \sum_{g \in G} p_* \mathbf{1}_{\mathrm{Fix}(g)}.$$

Apply the naturality of Chern–Schwartz–MacPherson classes used in Proposition 3.12. A ball of codimension r fixed by a finite reflection group has Chern polynomial $(1 - Lz)^{7-r}$. The triangular identities in codimensions one and two give respectively C_1, C_2 as defined above, and

$$48C_2 = 12Z_{A_1^2} + 16Z_{A_2} + 18Z_{B_2}, \qquad 4A_0 = 4Z_{A_2} + 9Z_{B_2}. \tag{4.8}$$

Here Z_P are ordinary reduced fixed cycles. For clarity, the first coefficients are 48 times the proportions $1/4, 1/3, 3/8$ of elements with no nonzero fixed vector. For the second identity, the coefficients of $B^2 + 2LB$ are $2, 4, 6$, obtained from $4\nu(\nu - 1)/|P|$. The mirror normal class is $-L$, so its diagonal self-intersection cancels with $2LB$. These observations derive both identities from the same formula for Chern–Schwartz–MacPherson classes.

On a closed locus $Z_I = \mathbb{P}^k$ where the sections in I coincide, put $q = |I|$, $\alpha = 9 - 2q$, and $L|_{Z_I} = \lambda H_0$. The exact sequence (2.1), written on W, gives $c_1(T_W)| = \alpha H_0$ and $c_2(T_W)| = (\alpha^2 - 7)H_0^2/2$. A plane in Z_I is the intersection of $q - 1$ divisors where pairs coincide and $k - 2$ first Chern classes of the bundles M_j with $j \notin I$. Thus (4.8) pairs integrally with this plane, including excess intersections. For $b = 14\lambda \in \mathbb{Z}$, substitution gives

$$48 \int_{\mathbb{P}^2} C_2 = 24(\alpha^2 - 7) + \frac{36}{7} b(b + 4\alpha), \qquad 4 \int_{\mathbb{P}^2} A_0 = 84 + \frac{6}{7} b(b + 3\alpha).$$

Their difference gives $7 \mid 6\alpha b$ and then $7 \mid 6b^2$, hence $7 \mid b$. The K_W-negative lines in the loci Z_I with $q = 2, 3, 4$ have H-degrees $5, 3, 1$ and L-degrees at least $1/2$. The corresponding ratios are at most $10, 6, 2$. Coordinate curves have ratio at most seven by Lemma 4.1. Proposition 2.8 proves the result; for $W = \mathbb{P}^6$ use Proposition B.2. The divisibility argument applies to the restrictions to the loci Z_I and does not assert descent of $2L$ on the compact W. □

Theorem 4.6. No nine-point quotient under consideration is a compact arithmetic ball quotient.

Proof. By Proposition 2.11, stabilizers are reflection groups, so the compact coarse quotient is smooth and π is an isomorphism. Replace an irreducible factor successively by larger irreducible factors of full parabolic stabilizers. If this replacement stops below rank 6, Proposition 2.14 gives basic invariants of equal degree δ and a closed image $\mathbb{P}^{6-s}$ with normal bundle $\mathcal{O}(-1)^s$ and $\delta L = \mathcal{O}(1)$. The classification of groups whose basic invariants have equal degrees first gives $s \leq 2$; no analogue of Lemma 2.13 for four labels is needed. For $s = 1$, $\delta \geq 2$ contradicts the bound $\lambda \geq 2/3$ for $k = 5$ in Proposition B.4. For $s = 2$ the groups are $G(m, 2, 2)$, with even $m \geq 4$, and G_7, G_{11}, G_{19}, all with $\delta \geq 4$, contradicting its bound $\lambda \geq 1/3$ for $k = 4$. Thus this replacement, started from any reflection, reaches rank 6. Lemma 3.15 excludes $G(m, p, 5)$ for $m \geq 4$ or $(m, p) = (3, 1)$ before any bound on a/t is used. Lemma 4.5 now gives $a \leq 10t$ on every complete curve. Lemma 4.2 excludes G_{34}, and Proposition 4.4 excludes E_6 through the disk whose full generic stabilizer is its parabolic subgroup D_5. Next $B_6, D_6, G(3, 3, 6)$ give full $B_5, D_5, G(3, 3, 5)$ disks, whose points with a primitive stabilizer have just been excluded; Lemma 3.15 excludes them. Only A_6 remains. The nontrivial lattice generated by reflections has a reflection, so replacing it successively by larger irreducible factors of full point stabilizers supplies a full A_5 disk. Proposition 3.11 gives $a \geq 11t + \sigma > 10t$, a contradiction. We did not assume that the coefficient field is imaginary quadratic or that $2L$ descends on the compact W. □

4.4 The nonuniform proof: bounds fixed once

Now suppose that the arithmetic quotient is nonuniform. At a genuine wall vertex the local slice is the cone over $\mathbb{P}^a \times \mathbb{P}^b$, $a+b = 5$, $a, b \geq 1$. The smooth GIT quotient $W = X_\nu$ obtained by perturbing μ to weights ν in a nearby chamber retains the smaller factor, so $\dim \mathrm{Exc}(\pi) \leq 2$ and its codimension is at least four. It is an isomorphism near every smooth cusp. Proposition 2.11 applies since $12 > 7$; the coefficient field K is imaginary quadratic and $w_K L$ is an ample line bundle on X, $w_K \in \{2, 4, 6\}$. After reflections of order greater than two are excluded below, it shows that $2L$ descends to a line bundle on the original X.

We use Lemma 2.10 in the following form throughout. For an ample line bundle $A_X = cL$, put $b = ct$. Then

$$a \leq 7b; \qquad \mathrm{Hom}(E, Q) \neq 0,\ \deg Q \leq 0 \quad \Longrightarrow \quad a \leq 5b. \tag{4.9}$$

If the common negative degree m_i of the coincident labels is $-e$, the same lemma gives

$$a = (9 - 2\ell)e + 2\delta, \quad e + \delta \leq b,$$

and every target of degree less than e kills the summand F of Proposition 2.3. Thus, after $2L$ descends, the two bounds in (4.9) are $a \leq 14t$ and $a \leq 10t$. These statements hold for each complete curve separately, also for $W = \mathbb{P}^6$; no nefness of a divisor on W is used.

Lemma 4.7. Every irreducible reflection factor lies in a chain of successively larger irreducible factors of full point stabilizers reaching rank 5 or 6. Every disk fixed by an irreducible group of rank 5 avoids genuine wall vertices, so all its ends are smooth ambient cusps.

Proof. If the chain of larger irreducible factors of full point stabilizers stops in rank $s \leq 4$, every full interior stabilizer along the corresponding fixed ball has the form $T \times J_x$ required in Proposition 2.14. Its image is a closed $Z_X = \mathbb{P}^{6-s}$ with equal degrees δ, and classification gives $s \leq 2$. The strict transform $Z \to Z_X$ has no exceptional divisor since $\dim Z \geq 4$ and $\dim \operatorname{Exc}(\pi) \leq 2$. A projective small morphism to a factorial target is an isomorphism: push down a relatively ample Cartier divisor and pull it back; their difference would be an exceptional divisor. Thus $Z = Z_X$, $N_{Z/W} = \mathcal{O}(-1)^s$, and $\delta L|_Z = \mathcal{O}(1)$. Its fixed Hermitian space is defined over the original imaginary quadratic field and has dimension at least three. Meyer supplies a rational isotropic line in that space, so the quotient $\mathbb{B}(V_0)/\Pi$ of Lemma 2.12 is nonuniform. Descent of the original $w_K L$ then gives $\delta \mid w_K$. The lowest invariant term of its ideal gives a generically invertible map

$$L^{-\delta} \otimes U_\delta \longrightarrow N_{Z/W}.$$

After canceling equal degrees it is a map from a finite flat bundle to $\mathcal{O}_Z^s$. On complete curves away from cusps its coefficients become constant after a finite cover. Interior inertia is therefore trivial; simple connectedness and Hartogs make the matrix constant and invertible on Z. At a smooth ambient cusp every vanishing function and the lowest coefficient of its expansion in the normal directions are divisible by q_*, a contradiction. All cusps of Z consequently lie at genuine wall vertices. Surjectivity on local rings and the generators of degree one in Theorem 2.19 force the full cusp stabilizers of Z and of the ambient quotient to contain the same central translations: a ratio greater than one between these periods would kill every nonconstant restricted generator. The induced smooth cusps therefore have only generators of degree one and trivial character on the fiber of $\mathcal{V}_*$.

In rank 2, the possibilities are $G(4,2,2), G(6,2,2)$. The two lowest directions require every larger point stabilizer to have two basic invariants of degree δ. The lift fixing the point for the full finite group permits this test at a point of a cusp that is a genuine wall vertex. Standard smooth abelian quotient actions have point groups that are products of symmetric groups and $G(e,1,k)$; within each irreducible factor, the degrees of the basic invariants on the reflection representation with the fixed part removed, $2, \dots, k+1$ for S_{k+1} and $e, 2e, \dots, ke$ for $G(e,1,k)$, are distinct. The sole remaining surface action is $G(4,2,2)$, excluding $\delta = 6$. For $\delta = 4$, the induced groups on $\mathbb{P}^4$ are $A_3, B_3, G(4,1,3)$. The generic branch groups are $G(4,2,2) \times \mu_e$, $e = 2, 4$. Let B_2, B_4 be the reduced branch divisors corresponding to these additional factors μ_2, μ_4, respectively. With these branch divisors, the formula for the canonical class of $\mathbb{P}^4$ becomes $2B_2 + 3B_4 = 25H_0$, so $B_4 \neq 0$. Meyer, applied in that quotient of a fixed ball of higher dimension by its full normalizer, yields a $G(4,1,3)$ cusp and the disk at this cusp fixed by the full group $G(4,1,3)$. Writing $u = 4t$, its equations are

$$46t = 4s + 15R/4 + \sigma, \qquad -12t + s + R \geq 4t.$$

The first right side is at least $15(s+R)/4 \geq 60t$, impossible.

For rank 1, $\delta = 4, 6$ are excluded by Propositions B.6 and B.5. For $\delta = 2, 3$, put $u = \delta t$ on $Z = \mathbb{P}^5$. All induced cusps have five generators of degree one. The field and cusp classifications leave A_4 and $G(m,1,4)$ with $m = 2, 3, 4$ for $\delta = 2$, and $m = 2, 3$ for $\delta = 3$. The case $m = 6$ is excluded by the general inequality for P_0 in rank 4 in the proof of Proposition B.5:

$$a \geq (17p/6 + 1)h_P + 30t + R + \sigma > 6u \qquad (h_P \geq u,\ t = u/\delta,\ \delta = 2, 3).$$

This uses that inequality itself, rather than the conclusion of the proposition for $L = H/6$. A G_{32} disk has no ends, since the actions of the groups F_P whose rings of invariant sections are generated in degree one have no such full stabilizer, and has no point with an irreducible stabilizer of rank 5. Equation (3.1) and the Riemann–Hurwitz formula for the disk give $a = 2 - 2g - 84t < 0$, since $t = u/\delta \geq 1/3$, contrary to $a = 6u > 0$. For $G(m,1,4)$, equation (3.4)

and Lemma B.1 give

$$(10m + 2 + 6\delta)t = 5s + (5 - 1/m)R + \sigma, \qquad -4mt + s + R \geq \delta t.$$

The difference between the forced lower coefficient and the left coefficient is

	$m = 2$	$m = 3$	$m = 4$
$\delta = 2$	11	64/3	63/2
$\delta = 3$	19/2	20	absent

and is positive. Curves of types $D_4, G(3,3,4), G(4,2,4), G(4,4,4), F_4$ have no ends, by a check of the orders and characters of their groups against the smooth action of F_P. For F_4, $(28{+}6\delta)t = 2-2g$ contradicts $t \geq 1/\delta$. For D_4, with R, M counting G_{33}, D_5 points, the two degree formulas give

$$g = 0, \quad u \leq 2, \qquad (36 + 96/\delta)u - 12 = 28R + 21M.$$

Modulo seven excludes $\delta = 2$; $\delta = 3$ leaves $(u, t, R, M, \sigma) = (1, 1/3, 2, 0, 1)$. Here the image is a line with normal bundle $\mathcal{O}(1)^4$. The four line bundles for the invariants on the coarse quotient have degree -2, with zero orders $(1,0,2,1)$ and $(1,2,0,1)$ at the two G_{33} points; the other inertia consists of two points with inertia of order two acting trivially on the root space. This is exactly the row (B.2), whose following calculation uses only these data. Its identity $S_{4a}S_{4b} = S_2S_6$ makes the two products of exchanged ratios equal to $1/12$; the determinant of the four linear columns is therefore zero. This excludes D_4, G_{33}, D_5 before the pure A_4 formula is used. For $G(m,p,4)$ with $p > 1$, put $D_0 = 6m + 4m/p$, $k = 3m$, and $\beta = 4 + 1/p - 1/m$. The equations for a curve without ends give

$$(D_0 + 2 + 6\delta)t = \beta R + \sigma, \qquad R \geq (k + \delta)t.$$

For $\delta = 2$, the differences $\beta(k+\delta) - (D_0 + 2 + 6\delta)$ are $8, 27/2, 14$. These correspond respectively to $(m,p) = (3,3), (4,2), (4,4)$. For $\delta = 3, (m,p) = (3,3)$ the difference is six. All are positive. Irreducible factors of smaller rank satisfying the condition on full interior stabilizers in Lemma 2.12 contradict the ample tangent bundle of $\mathbb{P}^5$, so a full A_4 disk remains. Its target associated with the invariant of greatest degree satisfies $5 \deg Q = (6\delta - 9)t - \sigma < 5\delta t$, contrary to Lemma B.1. This excludes every proper irreducible factor satisfying the condition on full interior stabilizers in Lemma 2.12.

Using the restrictions on fields of definition in the classification of finite reflection groups, we obtain

rank 5	$A_5,\ G(m,p,5),\ G_{33}$	$m \in \{2,3,4,6\}, \quad p \mid m.$
rank 6	$A_6,\ G(m,p,6),\ E_6,\ G_{34},$	

Every group of rank 6 has a parabolic subgroup of rank 5. An irreducible root space of rank 5 cannot lie in one of the factors $\mathbb{C}^a, \mathbb{C}^b$ of $\mathbb{C}^r$ at a genuine wall vertex, for which $\{a, b\}$ is $\{1,4\}$ or $\{2,3\}$. Thus every end of a curve fixed by a group of rank 5 is a smooth ambient cusp. □

4.5 Remove $G(6,p,5)$, $G(4,1,5)$, $G(3,1,5)$ before using $2L$

Lemma 4.8. There are no full disks or smooth cusps of types $G(6,p,5)$, $G(4,1,5)$, or $G(3,1,5)$. After $2L$ descends, the same holds for $G(4,2,5), G(4,4,5)$.

Proof. Apply Proposition 3.4 with the periods at the ends, polarizations, and ages of the characters on the fibers. For $p > 1$, the degrees of the target line bundles for the elementary symmetric invariant of greatest degree and for P_0 are

$$h = -4mt + R + \nu, \qquad h_P = (-5mt + R + \nu)/p, \qquad \nu = \sum_c \nu_c;$$

for $p = 1$ the target associated with the invariant of greatest degree is $h = -5mt + R + \nu$. The age of the line bundle corresponding to P_0 is retained separately.

For $m = 6$, put $j = 6t \in \mathbb{Z}_{>0}$ and use $A_X = 6L$. At every end,

$$(B_c - \nu_c/p, \nu_c) \in \{(5,1),(9,2),(13,3),(23,6)\}, \qquad \mathrm{age}_c = \lceil \nu_c/p \rceil - \nu_c/p.$$

It follows from the degrees $(2, 3, 6/p, 6, 6, 6)$ of the generators before normalization, with their character congruences. Set $\gamma = 23/6 + 1/p$. Since $B_c \geq \gamma \nu_c$, equation (3.4) gives

$$a \geq 53j/6 + R + \sigma + p\gamma h_P.$$

Formula (4.9) first gives $a \leq 7j$, then $h_P < 0$ and $a \leq 5j$. Every nonzero line target has degree at least $-j$. The Riemann–Hurwitz formula for the disk, using $\nu_c \leq 6$, gives

$$(\bar{k} - 3)j + 4R + 6\sigma \leq 12(1 - g), \qquad \bar{k} = 5 \ (p = 1), \quad \bar{k} = 4 \ (p > 1).$$

Thus $g = 0$, $j \leq 6$ for $p = 1$ and $j \leq 12$ otherwise; also $s \leq \lfloor 2 + j/3 \rfloor$ and $5R/6 + s \leq 2 + j/3$. Here $\sigma = 0$ or $\sigma \geq 1/2$, with no bound on the order of a product factor.

We give the finite check of the remaining cases to specify its scope. Let $V = \sum V_c$ be the sum of the orders of the zeros initially required at the cusps: $V_c = 1$, except that the two $p = 1$ end sums $11, 16$ require $V_c = 2, 3$. For a negative target $q = \min(h, h_P)$, the successive symmetric bounds are $q + ij - R - V$. The product bound is $q + (5j - R)/p + \sum \mathrm{age}_c - V$. Require $h, h_P \in \mathbb{Z}$, $h, h_P \geq -j$, and integrality of the degree of the line bundle corresponding to P_0

$$-5j/p - (1 - 1/p)R - \sum \mathrm{age}_c \,.$$

For $-q \leq e \leq j$ and $2 \leq \ell \leq 6$, require $a = (9 - 2\ell)e + 2\delta$, $e + \delta \leq j$, and $\delta \geq 0$ integral. The coefficient bounds must allow the evaluation map from the Hom space to have rank $\ell - 1$ and the Hom space to have dimension $(\ell - 1)(e + q + 1)$. The independent inverse directions on F also impose

$$\#\{0 \leq i < 4 : h + ij < e\} + \mathbf{1}_{h_P < e} \leq \ell - 1$$

for $p > 1$, with the directions of five elementary symmetric invariants and no term for P_0 for $p = 1$. Substitution in this proved finite range gives no row for $p = 1$. All remaining rows have $R = g = 0$, $a = e = j$, $\ell = 4$, $h = -j$, $\delta = 0$; their complete end table is

p	j	$\{B_c\}$	σ	h_P
2	2	$\{26\}$	$5/3$	-2
2	3	$\{29/2, 26\}$	1	-3
2	3	$\{11/2, 10, 26\}$	0	-3
3	6	$\{29/3, 29/3, 25, 25\}$	0	-4
6	6	$\{24, 24, 24\}$	1	-2
6	9	$\{27/2, 24, 24, 24, 24\}$	0	-3.

This includes curves without ends and every sign of a; positivity was not imposed. For $p = 3, 6$, the line bundle corresponding to P_0 is a direct summand because $R = 0$. Its degree strictly between $-j$ and zero contradicts $E = \mathcal{O}(-j)^3 \oplus F$, whose other summands have degree at least j. For $p = 2$, every row has an end with sum 26. Choose a local coordinate z at the elliptic fixed point and a local frame in which μ_6 acts by a character for the elliptic line bundle. For its sections u, x, y of degrees $1, 2, 3$, write $u = z(u_0 + O(z^6))$, $x = x_0 + x_1 z^6 + O(z^{12})$, and $y = y_0 + y_1 z^6 + O(z^{12})$, where $u_0 x_0 y_0 \neq 0$. The elliptic sections obey $y^2 = 4x^3 - g_3 u^6$, $g_3 \neq 0$, and

$$2y_1/y_0 - 3x_1/x_0 = -g_3 u_0^6 / y_0^2 \neq 0.$$

The radial and lowest symmetric derivatives are independent. Once the coefficient for the first elementary symmetric invariant vanishes, all coefficients for the elementary symmetric invariants are divisible by the sixth power of q_*. Only the direction of one elementary symmetric invariant and that of P_0 can remain, so the evaluation map from the Hom space has rank at most 2 instead of $\ell - 1 = 3$.

For $m = 4, p = 1$, use $j = 4t$ and $A_X = 4L$. The degrees $(d_1, d_2, 2d_3, 4, 4, 4)$ of the generators, $d_1, d_2 \mid 4$, $d_3 \mid 2$, give $B_c \geq 19\nu_c/4$ for every torsor and linearization. Consequently

$$a \geq 19h/4 + 33j/4 + R + \sigma.$$

Starting with $a \leq 7j$ gives $h < 0$, then $a \leq 5j$, $h \leq -13j/19$, and $e \geq 13j/19$. On F the equations $dZ = v\,df_5$, $dg_5 = df_5 + v\,df_4$ reduce the order of the next pole of the inverse differential, giving a second independent target of degree $h + j < e$. Thus $\ell \geq 3$ and $a \leq 3j$, but $h \geq -j$ gives $a \geq 7j/2$.

For $m = 3, p = 1$, put $j = 3t$. The integrality of j comes from the line bundles for the elementary symmetric invariants, not from $6L$ alone. After the case $m = 6$ has been excluded, the degrees of the generators at the ends are $(1^6), (1^2, 3^4), (1, 3^5)$, with counts x, y, z. Set $h = -5j + R + x + 3y + 3z$. The equations give

$$a - 5h = 10j + 2 - 2g - 2y > 4j, \qquad j + R + 3\sigma \leq 6(1 - g).$$

Here $y \leq 2 + 2j/3 - 2g$ proves the strict inequality. Lemma 2.10 gives $g = 0$, $1 \leq j \leq 6$, $h < e \leq 2j$, and factorization through the quotient D of Proposition 2.2. After division by the zeros forced by the differential, the coefficient for the first elementary symmetric invariant has degree $-4j + 2y < 0$; its sole possible exception $j = 1, y = s = 2, R = 0, e = 2$ gives $a = 13 > 10j$. At $(1^2, 3^4)$, use the cubic $E = \{u^3 - v^3 = w^3\} \subset \mathbb{P}^2$. Let u_i, v_i, w_i be its coordinate sections on the ith factor of the finite étale cover E^5 of the abelian variety A_P. The sections s_u, s_v pull back to $\prod_{i=1}^5 u_i, \prod_{i=1}^5 v_i$, respectively, and $z_i = w_i/v_i$ near the fixed point, where all v_i are nonzero. The local equation $(s_u/s_v)^3 = \prod_{i=1}^5 (1 + z_i^3)$ makes the radial and first symmetric derivatives independent at every fixed point. At $(1, 3^5)$, the section of degree one for width one is nonzero. Thus all coefficients for the elementary symmetric invariants are divisible by the third power of q_*; after this division their degrees are $(i - 5)j$. Hom has dimension one, so $\ell = 2$, $h = -e$, and $5e + \delta = 5j + 1 - y$. The only integer case with $2e \leq j$ is $(j, e, y, R) = (2, 1, 3, 0)$, forcing $\sigma = 1/3$, impossible for product contributions. Hence $2e > j$ and the next inverse target $h + j = j - e < e$ gives two independent rows through a quotient of rank 1 constructed from the coincident marks.

These exclusions remove every reflection of order greater than two, including those in factors of F_P of smaller rank by Lemma 4.7 and the lift fixing the point for the full group of Theorem 2.19. The normalized lifts now have determinant ± 1, so a scalar zI_7 in their kernel satisfies $z^{14} = 1$. Its order also divides w_K, hence divides two. Proposition 2.11 shows that $2L$ descends to a line bundle, with the full cusp stabilizer included.

For $m = 4, p = 2, 4$, we may therefore put $j = 4t \in 2\mathbb{Z}_{>0}$, $b = j/2$. The line bundle corresponding to P_0 gives $h_P = (h - j)/p$. The degrees of the generators before normalization imply $5\nu_c + 1 - B_c \leq 4$ for $p = 2$ and ≤ 5 for $p = 4$. For $p = 2$ the stronger bound uses the even degree of the unnormalized product: if its maximum exponent is four, the opposite branch of ramification index four has odd exponent and the sum is at least seventeen. Elimination of s gives

$$a - 5h \geq 11j/2 - 6 + 6g + 5R/2 + 4\sigma \quad (p = 2),$$

$$a - 5h \geq 25j/4 - 8 + 8g + 3R + 5\sigma \quad (p = 4).$$

Both exceed $2b = j$. Thus $h < e \leq j/2$, and the independent target line bundle corresponding to P_0 also has degree less than e; hence $\ell \geq 3$ and $a \leq 3j/2$. For $p = 2$, $h_P \geq -j/2$ gives

$h \geq 0$, contradicting the first bound. For $p = 4$, $h \geq -j/2$ gives $a - 5h \leq 4j$, contradicting the second bound for $j \geq 4$. For $j = 2$, $h = -1, 0$ and $(h-2)/4$ is not integral. All corresponding exclusions of cusps and groups of rank 6 use the lifts fixing the points for their full finite groups, preserving period and character. □

4.6 The remaining five types

Lemma 4.9. There are no G_{34} points, full G_{33} cusps, or full G_{33} disks.

Proof. The remaining groups F_P are $A_5, B_5, D_5, G(3,3,5), G_{33}$. Primitivity forces every G_{33} end to have full group G_{33}, trivial character on the fiber of $\mathcal{V}_*$, and width one. The degrees of its generators are $(1,1,3,3,4,6)$. The four fixed points form the kernel of the polarization; lifts of translations to the theta group preserve the characters at all four, by Lemma A.1. The only larger interior groups are $G_{34}, G_{33} \times A_1$. Let R count the former and let σ be the latter contribution, so $2\sigma \in \mathbb{Z}_{\geq 0}$. The equations are

$$2g-2+s+5R/6+\sigma = 2t, \quad a = 36(1-g)-16t-3R/2-17\sigma, \quad h = 12(1-g)-6t-R-6\sigma. \tag{4.10}$$

Before correcting at the cusps, we use line bundles of degrees $(-4t + R/3, -6t, -10t + R/3, -12t, -18t)$ for the basic invariants of the normal representation. To bound the dimension of the Hom space, we successively divide the coefficients by powers of q_* with exponents $(1,3,3,4,6)$. If $g \geq 1$, after division by the zeros forced by the differential, the first two coefficients for the target associated with the invariant of greatest degree have degrees

$$10(1-g) - 4t - 3R/2 - 5\sigma, \qquad 10(1-g) - 2t - 13R/6 - 5\sigma.$$

These degrees and that of the coefficient on the tangent direction of the curve are negative. The remaining three directions bound the rank $\ell - 1$ of the quotient from the ℓ coincident labels in Proposition 2.2, so $\ell \leq 4$, thus $a > 0$, contrary to (4.10). Hence $g = 0$.

For $R > 0$, put $j = 2t$, $R = 3r$, $q = 2\sigma$; integrality of the degrees of the line bundles on the coarse curve makes these integers. Then

$$r + q \equiv 0 \pmod 2, \quad s = 2 + j - (5r+q)/2 \geq 0, \quad a = 36 - 8j - (9r+17q)/2 \leq 7j.$$

If $r = 1$, then q is odd and $j \geq 2$; if $r \geq 2$, then $j \geq 3$. After division by the zeros forced by the differential, the degrees of the first two coefficients are strictly negative, and when $h < 0$ the same bound $\ell - 1 \leq 3$ from Proposition 2.2 gives $a > 0$. When $h \geq 0$, use $a = 3h+j+9r/2+q/2 > 0$ instead. For $r \geq 3$, the inequality for s gives $a \leq 52-49r/2-25q/2 < 0$. Positivity leaves exactly

$$(t, R, s, \sigma, a, h) = (3/2, 6, 0, 0, 3, -3), \qquad (1, 3, 1, 1/2, 7, 0).$$

In the first row a third coefficient vanishes, so $\ell \leq 3$, $e \geq 3$, and $a \geq 3e \geq 9$. In the second, after division by the zeros forced by the differential, the five coefficients for a target $\mathcal{O}(-1)$ have degrees $(-2, -4, -1, -2, -1)$, so no quotient onto a line bundle of negative degree exists. All degrees m_i have the same parity and sum seven. They cannot all be nonnegative, since then all nine would be positive odd integers. Two negative degrees m_i give a quotient onto a line bundle of negative degree. With one, $7 = 7e + 2\delta$ gives $e = 1$, $\delta = 0$, and $E = \mathcal{O}(2) \oplus \mathcal{O}(1)^{\oplus 5}$, which cannot map nontrivially to Q_0. Thus G_{34} is excluded.

Now $R = 0$. The bound $a \leq 10t$ follows from $a \leq 14t$: it is automatic for $t \geq 3/2$, and for $t = 1/2, 1$ the Riemann–Hurwitz formula for the disk leaves respectively $a \leq 0, 3$. The central scalar ω has characters $(\omega, 1, \omega, 1, 1)$ on the normal directions. Its projector preserves the local lattices of the normal bundle on the coarse curve and splits the quotient N_f^{sat} of f^*T_W by the saturated image of df as $U \oplus V$ of ranks 2 and 3. We have

$$\deg U = -14t + 4s = 8 - 6t - 4\sigma < 0, \qquad 0 \longrightarrow E_4(S) \longrightarrow U \longrightarrow E_{10}(3S) \longrightarrow 0.$$

The splitting of Proposition 2.3 forces $U = \mathcal{O}(-e)^{\oplus 2}$; a summand of U of positive degree would make its determinant nonnegative. Thus $e = 3t + 2\sigma - 4 > 0$ and $t + \sigma \leq 2$. Integrality leaves $(t, s, \sigma, a, h, e) = (1, 3, 1, 3, 0, 1), (2, 6, 0, 4, 0, 2)$. After division by the zeros forced by the differential, the coefficients of maps $V \to \mathcal{O}(-e)$ have degrees $(-4, -1, -1)$ or twice these values, so $\ell = 3$. The second row contradicts $a \geq 3e$; the first gives $V = \mathcal{O}(1)^{\oplus 3}$ and a nonzero map $V \to \mathcal{O}$ associated with the invariant of greatest degree, also impossible. A disk without ends has $h = -18t$ and $a = 2 - 2g - 50t < 0$, since $2t$ is a positive integer. Only its last coefficient can survive, giving $\ell = 2$ and $a \geq 90t > 0$. □

Lemma 4.10. There are no full B_5 disks or smooth cusps.

Proof. The remaining groups F_P force full B_5 ends with trivial character and width one. The complete pairs are $(B_c, \nu_c) = (6, 1), (9, 2), (10, 2), (11, 2)$; the only larger irreducible stabilizer is B_6. For $h = -10t + R + \sum \nu_c$, equation (3.4) gives

$$a \geq 9h/2 + 13t + R + \sigma.$$

First $a \leq 14t$ gives $h \leq 2t/9$ and $a - 5h \geq 116t/9 > 4t$. Then Lemma 2.10 gives $h < e$, $\ell \geq 2$, $a \leq 10t$, and $h \leq -2t/3$. After division by the zeros forced by the differential, the lowest two coefficients have degrees $-8t+\nu-s, -6t+\nu-s$, with $\nu - s \leq s \leq 2+2t-2g$. They are negative for $t \geq 1$; for $t = 1/2$ use integrality of $h < 0$ to get $R + \nu \leq 4$. In each of the three models with $\nu_c = 2$, the sections of degree one vanishing at a fixed point project injectively to the directions of the first two elementary symmetric invariants. Use the four sections $x_1, \dots, x_4$, with $x_j^2 = \ell_j(T, U)$, where $[T : U] \in \mathbb{P}^1$ and the four linear forms have distinct zeros. At the chosen fixed point, let $x_j = 0$, choose $x_b \neq 0$, set $z = x_j/x_b$, and let c, d be the other two indices. Set $\gamma_c = (x_c/x_b)(0)$ and $\gamma_d = (x_d/x_b)(0)$, and define a_c, a_d by $\ell_c/\ell_b = \gamma_c^2(1 + a_c z^2)$ and $\ell_d/\ell_b = \gamma_d^2(1 + a_d z^2)$. Their ratios are $x_c/x_b = \gamma_c(1 + a_c z^2)^{1/2}$ and $x_d/x_b = \gamma_d(1 + a_d z^2)^{1/2}$, with $a_c a_d(a_d - a_c) \neq 0$. Their three product sections have an invertible matrix of values and derivatives in degrees two and four, and this remains injective on the descending subspace. Theta translation retains all torsors and linearizations at every fixed point. Thus the two vanished coefficients give a second division by powers of q_*; the $\nu_c = 1$ model needs only the first such division. The five remaining degrees are $(2i - 10)t$, so Hom has dimension one and $\ell = 2$. Formula (2.3) then yields

$$h + 2t \leq 2e/3 - 2(R + \sigma)/9 < e.$$

On F, the row associated with the invariant of greatest degree vanishes and the B_6 equations $dg_6 = v\, df_5$, $dg_5 = df_5 + v\, df_4$ leave only a simple pole in the next row. Its target has degree $h + 2t < e$. Two independent rows cannot factor through the part of rank 1 constructed from the coincident marks. The lift fixing the point for the full group excludes the cusps by the same argument. □

Lemma 4.11. There are no full E_6 points.

Proof. Their full D_5 disks have precisely the models of Lemma 4.3: G_{33} has no reflection subgroup of full rank D_5, and the other possibilities for F_P have been excluded. Since $R > 0$, that lemma gives $h > 0$ and E nef. If all $m_i \geq 0$, the decomposition $[f_* C] = \frac{1}{2} \sum_i m_i[\gamma_i]$ of Proposition 2.8 and integrality of $(2L)\gamma_i$ give $a \leq 4t$, contrary to $a = 4h+5t+D+\sigma$. Otherwise nefness leaves one negative degree $m_i = -e$, and the columns for the other marks generate E, giving $h \geq e$. We need $(2L)\beta_i \geq 2$. If this degree were one, on the open set where mark i is distinct the polarization would be $\mathcal{O}(1)$ on $\mathbb{P}^6$ minus linear subspaces of codimension two. Represent $2L$ by multidegrees c_i, $\sum c_i = 2q$, so $q - c_i = 1$. For $j, k \neq i$, we have $m_j, m_k \geq e$ by (2.1), so the sections σ_j, σ_k either coincide or have a bracket of positive degree $(m_j + m_k)/2$;

in both cases they meet in some fiber, whose stability makes (j,k) short. Hence, for a short pair (i,j), every pair containing j is short, so the coordinate curve γ_j of Proposition 2.8 exists; since $K_W \cdot \gamma_j = -2$ and K_W is π-nef, π does not contract it, and $c_j = (2L) \cdot \gamma_j \geq 1$. For the other pairs, the bracket $[ij]$ is a unit on the stable locus; adding its multidegree makes all $c_j \geq 1$, preserving $q - c_i = 1$. For $j,k \neq i$, the global invariant

$$[jk] \prod_{\ell \neq i} [i\ell]^{c_\ell - \delta_{\ell j} - \delta_{\ell k}}$$

restricts to $z_j - z_k$. Normality and multiplication identify the section ring with a polynomial ring on seven generators of degree one. Hence $(X, 2L) = (\mathbb{P}^6, \mathcal{O}(1))$, excluded independently by Proposition B.2. The small projective map to factorial $\mathbb{P}^6$ is an isomorphism, so this exclusion applies to X as well as W. The first decomposition in (2.5) now gives $t \geq e + \delta/2$, and

$$4h = 7e + 2\delta - 5t - D - \sigma \leq 2e - \delta/2 - D - \sigma < 4e,$$

contrary to $h \geq e$. □

Lemma 4.12. There are no full D_5 disks or smooth cusps.

Proof. Only points with stabilizer D_6 or $D_5 \times \mu_e$ remain, and every cusp has full group D_5 and width one. The line bundle E_P associated with P_0 has degree $-5t - D/2$ and defines a direct summand

$$E = N_P \oplus E', \qquad N_P = E_P(D + S), \qquad p = \deg N_P = -5t + D/2 + s.$$

At D_6, P_0 pulls back to $Z = v\psi$ in the frame P/u on the coarse quotient. At an end its column is the odd section $q\theta_P$. Divide that column and the corresponding row of the inverse differential by the same parameter; their composite is the identity. Theta translation proves this at all sixteen fixed points, so the splitting is global. The formulas become

$$2g - 2 + s + D/2 + \sigma = 2t, \quad p = 2(1-g) - 3t - \sigma = h/2 - t, \quad a = 8p + 13t + D + \sigma.$$

Every nonzero target has degree at least $-2t$, so $a - 5p \geq 7t > 4t$. Lemma 2.10 gives $p < e$, $\ell \geq 2$, $a \leq 10t$, and $p < 0$. The conclusion on direct summands in Proposition 2.3 gives $p = -e$. Therefore $g = 0$ and $k = 2 - t - \sigma = 2t - e \in \{0, 1\}$. For $k = 0$, $e = 2t$, $\delta = 0$, $h = -e$, and $s + D/2 = 3t$. After division by the zeros forced by the differential, the coefficients for the first two elementary symmetric invariants have degrees $(2i-8)t + s$, $i = 1, 2$, which are negative. The projections of the first derivatives of invariant sections give the second division by powers of q_* at the cusps; the degrees of the coefficients for the four elementary symmetric invariants become $(2i-8)t$. The directions of P_0 and of the elementary symmetric invariants contribute to a Hom space of dimension at most two, hence $\ell = 3$. For the next symmetric invariant, the inverse differential restricted to F has a row with only simple D_6 poles and target $h + 2t = 0 < e$. Three independent rows would factor through the part of rank 2 constructed from the coincident marks. For $k = 1$, $(t, \sigma, e, p, h) = (1, 0, 1, -1, 0)$. The independent rows associated with P_0 and the elementary symmetric invariant of greatest degree give $\ell \geq 3$. The equations $D = 8 - 2s$, $a = 13 - 2s = (9 - 2\ell) + 2\delta$, $\delta \leq 1$, leave $(s, D, \ell, \delta) = (4, 0, 3, 1)$. For target $N_P = \mathcal{O}(-1)$, after division by the zeros forced by the differential, the first two coefficients have degrees $-3, -1$, and their degrees after the second division are $-7, -5, -3, -1$. Only the direction of P_0 contributes to Hom, of dimension one; the part of rank 2 constructed from the coincident marks requires dimension two. The lift fixing the point for the full group again excludes all cusps. □

Lemma 4.13. There are no full $G(3,3,5)$ disks or smooth cusps.

Proof. Only points with stabilizer $G(3,3,6)$ or $G(3,3,5) \times \mu_e$ remain; the cusps have width one and, for the full group, generators of degrees $(1^3, 3^3)$. Integrality of the degrees of the line bundles on the coarse curve gives $3t \in \mathbb{Z}$, $-5t - 2R/3 \in \mathbb{Z}$. Together with $2t \in \mathbb{Z}$, they imply $t \in \mathbb{Z}_{>0}$ and $3 \mid R$; the Riemann–Hurwitz formula for the disk makes σ integral. The formulas are

$$2g - 2 + s + 2R/3 + \sigma = 2t, \quad a = 24(1-g) - 13t - 3R - 11\sigma, \quad h = 6(1-g) - 6t - R - 3\sigma.$$

Since $h \geq -2t$, we have $4t + R + 3\sigma \leq 6(1-g)$. Only $(g, t, R, \sigma, s, a, h) = (0, 1, 0, 0, 4, 11, 0)$ remains. By the nonzero map to a line bundle of degree zero associated with the invariant of greatest degree, we have $a \leq 10t$, contrary to $a = 11$. □

Theorem 4.14. No nine-point quotient under consideration is the Baily–Borel compactification of a nonuniform arithmetic ball quotient.

Proof. The nontrivial lattice is generated by reflections by Proposition 2.11. Start with a reflection and pass to larger irreducible factors of full point stabilizers by Lemma 4.7. The preceding exclusions force A_5 or A_6, and the latter has a parabolic subgroup A_5 that is the full generic stabilizer of a disk. All ends of this full A_5 disk are smooth, full A_5 ends, with width one, trivial character on the fiber of $\mathcal{V}_*$, and six generators of degree one. Proposition 3.11 applies without genus or immersion assumptions. For the invariant of greatest degree, the nonzero target line bundle satisfies $6h = a - 14t - \sigma$. First $a \leq 14t$ gives $h \leq 0$, and then (4.9) gives $a \leq 10t$. The quotient $\mathcal{Q}$ of rank 2 constructed in the proof of Proposition 3.11 gives $a \geq 11t + \sigma$, a contradiction. Combining this with Theorem 4.6 proves the case of nine points with arbitrary weights. □

5 Ten points

Here the ball has dimension seven. We use one argument for the three arithmetic cases, treating them separately only until all their stabilizers have been shown to be of ADE type. For a curve obtained from the full normalizer put $a = -K_W \cdot C$, $t = L \cdot C > 0$, and $g = g(C)$. Propositions 2.2 and 2.3 give, whenever a nonzero row of the inverse differential has a target of negative degree,

$$a = (10 - 2\ell)e + 2\delta, \qquad e \in \mathbb{Z}_{>0}, \quad \delta \in \mathbb{Z}_{\geq 0}, \qquad f^*T_W \twoheadrightarrow M^{\oplus(\ell-1)}, \quad \deg M = -e. \tag{5.1}$$

Moreover a is even, since the anticanonical class descends from $\mathcal{O}(2, \dots, 2)$. The finite enumerations in this section include the cases $a \leq 0$ and $s = 0$, and retain every integer e allowed by (5.1).

5.1 Initial degree bounds

Lemma 5.1. In the compact case $K_W + (32/5)L$ is nef. In the nonuniform Gaussian case $K_W + 8L$ is nef. In the remaining nonuniform cases $K_W + 4wL$ is nef, where $w = |\mu(K)| = 2$ or 6.

Proof. Proposition B.7 excludes $W = \mathbb{P}^7$. In that case $\pi^* A_X = \mathcal{O}(k)$ with $k > 0$ is ample, so π is finite birational onto normal X_μ and hence is an isomorphism. The K_W-negative rays of Proposition 2.8 have respective degrees against $-K_W$ $2, 6, 4, 2$: a coordinate curve, and a line where $q = 2, 3, 4$ labels coincide. The closed loci where these labels coincide are $\mathbb{P}^{8-q}$ with normal bundle $\mathcal{O}(-1)^{q-1}$. Relative nefness prevents their contraction, and the maps of these loci to their normal images are isomorphisms. Lemma B.3 and Proposition B.2 give L-degree at least one for $q = 2$. For cocompact arithmetic lattices, Proposition B.4 gives the bounds

$1, 2/3, 1/3$ on the lines where $q = 2, 3, 4$ labels coincide. Their ratios are at most six, but Lemma 3.9 gives $L \cdot \gamma \geq 5/16$ on a coordinate curve, hence ratio at most $32/5$. For nonuniform groups wL is a line bundle. This gives ratios at most $2w, 6, 4w, 2w$. When $w = 4$, a slope $1/4$ on the locus $\mathbb{P}^5$ where three labels coincide would induce a quotient by Lemma B.3. The induced quotient is nonuniform: its fixed Hermitian space over the original imaginary quadratic field has dimension at least three, so Meyer supplies a rational isotropic line. Proposition B.6 excludes it. Therefore $L|_{\mathbb{P}^5} \geq H/2$, and all four ratios are at most eight. □

5.2 Stabilizers and nefness in the nonuniform case

Lemma 5.2. If the arithmetic quotient for ten points is nonuniform, all point stabilizers are products of ADE groups, all ends relevant to D_6 and E_6 have width one, and $4L$ is integral. If the field is not Gaussian, $2L$ is integral.

Proof. Apply Proposition 3.4, using Lemma 5.1, before reducing any multiple of L in that lemma. The domains are given below for rank 6 and in Appendix B.1 for the auxiliary projective spaces; the shared lists of cusps and the conditions on the common degree $-e$ of the coincident labels are in Appendices C.1–C.2. For clarity, we give the reductions determining those domains. The primitive group G_{34} is treated separately before Proposition 3.4 is applied to the monomial groups. Its polynomial invariants have degrees $(d_1, \ldots, d_6) = (6, 12, 18, 24, 30, 42)$, and the generators of its ring of invariant sections have degrees $(1, 3, 4, 6, 7, 9, 12)$, with no torsor and width one. Its principal theta divisor is irreducible, and its multiplicity is at most four [22, Theorem 1.4], below the first invariant degree six; hence its invariant generator of degree one is nonzero at the origin. After imposing three zeros at each end, the coefficient associated with the invariant of degree d_i has degree $(d_i - 42)t + 9s$. We have $h = -42t + 12s$, $a = 84(1 - g) - 50t - 41\sigma$, with even s and $j = 6t$ integral. The weak inequality $a \leq 24t$ gives $g = 0$, $j \leq 8$ for $h \geq 0$ and $j \leq 11$ for $h < 0$. The Riemann–Hurwitz formula for the disk and (5.1) exclude every row; the sole remaining value $1/3$ for the contribution of disk inertia is not an allowed sum of $1 - 1/e$. For $m = 6$, set $j = 6t$, $h_P = (-36t + R + \nu)/p$ and $h = h_P$ for $p = 1$, $h = ph_P + 6t$ otherwise. The line spanned by P_0 is preserved since $6^5 > 6p$. The degree and rank bounds reduce the calculation to $g = 0$, $1 \leq j \leq 9$, $s \leq \lfloor 2 + j/3 \rfloor$, $R \leq \lfloor (6/5)(2 + j/3 - s) \rfloor$. The twelve remaining rows have respectively $p = 1$ (two rows), $p = 2$ (five), or $p = 3, 6$ (five). The first two violate $a \geq 6(-h)$; the five $p = 2$ rows require $a = 4j$, with three further contradicted by a nonzero map from the kernel in Proposition 2.3 to a line bundle of degree zero, below $e = j$; the last five have $h < h_P < 0$, contrary to the splitting of Proposition 2.3. The tests for $m = 3$ use $G(3, 1, 6)$ with degrees of the generators at the ends $(1^7), (1^2, 3^5), (1, 3^6)$ and $1 \leq 3t \leq 4$, followed by $G(3, 3, 6)$ with $(1^3, 3^4)$ and $1 \leq 3t \leq 3$. They leave no row. In the Gaussian case use $G(4, p, 6)$, $p = 1, 2, 4$, and $j = 4t$. The negative coefficient rank bound gives $j \leq 5, 6, 8$, respectively. For $p = 4$, it suffices to retain the following end types, where α is the tuple of degrees of the generators and κ the width:

type	α	κ
A	$(1^3, 2^4)$	$1, 2$
B	$(1^2, 2^5)$	$1, 2$
C	$(1, 2^2, 4^4)$	$1, 2$
D	$(1, 2, 4^5)$	$1, 2$
E	$(1^2, 2, 4^4)$	1
X	$(1^2, 4^5)$	$1.$

The type X is the nontrivial torsor; width two would require a point fixed by all of $G(4, 2, 6)$ and is therefore excluded. For each multiset of s ends, put $B = \sum_c \sum_i \alpha_{c,i}/\kappa_c$ and $\nu =$

$\sum_c \max_i \alpha_{c,i}/\kappa_c$. The complete necessary numerical conditions are

$$1 \le j \le 8, \quad 0 \le s \le \lfloor 2 + j/2 \rfloor, \quad 0 \le R \le \lfloor (4/3)(2 + j/2 - s) \rfloor,$$
$$\sigma = 2 + j/2 - s - 3R/4 \in \{0\} \cup [1/2, \infty),$$
$$a = B - s + 2 - 33j/2 + 21R/4 \in 2\mathbb{Z}, \quad a \le 2j, \qquad h = -5j + \nu + R.$$

There is no solution with $h \ge 0$. For $h < 0$, put $u = \max(0, \min(5, \lfloor (\nu - s)/j \rfloor + 1))$, and use $2 \le \ell \le \min(7, 1 + u + \epsilon_P)$. Here ϵ_P is the number of possible coefficients for P_0, so $\epsilon_P \le 1$. When all widths are one, the degree $E_P = -3j/2 - 3R/4$ must be integral and $h - E_P - s < R$ forces $\epsilon_P = 0$. Finally require $e \ge \max(1, -h)$ and $a - (10 - 2\ell)e \in 2\mathbb{Z}_{\ge 0}$. Substitution in these bounded sets leaves only the following signatures; all have $R = 0$, $\ell = 5$, and at most two ends of width two:

j	s	ν	h	a	σ
2	3	7	-3	$2, 4$	0
3	3	9	-6	0	1/2
3	3	10	-5	$0, 2, 4, 6$	1/2
4	4	12	-8	0	0
4	4	13	-7	$0, 2, 4, 6$	0.

At such an end the least degree of a generator that does not vanish there is the width; thus the coarse curve is immersed at the end, with no fiber correction. Condition (3.9) for the injection (3.8), for degrees $(4, 8, 12, 16, 20, 6)$, requires

$$2a - 4 + 45j - 12\nu \le 0.$$

The five rows give $2a + 2$, $2a + 23$, $2a + 11$, $2a + 32$, $2a + 20$, all positive. This excludes the remaining groups and cusps with $m = 4$. For an irreducible factor of rank at most 5 satisfying the condition on full interior stabilizers in Lemma 2.12, apply Meyer to its fixed Hermitian space of dimension at least three; the induced quotient isomorphic to projective space is nonuniform over the same field. The reduction for these factors, Proposition B.2, and Propositions B.5–B.6 now reduce the possibilities to parabolics of A_7, B_7, D_7, E_7. For B_6, the four tuples of degrees of the generators at cusps of width one are $(1^7), (1^3, 2^4), (1^2, 2^5), (1, 2^6)$. With $\nu = \sum_c \max_i \alpha_{c,i}$, its equations are

$$2g - 2 + s + R/2 + \sigma = 2t, \quad a = \sum_c (B_c - 1) + 2 - 2g - 42t + 6R, \quad h = -12t + \nu + R.$$

The integrality of $42t$ and $4t$ gives $2t \in \mathbb{Z}_{>0}$. For $h \ge 0$ one obtains $t = 1/2$, $g = \sigma = 0$, and $a \ge 17 - 2s \ge 11 > 8t$. For $h < 0$ only $g = 0$, $2t = 1, 2$ remain, and substitution of the four tuples in (5.1) gives no row. The stabilizers are therefore ADE products. Normalized reflection lifts in dimension eight have determinants ± 1; the group of scalars among them has order dividing $\gcd(16, w)$. The descent at cusps uses the invariant corresponding power of the frame $h(v, \ell) = 1$. It gives $2L$ outside the Gaussian case and $4L$ in that case. □

Lemma 5.3. Once the ambient stabilizers are ADE products, $K_W + 6L$ is nef.

Proof. On the closed locus $Z = \mathbb{P}^5$ where three labels coincide, a restriction $L|_Z = \lambda H$ with $\lambda < 1$ induces a quotient by Lemma B.3. Its pointwise stabilizer is A_2 or $2A_1$. The effective restrictions of the normalizers to the fixed spaces are products of classical A, B, D groups; for exceptional J they are

J	F	$N_J(F)/F$
E_7	A_2	$A_5 \times \mathbb{Z}/2$
E_7	$2A_1$	$A_1 + B_4$
E_6	A_2	$A_2 \wr (\mathbb{Z}/2)$
E_6	$2A_1$	B_3

.

If $F = 2A_1$ has one factor in E_6 and the other in a separate direct factor of J, the effective action of $N_{E_6}(A_1)/A_1$ on the fixed hyperplane is A_5. Smoothness shows that the effective restrictions are generated by reflections. The kernel of the character on the fiber of $\mathcal{V}_*$ at the cusp is a finite group of this type and rank at most 4, hence of order at most 384. This excludes the cusp types other than A_4, B_4, and D_4: the Gaussian $G(4,4,4)$ lift has order 1536; $m = 3$ or $m = 6$ gives a nonreal character; for F_4 the kernel has order at least 576; G_{32} lifts in its entirety; and the G_{29}, G_{31} kernels have order at least $480, 2880$, respectively. For the G_{31} torsor the residual orbit has size six, and the kernel of the character on the fiber of $\mathcal{V}_*$ has order at least 480. Thus only A_4, B_4, D_4 cusp types remain. The integrality of $2L$, or of $4L$ and $24c_2^{\mathrm{orb}} = 360L^2$, makes $\lambda = 1/2$ the only possible value below one. For $\lambda = 1/2$ put $t = u/2$ and $a = 6u$. For B_4, the bound $h \geq 2t$ from Lemma B.1 and $h \leq -4t + 4(1-g) - 2\sigma$ leave $g = 0, u = 1$; the four possible tuples of degrees of the generators at the ends give $h = 0 < u$. For D_4 the equations give

$$7u + 2s + 7\sigma = 16(1-g), \qquad (u, t, s, R, \sigma) = (1, 1/2, 1, 2, 1), (2, 1, 1, 6, 0).$$

The curve is a line or smooth conic, whose normal bundle has largest summand 1 or 4. The column of the differential of the quotient map associated with the quadratic invariant gives a nonzero map into that normal bundle from a line bundle of degree $-2t + R + s = 2$ or 5, a contradiction. Finally A_4 gives $5h = 3t - \sigma < 5u$. Consequently $L|_Z \geq H$. In the Gaussian case, on the closed locus $\mathbb{P}^4$ where four labels coincide, the pointwise stabilizer, an ADE group of rank 3, has a normalizer whose effective restriction to the fixed space is real. The induced group is generated by reflections and the group of scalars among its lifts in dimension five has order dividing $\gcd(10, 4) = 2$. The same frame at the cusp therefore proves that the restriction of the original $2L$ is integral. Hence $L|_{\mathbb{P}^4} \geq H/2$. The lower bound $5/16$ for the L-degree of a coordinate curve, combined with integrality of $2L$ or $4L$, gives L-degree at least $1/2$ on coordinate curves. The four degrees against $-K_W$ $2, 6, 4, 2$ now have L-degree bounds $1/2, 1, 1, 1/2$, so Proposition 2.8 proves the assertion. □

5.3 Fixed curves with full generic stabilizer E_6 and the conclusion

Lemma 5.4. For a curve in the quotient of ten points obtained from a disk fixed by a finite reflection group, with full generic stabilizer E_6, points with stabilizer $E_6 \times \mu_e$, cusps with full group E_6 and width one, and possibly R points of type E_7, the bound $a \leq 32t/5$ excludes the curve.

Proof. For $R > 0$ the coordinates of odd degree on the coarse quotient in the normal directions are $p_5/u, p_9/u$, with $v = u^2$. Their residues give

$$2g - 2 + s + R/2 + \sigma = 2t, \quad a = 12s + 27R/2 + \sigma - 44t, \quad h = -12t + 3R + 3s.$$

Here $2t \in \mathbb{Z}_{>0}$, $2t + R \in 2\mathbb{Z}$, and a is even. The case $h \geq 0$ reduces to $2t = 1, 2, 3$ and has no row satisfying the parity conditions. For $h < 0$, after division by the zeros forced by the differential, the first five coefficients have degrees

$$-10t + R + 2s,\ -7t + 3R/2 + 2s,\ -6t + R + s,\ -4t + R + s,\ -3t + R/2 + s.$$

They are negative for $g > 0$. For $g = 0$, one of these degrees can be nonnegative only if $t \leq 2$: the first, third, and fourth degrees are negative, the second is less than $2 - t$, and the fifth is at most $2 - t$. All vanish except at $(t, R, s, \sigma, a, h) = (3/2, 1, 4, 1/2, -4, -3)$, where the rank of the evaluation map from the Hom space violates (5.1). For the target associated with the invariant of greatest degree, the Hom space has dimension one and $e \geq -h$. On its kernel, the row associated with the invariant of degree nine extends to a line bundle of degree $h_9 = -9t +$

$5R/2 + 2s$. Since $h + h_9 \le -R/8 - 5\sigma/12 < 0$, one has $h_9 < e$, contradicting Proposition 2.3. For $R = 0$ the degree equations are $a = 24(1-g) - 20t - 11\sigma$, $h = 6(1-g) - 6t - 3\sigma$, and $t \in \mathbb{Z}_{>0}$. After two successive divisions by powers of q_* at the ends, the degrees $(d-12)t + s$ exclude every case except $(g, t, s, \sigma, a, h) = (0, 1, 4, 0, 4, 0)$. Here the normal bundle splits under the involution induced by $-I$ as $N^+ \oplus N^-$, with $\operatorname{rank} N^- = 2$ and $\deg N^- = -14t + 3s = -2$. A quotient line bundle of negative degree exists, but the flags defined by the theta functions give $\operatorname{Hom}(N^+, \mathcal{O}(-1)) = 0$. The splitting in Proposition 2.3 then forces every summand of N^+ to have positive degree. The nonzero map to $E_{12}(3S) = \mathcal{O}$ associated with the invariant of greatest degree is impossible. □

Theorem 5.5. For every ten weights satisfying (1.1), the quotient is not the Baily–Borel compactification of an arithmetic ball quotient.

Proof. For cocompact arithmetic lattices, Lemma 3.15 and the reduction for maximal irreducible factors satisfying the condition on full interior stabilizers in Lemma 2.12 remove the monomial factors outside the parabolics of A_7, B_7, D_7, $G(3,3,7)$, E_7, G_{34}. Factors of rank at most 5 are excluded by Propositions B.2 and B.4 and equality of the degrees of the basic invariants. For cocompact arithmetic lattices, Lemma 3.15 includes the case of full generic stabilizer B_6. At this stage $a \le 32t/5$ by Lemma 5.1. In the nonuniform cases, Lemmas 5.2 and 5.3 give ADE stabilizers and $a \le 6t < 32t/5$. In both cases, Lemma 5.4 excludes E_7, then the remaining E_6 curves. For cocompact arithmetic lattices, G_{34} gives $a \ge 252t$, and Proposition 3.4 then gives $a \ge 74t/5$ for $G(3,3,6)$ and $a \ge 44t/5$ for D_6. Each contradicts the bound $a \le 32t/5$ for cocompact lattices. For nonuniform D_6, the cusps have width one and generators of degrees $(1^4, 2^3)$, and the equations of Proposition 3.4 are

$$2g - 2 + s + R/2 + \sigma = 2t, \quad a = 9s + 2 - 2g - 36t + 11R/2, \quad h = -10t + R + 2s.$$

They reduce to $g = 0$, $j = 2t = 1, 2, 3$. Equation (5.1), with every allowed integer e retained, leaves only

$$(j, s, R, \sigma, a, h) = (1, 2, 0, 1, 2, -1), (2, 2, 4, 0, 6, -2), (2, 3, 2, 0, 4, -2), (2, 4, 0, 0, 2, -2).$$

The product contributes one map. The flag for the minimal polarization of Proposition 3.13, after two successive divisions by powers of q_* at the ends, leaves degrees $h + ij - 2s$ for the coefficients of the elementary symmetric invariants, less than R for $i < 5$ and equal to R for $i = 5$. The interior zeros therefore leave at most one symmetric map. But (5.1) requires $\ell \ge 4$ and at least three independent maps, a contradiction. Finally Proposition 3.11 gives $a \ge 33t/2 + \sigma$ for the last A_6 disk. This contradicts the upper bound for a available in each case and completes the ten-point case without any assumption on the support of the branch divisor or on cusp width. □

6 Eleven or more labels

Write $d = n - 3$, $r = d - 1$, and $N = d + 1$. On the smooth GIT quotient $W = X_\nu$ obtained by perturbing μ to weights ν in a nearby chamber, with the small projective morphism $\pi : W \to X_\mu$, put $L = \pi^*\mathcal{L}$, and in the nonuniform case put $w = w_K = |\mu(K)|$ for the coefficient field K of Proposition 2.11. For a curve $f : C \to W$ obtained from the quotient of a disk fixed by a finite reflection group by its full arithmetic normalizer, put

$$a = -K_W \cdot f_*C, \qquad t = L \cdot f_*C > 0, \qquad g = g(C).$$

No sign condition on a will be used before it follows from the comparison using the tangent bundle. We use strong induction on the number of labels; the cases of nine and ten labels

are proved in Sections 4 and 5. If an irreducible factor of rank at most $d-2$ satisfies the condition on full interior stabilizers in Lemma 2.12, Proposition 2.14 identifies the Baily–Borel compactification of the quotient of its fixed ball by the full normalizer with the GIT quotient of the original merged weights on $n-s$ labels, where $s \le 2$. Thus $9 \le n-s < n$, and the induction hypothesis excludes every such factor. This uses the original weights and the quotient of the fixed ball by the full normalizer; no ball structure is transferred to W. Each nefness statement below for K_W plus a smaller multiple of L is established only after the stabilizers it requires have been excluded. After the common calculation, we complete the induction in the order eleven labels, twelve labels, and $n \ge 13$.

6.1 The common monomial calculation

For $n = 11, 12$, Proposition B.7 excludes $W = \mathbb{P}^d$. If the source W is projective space, then the pullback of an ample line bundle A_X is $\pi^* A_X = \mathcal{O}(k)$ with $k > 0$, hence is ample. Thus π has finite fibers, and a finite birational morphism onto normal X_μ is an isomorphism. Corollary 2.9 then gives $a \le rwt$ in the nonuniform case. For $n \ge 13$, we retain the weaker bound $a \le (r+2)wt$ of Lemma 2.10 for each curve until a homomorphism from the tangent bundle to a line bundle of nonpositive degree has been found. Thus the argument for large n needs no auxiliary theorem for projective space.

For a disk with full generic stabilizer $G(m,p,r)$, use

$$D = \frac{mr(r-1)}{2} + \frac{mr}{p}, \quad \eta = r - 1 + \frac{1}{p}, \quad k = \begin{cases} mr & p = 1, \\ m(r-1) & p > 1. \end{cases}$$

At an end let α_c be the normalized tuple of degrees of the generators of its ring of invariant sections for the full stabilizer, κ_c the order of its character on the fiber of $\mathcal{V}_*$, $B_c = \sum_i \alpha_{c,i}/\kappa_c$, and $\nu_c = \lfloor \max_i \alpha_{c,i}/\kappa_c \rfloor$. Write R for the number of points with a larger irreducible stabilizer, s for the number of ends, and σ for the sum of $1 - e^{-1}$ over the remaining points of the disk, where e is the effective inertia order at the point. Proposition 3.4 gives

$$2g - 2 + s + (1 - 1/m)R + \sigma = 2t, \tag{6.1}$$

$$a = \sum_c (B_c - 1) + 2 - 2g - Dt + \eta R, \tag{6.2}$$

$$j = mt \in \mathbb{Z}_{>0}, \qquad h = -kt + \sum_c \nu_c + R. \tag{6.3}$$

Here h is the degree of a target line bundle of the inverse differential, with its finite character and its correction on passing to the coarse curve retained. These equations are used together with the homomorphism from the tangent bundle to a line bundle of negative degree

$$a = (n - 2\ell)e + 2\delta, \qquad e \ge -h, \quad e > 0, \quad \delta \in \mathbb{Z}_{\ge 0}. \tag{6.4}$$

The bounds on ℓ obtained by counting coefficients, table of the degrees of the generators at the cusps before normalization, and exact integer ranges are recorded once in Appendix C.2.

Proposition 6.1. If $r = n - 4 \ge 9$, no disk with monomial full generic stabilizer or irreducible smooth monomial cusp occurs.

Proof. Disks without ends are excluded by Lemma 3.15. For a nonuniform arithmetic lattice, the trace of a normalized reflection of order m lies in the imaginary quadratic field K, which restricts m to $2, 3, 4, 6$. Descending in m excludes smaller values of m at the ends, and every end has group $G(m, q_c, r)$, $q_c \mid p$, with $\kappa_c \mid p/q_c$. The bounds

$$(r-1)\max \alpha_c \le \sum_i \alpha_{c,i} \le rm + 1$$

of (C.2) and the bound $a \le (r+2)wt$ of Lemma 2.10 give

$$(r-1)h \le \{(r+2)w + D + 2 - (r-1)k\}t - (1 + 1/p - 1/m)R - \sigma,$$
$$a \le (2rm+2)(1-g) - \{r(r-5)/2 + r/p\}j + \{r - 1 + 1/p - r(m-1)\}R.$$

For $m \ge 3$, the first coefficient is at most $-3r^2/2 + 21r/2 + 14$ if $p = 1$, or $-3r^2/2 + 12r + 11$ otherwise. For $m = 2$, $h \le -(r-3)j + 4(1-g) \le -2$. Thus $h < 0$ without a prior assumption on a or $W \ne \mathbb{P}^d$. Lemma 2.10 now strengthens the bound to $a \le rwt$, so the finite range of parameters obtained below is unchanged. Counting coefficients as in Appendix C.2 gives $\ell \le 4$ for $g \ge 1$, or $\ell \le 6$ for $j \ge 6$, and then (6.4) gives $a > 0$. The bound $a \le (2rm+2)(1-g) - \{r(r-5)/2 + r/p\}j + \{r - 1 + 1/p - r(m-1)\}R$ excludes $g \ge 1$ and $j \ge 6$, the latter by $3r^2 - 26r - 1 \ge 8$. For $r \ge 29$, the remaining bound $\ell \le 14$ contradicts $a \le -r^2/2 + 43r/3 + 2 < 0$. For $r \ge 10, j \ge 5$, the contradiction is $(15r^2 - 142r - 6)/6 > 0$. The remaining range of parameters is therefore $9 \le r \le 28$, with $j \le 5$ at $r = 9$ and $j \le 4$ otherwise. The 170 rows in Appendix C.3 are excluded by Proposition 3.7, the width given by (C.3), and the absence of points of the Gaussian torsor fixed by all of $G(4,2,r)$. The lifts of the entire groups at the cusps in Appendix C.1 then exclude the corresponding cusps. □

6.2 Eleven labels

Proposition 6.2. The nonexistence theorem holds for eleven labels.

Proof. Here $d = 8, r = 7$. First use only the bound $a \le 7wt$. The monomial equations give $g = 0, h < 0, 1 \le j \le 7$; Appendix C.5 records all 413 rows of the finite range. By the tests in Appendix C.5 that use only $a \le 7wt$, Lemma 3.15 except for D_7 with possible points whose stabilizer is primitive, and the induction hypothesis for irreducible factors satisfying the condition on full interior stabilizers in Lemma 2.12, every irreducible finite factor is a parabolic subgroup of one of the groups

$$A_8,\ B_8,\ D_8,\ G(4,2,8),\ G(4,4,8),\ E_8.$$

Among these parabolic subgroups, all reflections have order two, and products of total rank at most 3 have orders dividing 384. Thus $T = 18L = 2K_W + B$ is integral and $384c_3^{\mathrm{orb}} = -32256L^3$ extends across the cusps and the exceptional locus $\mathrm{Exc}(\pi)$ of the small morphism π. On a closed locus $Z_J = \mathbb{P}^k$, $k \ge 3$, with $T = bH$, this says

$$\frac{448b^3}{81} \in \mathbb{Z}, \qquad 9 \mid b, \qquad L \cdot \ell \ge \frac{1}{2}.$$

For a maximal short pair J, the corresponding closed locus is the divisor $E = Z_J \simeq \mathbb{P}^7$. Relative nefness of K_W makes $\pi|_E$ finite, and normality of its image, the GIT quotient of the original merged weights, makes this restriction an isomorphism. The divisor E cannot be a branch component, by Proposition B.7; its normal line is $\mathcal{O}(-1)$, so Lemma 3.10 gives $L|_{\mathbb{P}^7} \ge H$. The lines in the remaining closed loci Z_J have degree against $-K_W$ at most five, and axes have $t \ge 1/3$. The generators of $\overline{\mathrm{NE}}(W)$ in Proposition 2.8 therefore give

$$K_W + 10L \text{ nef.} \tag{6.5}$$

This argument precedes the exclusion of E_8.

At an E_8 point take its E_7 parabolic disk, and let $R > 0$ count the E_8 points on this disk. Its cusps of width one and the characters of the full groups give

$$2g - 2 + s + R/2 + \sigma = 2t, \quad 2t \in \mathbb{Z}_{>0}, \quad a = 18s + \frac{57}{2}R + \sigma - 72t.$$

The target associated with the invariant of greatest degree has $h = -18t + 6R + 4s$, but (6.5) gives

$$\frac{57}{2}R + 18s + \sigma \le 82t. \tag{6.6}$$

If $h \ge 0$, these inequalities give $5t/2 - 2 \le R \le 2t/3$, hence $t \le 12/11$. If $R + s \ge 4t$, they give $4t - 4 \le R \le 20t/21$, hence $t \le 21/16$. In either case $2t \in \mathbb{Z}$ leaves $t = 1/2, 1$, forcing $R < 1$. Therefore $h < 0$ and $R + s < 4t$. The flags of ideals in the local algebras and the derivatives of the theta functions associated with the root lattice give the following degrees of the coefficients after division by the zeros forced by the differential:

d_i	deg	d_i	deg
2	$-16t + 3R + 3s$	10	$-8t + 2R + 2s$
6	$-12t + 3R + 2s$	12	$-6t + 2R + s$
8	$-10t + 3R + 2s$	14	$-4t + R + s$

All are negative; use $3R + 2s \le 82t/9$ and $2R + s \le 328t/57$ for the middle entries. The coefficient of the invariant of degree eighteen is constant, so the Hom space has dimension one. Hence $\ell = 2$ by Proposition 2.2, and by Proposition 2.3 the kernel $F = f^*T_{Z_I^s}$ of $\psi : f^*T_W \twoheadrightarrow M$ has no quotient of degree less than $e \ge -h$. But the projection of F to the invariant of degree fourteen has degree $h_{14} = -14t + 5R + 3s$, and

$$h_{14} + h = 11R + 7s - 32t \le \frac{7}{18}\left(\frac{57}{2}R + 18s\right) - 32t \le -\frac{t}{9} < 0.$$

This contradiction excludes E_8 in every genus and CM field.

Now the tests in Appendix C.5 that use (6.5) exclude all monomial factors of ranks 7 and 8 and their full cusps. For D_7, no stabilizer E_7 containing D_7 is possible, since $\det D_7 / \det E_7 = 2$ is not the square of an integer; its possible points with stabilizer E_8 have just been excluded. Its last nonuniform row has $a = 6, t = 1/2$, contrary to $a \le 10t$, and Lemma 3.15 now applies. An E_7 disk has no point with a larger irreducible stabilizer left. Without ends its projection of the inverse differential to the coordinate of greatest invariant degree gives $a \ge 126t$. With ends,

$$a = 36(1 - g) - 36t - 17\sigma, \quad h = -18t + 4s, \quad 2t \in \mathbb{Z}_{>0}.$$

After two successive divisions by powers of q_* at the cusps, at most two directions remain for $g \ge 1$, or four for $g = 0, t \ge 1$, forcing $a > 0$ when $a = 36(1 - g) - 36t - 17\sigma$ gives $a \le 0$. At $t = 1/2$, $s = 1, 2, 3$ give $(a, h) = (-16, -5), (1, -1), (18, 3)$. Only $s = 2$ survives positivity and (6.5). Then (6.4) forces $\ell = 5, e = 1, \delta = 0$. Four independent maps to $\mathcal{O}(-1)$, after two divisions, have coefficients of degrees $(-4, -2, -1, 0, 1, 2, 4)$. After the constant coefficient of the invariant of degree ten vanishes, three independent covectors remain; the next projection defined by the theta functions associated with the root lattice forces a third division by powers of q_* and leaves only degrees fourteen and eighteen, a contradiction. All irreducible factors of smaller rank satisfying the condition on full interior stabilizers in Lemma 2.12 are excluded by the induction hypothesis, including type A. Thus an A_7 disk exists. Proposition 3.11 gives $a \ge 23t + \sigma > 10t$, contradicting (6.5). □

6.3 Twelve labels

Proposition 6.3. The nonexistence theorem holds for twelve labels.

Proof. Here $d = 9, r = 8$, and descent of $\mathcal{O}(1, \dots, 1)$ gives $a \in 2\mathbb{Z}$. An E_8 disk has polynomial degrees $(2, 8, 12, 14, 18, 20, 24, 30)$, degrees $(1, 2, 2, 3, 3, 4, 4, 5, 6)$ of the generators of the ring of invariant sections, and

$$2g - 2 + s + \sigma = 2t, \quad a = 29s + 2 - 2g - 128t, \quad h = -30t + 6s, \quad 2t \in \mathbb{Z}_{>0}.$$

These equations and parity give $h+2t<0$. Thus the projection of the inverse differential to the coordinate of greatest invariant degree kills both df and the quadratic invariant. We use the normalized Eisenstein series E_2, E_4, E_6, with E_2 quasimodular, and the modular discriminant $\Delta = (E_4^3 - E_6^2)/1728$. The unique theta function of degree one of E_8 satisfies

$$\Theta = E_4 + c(E_2E_4 - E_6)p_2 + O(z^4), \qquad c \neq 0.$$

If $E_4 = 0$, then $E_4^3 - E_6^2 = 1728\Delta \neq 0$ makes its quadratic coefficient nonzero. Consequently the equation in the radial direction or, when $E_4 = 0$, the equation from the quadratic coefficient of Θ forces a second zero at the cusps in every coefficient for the directions of A_P. After division by the zeros forced by the differential, the degrees of the coefficients for the directions of A_P are $(d_i - 30)t + 4s$. For $g \geq 1$ at most two directions remain; for $g = 0, t \geq 1$ at most five remain. Thus $\ell \leq 3$ or 6, contradicting (6.4) and $a \leq 60(1-g) - 70t$. At $t = 1/2$, the possible $s = 0, 2$ give $a = -62, -4$ and again too few directions. This excludes E_8 points, disks, and cusps before the monomial groups are treated.

Lemma 3.15 excludes disks without ends with monomial full generic stabilizer. For nonuniform disks with monomial full generic stabilizer, (6.1)–(6.3), $a \leq 8wt$, and counting coefficients as in Appendix C.2 imply $g = 0, h < 0, 1 \leq j \leq 7$. Appendix C.4 excludes all 117 rows. The induction hypothesis excludes irreducible factors of smaller rank satisfying the condition on full interior stabilizers in Lemma 2.12, so every remaining finite stabilizer is a product of type A groups. For $T = 20L$ restricted as bH to a closed locus $Z_J = \mathbb{P}^k$, $k \geq 3$, Proposition 3.12 gives

$$\frac{27b^2}{20}, \ \frac{9b^3}{25} \in \mathbb{Z}, \qquad 10 \mid b.$$

Thus its lines have $t \geq 1/2$; their degree against $-K_W$ is at most eight. Axes have $a = 2, t \geq 7/20$ by Lemma 3.9. The generators of $\overline{\mathrm{NE}}(W)$ in Proposition 2.8 therefore give $K_W + 16L$ nef, including the rays spanned by the lines β_J with $2|J| = n$, on which K_W has degree zero. For the resulting A_8 disk, Proposition 3.11 gives $a \geq 61t/2 + \sigma > 16t$, the required contradiction. □

6.4 Thirteen or more labels

Proposition 6.4. The nonexistence theorem holds for every $n \geq 13$.

Proof. By the induction hypothesis, no irreducible factor of rank at most $d-2$ satisfies the condition on full interior stabilizers in Lemma 2.12. A maximal factor outside type A must therefore lie in a monomial group of rank $d-1$ or d, and Proposition 6.1 excludes it. The classification of parabolic subgroups now gives an A_{d-1} disk, with larger groups A_d and products, and cusps of width one whose section degrees for the full group are 1^d. All finite stabilizers are products of type A. Put

$$b_N = \prod_{q|N,\ q \text{ prime}} q^{\lceil 2v_q(N)/3 \rceil}, \qquad C_N = \frac{2N^2}{b_N}.$$

For the integral $T = 2NL$, Proposition 3.12 gives $N^2 \mid b^3$ on every closed locus Z of coincident labels of dimension at least three, with $T|_Z = bH$. Thus $b_N \mid b$, and every K_W-negative line in such a locus has

$$\frac{-K_W \cdot \ell}{L \cdot \ell} \leq \frac{2N^2}{b_N} = C_N.$$

This includes the case $Z = W = \mathbb{P}^d$ of a singleton J, if it exists. Every such locus on whose lines K_W is negative has dimension at least five, so these Chern restrictions are legitimate. For coordinate curves, Lemma 3.9 gives

$$L \cdot \gamma \geq \frac{N-3}{2N}, \qquad \frac{-K_W \cdot \gamma}{L \cdot \gamma} \leq \frac{4N}{N-3} < C_N,$$

where the last inequality follows from $b_N \le N$. The generators of $\overline{\mathrm{NE}}(W)$ in Proposition 2.8 therefore give $K_W + C_N L$ nef. There is no preliminary exclusion of $W = \mathbb{P}^d$ in this argument.

Proposition 3.11, applied to the A_{d-1} disk, gives

$$a \ge \frac{N^2 - 4N + 1}{2} t + \sigma.$$

For $N = 9, 10, 11, 12, 13, 14$, one has $b_N = N$, and $C_N = 2N < (N^2 - 4N + 1)/2$. For $N \ge 15$, the inequalities $b_N \ge N^{2/3}$ and $N^{1/3} < N/6$ give

$$C_N \le 2N^{4/3} < \frac{N^2}{3} < \frac{N^2 - 4N + 1}{2}.$$

The last strict inequality is $N^2 - 12N + 3 > 0$, valid for $N \ge 15$. This contradicts $a \le C_N t$ for every $n \ge 13$. □

Proof of Theorem 1.1. The case of nine labels is proved by Theorems 4.6 and 4.14, and the case of ten labels by Theorem 5.5. Assume the result for every smaller number of labels, starting at nine. Proposition 2.14 excludes each irreducible factor of rank at most $d - 2$ satisfying the condition on full interior stabilizers in Lemma 2.12, because the Baily–Borel compactification of the quotient of its fixed ball by the full normalizer is the original quotient with $s = 1$ or 2 labels merged. Propositions 6.2, 6.3, and 6.4 exclude the remaining cases $n = 11$, 12, and $n \ge 13$, respectively. The smooth GIT quotient $W = X_\nu$ of Lemma 2.1, obtained by perturbing μ to weights ν in a nearby chamber, and the arguments avoiding the finitely many cusps include all stability walls. □

A Derivatives of theta functions and differentials of quotient maps

This appendix proves Propositions 3.13 and 3.14. The degrees of polynomial invariants, the degrees of generators of the ring of invariant sections, and the weights of Jacobi forms remain distinct throughout.

A.1 Projections defined by theta functions associated with the root lattice

Proof of Proposition 3.13. Write the basic Jacobi forms of [31, Theorem 3.6] as $\phi_0, \dots, \phi_r$, of weights $0, -d_1, \dots, -d_r$ and indices $1, m_1, \dots, m_r$. The relative and fiber statements of [31, Theorem 3.6(ii)–(iv)], at level three, identify the section ring on each elliptic fiber with the specialization of the polynomial algebra of Weyl invariant Jacobi forms. For linear coordinates z_a, set $\mathcal{J}_{0i} = m_i \phi_i$, $m_0 = 1$, and $\mathcal{J}_{ai} = \partial_{z_a} \phi_i$. By [30, Propositions 2.2–2.3, Theorem 3.1, Section 4], the Jacobian satisfies

$$\det \mathcal{J} = c \prod_{\alpha \in R^+} \frac{\vartheta(\tau, (\alpha, z))}{\eta(\tau)^3}, \qquad c \ne 0.$$

Induction on Taylor degree in the Jacobi transformation law kills terms of ϕ_i below degree d_i. The inhomogeneous term only uses lower terms already zero, leaving a holomorphic modular form of negative weight. If $\phi_0(\tau, 0) = 0$, the Jacobian would have order at least $|R^+| + 2$, contradicting $\det \mathcal{J} = c \prod_{\alpha \in R^+} \frac{\vartheta(\tau,(\alpha,z))}{\eta(\tau)^3}$. Put $r_i = \phi_i / \phi_0^{m_i}$, and write $r_i = \sum_k a_{ik}(\tau) p_k$ modulo the square of the invariant maximal ideal. The coefficient a_{ik} has modular weight $d_k - d_i$. Coefficients of negative weight vanish, coefficients of weight zero are constant, and the term of lowest degree in z of $\det \mathcal{J}$ makes each diagonal block of equal degree invertible. For type D_r, the index one form $\prod_i \vartheta(\tau, z_i)/\eta(\tau)^{3r}$ is odd under the change of sign of one coordinate.

Averaging separates $P_0 = x_1 \cdots x_r$ from a symmetric invariant of the same degree. Thus the indicated projections are triangular with invertible diagonal blocks. Polynomial generation shows that derivatives obtained from sections of degree at most j are spanned by the dr_i with $m_i \le j$. This proves the projection statement while retaining the components along the differentials of basic invariants of higher degree.

For the translation assertion, put $\lambda_\Theta(x) = t_x^*\Theta \otimes \Theta^{-1}$. Primitive simple roots give $A_\tau^{W(R)} = \ker \lambda_\Theta$. At a common fixed point where the Weyl group acts trivially on the fiber of Θ, a lift of t_x to the theta group commutes with the equivariant structure for the Weyl group: its scalar commutator is the ratio of the characters at x and zero. It therefore transports the section spaces and their projections. The same comparison preserves eigenspaces of any further automorphism fixing both points and acting trivially on their fibers. The modular action of $-I$ gives parity $(-1)^{d_i}$. □

For the cusps of type D_v used below, $v = 5, 6$, the actual polarization and trivial character on the fiber of $\mathcal{V}_*$ can be recovered without choosing an isogenous substitute. Adjacent simple roots give $Q(D_v)_{W(D_v)} = 0$ and $\mathrm{Pic}^0(A)^{W(D_v)} = 0$. Invariant positive integral alternating forms are integral multiples of the root form tensored with the elliptic form. Connected reflection divisors through the origin, together with generation of the entire group of lifts by reflections, force trivial character on the fiber of $\mathcal{V}_*$. The minimal section degrees are $(1^4, 2^{v-3})$ [26]. Every proper Veronese subring of the ring of invariant sections gives a cone with a singular vertex: no nonidentity scalar is a pseudoreflection on the four coordinates of degree one. Consequently the smooth cusp itself uses the minimal bundle. At a relevant fixed point, let $\mathfrak{n}$ be the maximal ideal of its image, let $F_1 \subset \mathfrak{n}/\mathfrak{n}^2$ be the span of the differentials of s/ϕ_0 for the invariant sections s of degree one, and let $F_1^{\mathrm{sym}} \subset F_1$ be the $+1$ eigenspace of the involution $\iota(x_1, \dots, x_v) = (-x_1, x_2, \dots, x_v)$, which normalizes $W(D_v)$ and sends $P_0 = x_1 \cdots x_v$ to $-P_0$. With $f_j = e_j(x_1^2, \dots, x_v^2)$, Proposition 3.13 gives

$$F_1 = \mathbb{C}\, dP_0 \oplus F_1^{\mathrm{sym}}, \qquad F_1^{\mathrm{sym}} \longrightarrow \langle df_1, df_2 \rangle$$

isomorphically, uniformly in the elliptic parameter and at all relevant translated fixed points.

A.2 Why the algebra comparison preserves the flags

Let G be a finite complex reflection group with well-generated irreducible factors and reflections of order two, and let $\pi : V \to V/G$. The product of the Saito structure makes $\operatorname{coker} d\pi$ a finite birational unital algebra on the reduced reflection arrangement. Here is the localization argument needed to use that product. Multiplication by the Euler vector field has matrix $U = x_r I + U_0(x_1, \dots, x_{r-1})$ and determinant the discriminant [18, Theorem 5.6 and its proof]. Its characteristic polynomial is the discriminant with x_r translated by the spectral variable. Generic roots on every discriminant component are distinct [5, Theorem 5.3]. There the product is diagonal in canonical coordinates and the quotient is a fold. The square differential matrix has nonzero determinant, so its image is isomorphic to its locally free source. Multiplication preserves this image at every point of codimension one by the fold description. The coefficients in a basis of the image therefore extend by Hartogs, as in [12, Proposition 2.3]. The square presentation of $\operatorname{coker} d\pi$ has reduced determinant, so $\operatorname{coker} d\pi$ is Cohen–Macaulay and torsion-free of rank 1 on each arrangement component.

For a finite birational unital S-algebra M, one has $\mathrm{End}_S(M) = M$. An endomorphism is multiplication in the total quotient ring, and its value at one belongs to M. An isomorphism of the underlying modules thus conjugates their endomorphism algebras; the original map differs from the resulting algebra isomorphism by a unit. At a point with stabilizer P, the quotient germ is the product of its fixed space and the normal P-quotient. The fixed factor contributes zero to the cokernel, and parabolic subgroups remain well-generated [5, Lemma 2.7]. This proves localization, up to a unit, for the normal quotient.

For the inclusions $D_5 \subset E_6$, $E_6 \subset E_7$, and $E_7 \subset E_8$, let $F(x,t)$ be the corresponding semiuniversal deformation with base B, discriminant D, and projection $\rho : \{F = d_x F = 0\} \to B$. Kodaira–Spencer identifies $\mathcal{T}_B / \operatorname{Der}_B(-\log D)$ with $\rho_* \mathcal{O}_{\{F=d_xF=0\}}$ [12, Introduction, (0.1)]. Multiplication by the Euler vector field has the same image [18, Lemma 4.6]. Apply the preceding endomorphism argument before taking fibers to compare these two products. Apply it also on the reflection arrangement to compare the product of [12] with the product of the Saito structure. Only then use the algebra fiber comparison of [12, Proposition 4.21, Theorem 4.23]. Thus the cokernel of the differential of the quotient map on the fiber is the local Tjurina algebra up to a unit. The sheaves on the deformation base and on the reflection space have not been identified as sheaves on the same space.

The complete flags consist of the following ideals, each defined by the algebra structure alone, with $\mathfrak{m}$ the maximal ideal:

$$\begin{array}{c|l} D_5 & \mathfrak{m}^3 \subset \mathfrak{m}^2 \subset \operatorname{Ann}(\mathfrak{m}^2) \subset \mathfrak{m} \subset A \\ E_6 & \mathfrak{m}^3 \subset \mathfrak{m}J \subset \mathfrak{m}^2 \subset J + \mathfrak{m}^2 \subset \mathfrak{m} \subset A \\ E_7 & \mathfrak{m}^4 \subset \mathfrak{m}^3 \subset \mathfrak{m}I \subset \mathfrak{m}^2 \subset I \subset \mathfrak{m} \subset A \\ G_{33} & \mathfrak{m}S \subset \mathfrak{m}^2 \subset S \subset \mathfrak{m} \subset A. \end{array}$$

The first three algebras and their ordered bases are

$$\begin{array}{c|c|c} D_5 & \mathbb{C}[x,y]/(xy, x^2+y^3) & (y^3, y^2, x, y, 1) \\ E_6 & \mathbb{C}[x,y]/(x^2, y^3) & (xy^2, xy, y^2, x, y, 1) \\ E_7 & \mathbb{C}[x,y]/(x^2+y^3, xy^2) & (y^4, y^3, xy, y^2, x, y, 1). \end{array}$$

For E_6, J is the closure of square-zero elements in $\mathfrak{m} \setminus \mathfrak{m}^2$, here the ideal (x); for E_7, $I = \operatorname{Ann}(\mathfrak{m}^3)$. Direct multiplication shows each flag consists of the spans of the initial segments of its ordered basis. It is therefore preserved both by algebra isomorphisms and by unit multiplication.

Brieskorn identifies the weighted deformation base with the Weyl quotient [8, p. 282]. For $E_7 \subset E_8$ and $E_6 \subset E_7$, after splitting off a square use respectively

$$\begin{array}{c|c} F_a & (h_i) \\ \hline y^3 + ayz^3 + z^5 & (z^4, z^3, yz, z^2, y, z, 1) \\ y^3 + az^4 + yz^3 & (yz^2, yz, z^2, y, z, 1), \end{array} \qquad a \neq 0.$$

The same ideals are the spans of the initial segments of these bases; in the second row $J = (y - z^2/(8a))$. The monomials multiplying the normal deformation parameters differ only in their first entries, yz^2 and z^3. Their classes have diagonal coefficients $-5/(3a)$ and $-3/(4a)$, respectively, and all other diagonal coefficients one.

For $D_5 \subset E_6$, use

$$\begin{aligned} F_T &= y^3 + z^4 + T_2 yz^2 + T_5 yz + T_6 z^2 + T_8 y + T_9 z + T_{12}, \\ T(u) &= (-3u^2, 3u^5, -3u^6/2, -3u^8/4, u^9, -3u^{12}/16). \end{aligned} \tag{A.1}$$

At its D_5 point put $Z = z - u^3/2$, $w = y - uZ$. Then $F_{T(u)} = w^3 + 3uw^2 Z + Z^4$, with basis $(Z^3, Z^2, w, Z, 1)$ of its Tjurina algebra and relations

$$Z^4 = 0, \qquad wZ = 2Z^3/(3u^2), \qquad w^2 = -2uwZ.$$

With T_2 held constant, the classes of $(yz, z^2, y, z, 1)$ are lower triangular in this basis, with diagonal $(2/(3u^2), 1, 1, 1, 1)$. At $u = 1$, comparison with this flag of ideals is upper triangular with nonzero diagonal. The resulting matrix of the normal differential is $K_0^{-1} U$, with K_0 lower triangular invertible and U upper triangular invertible, so all its leading principal minors are nonzero.

A.3 The differential for $G_{33} \subset G_{34}$

Use flat basic invariants x_i of G_{34}, of degrees $(6, 12, 18, 24, 30, 42)$, and put $e_i = \partial_{x_i}$. The invariant of degree six is nonzero on the axis [27, Theorem 3, Case ST34 (3)], so put $\lambda = x_1 = u^6$. The tangent sheaf with the product of the Saito structure, pulled back over $R = \mathbb{C}[\lambda]$, has basis $(e_1, \dots, e_6)$ with weights $(6, 5, 4, 3, 2, 0)$ after a common shift. The quadratic potentials in [17, Section 9], identified as G_{34} in [18, Remark 5.11], give the fiber of this algebra over $\lambda = 0$

$$A_{34} = \mathbb{C}[a, b]/(a^4, a^2b, b^2 - a^3), \quad a = e_5,\ b = e_4, \quad (a^3, ab, a^2) = (2e_1, e_2, 2e_3).$$

For G_{33} the algebra is

$$A_{33} = \mathbb{C}[A, B]/(A^3, A^2B, B^2), \quad (\partial_{p_4}, \partial_{p_6}, \partial_{p_{10}}, \partial_{p_{12}}, \partial_{p_{18}}) = (AB, A^2, B, A, 1).$$

Saturate the axis ideal in the pulled back algebra and let Q be the free quotient of rank 5. Subtract the coordinates of the support from a, b; they belong to R because each is one fifth of the trace of multiplication by the corresponding element a or b on Q. The augmentation ideal satisfies $\mathfrak{m}^3 = 0$ generically and hence everywhere by torsion-freeness. The kernel of $A_{34} \to Q/\lambda Q$ contains nonzero a^3 and has dimension one by flatness, so $Q/\lambda Q = A_{33}$. Graded Nakayama gives the basis $(1, a, b, a^2, ab)$. Weights and $\mathfrak{m}^3 = 0$ imply

$$b^2 = c\lambda^2 a^2 + d\lambda ab, \qquad d^2 + 4c = 0.$$

The second equation follows from the generic quadratic relation of rank 1. Replace b by $b - d\lambda a/2$; then $b^2 = 0$. The square-zero ideal is $S = \langle b, a^2, ab\rangle$. The inverse images of $\mathfrak{m}S, \mathfrak{m}^2, S, \mathfrak{m}, Q$, together with the axis ideal, are free direct summands, with special fibers the spans of the initial segments of $(e_1, \dots, e_6)$. Their homogeneous lifts can mix an e_i only with basis elements of lower weight multiplied by nonnegative powers of λ. Consequently each initial block remains invertible on the generic fiber. This argument with weights proves that all leading principal minors including the axis are nonzero; removing the axis then proves that all five leading principal minors in the normal directions are nonzero. The center contains μ_6, and the nonzero invariant of degree six bounds the order of the scalar action on the axis by six, so that order is exactly six. The coordinates on the coarse quotient in the normal directions are $u^2p_4, p_6, u^2p_{10}, p_{12}, p_{18}$, of effective weights $(1, 1, 2, 2, 3)$, against target weights $(2, 3, 4, 5, 7)$.

Completion of Proposition 3.14. For each inclusion, let D_i be the increasing ambient degrees after deleting the lowest. The nonzero leading minors just proved and homogeneity give

$$M(v) = \operatorname{diag}(u^{D_i}) B \operatorname{diag}(u^{\epsilon_i - k_i}),$$

where B is constant with every leading principal minor nonzero. Elimination of the tangent column uses ratios of derivatives of restricted invariants, holomorphic in v. Elimination below the diagonal uses $u^{D_i - D_j}$, $i > j$, also holomorphic in v, and changes only target coordinates. The diagonal orders are $(D_i - k_i + \epsilon_i)/e$, giving the four tuples of orders in Proposition 3.14. The last row of the inverse differential uses only the last transformed target coordinate. On its kernel that coordinate vanishes, so the preceding row has the asserted preceding pole. □

A.4 The image of the ambient local ring along the axis for $D_5 \subset E_6$ and an obstruction to smooth threefolds

The image of the ambient local ring along (A.1) is exactly $\mathcal{A} = \mathbb{C}\{u^2, u^5\}$; its only missing nonnegative exponents are one and three. In the local Tjurina algebra, the two lowest normal directions span $\mathfrak{m}^2 = \langle Z^2, Z^3\rangle$. We reduce the monomials multiplying the ambient deformation

parameters modulo this ideal in the basis $(w, Z, 1)$ and obtain the subspace of dimension three containing the tangent axis and those two directions:

$$dT_8 = -u^6 dT_2/4 - u^3 dT_5/2, \qquad dT_9 = -u^3 dT_6, \qquad dT_{12} = u^6 dT_6/4. \tag{A.2}$$

There is no smooth threefold through zero containing the axis and with these tangent spaces away from zero. For such a threefold, the limiting projection onto (T_2, T_5, T_6) would be invertible, making the threefold a holomorphic graph. Its derivative coefficients along the curve would then lie in $\mathcal{A}$, but the coefficient $-u^3/2$ in (A.2) does not. This obstruction applies to derivatives of actual ambient holomorphic functions; arbitrary covectors on the normalization need not have coefficients in $\mathcal{A}$.

A.5 The G_{33} theta characters and derivatives

Lemma A.1. For a cusp with $F_P = G_{33}$ and the minimal polarization, the generators of the ring of invariant sections have degrees $(1, 1, 3, 3, 4, 6)$. For a scalar ω of order three, their characters are $(1, \omega, 1, \omega, 1, 1)$. At such a cusp, the four common fixed points in the abelian variety equal the kernel of the polarization. At each relevant point with trivial character of the full stabilizer on the fiber of $\mathcal{V}_*$, a lift of the corresponding translation to the theta group transports these characters and the projections of the first derivatives of invariant sections. For the basic invariants of polynomial degrees $(4, 6, 10, 12, 18)$ on the normal representation, the successive divisions of coefficients use powers of q_* with exponents $(1, 3, 3, 4, 6)$.

Proof. Use the polarized abelian variety of [26, Sections 4.2 and 7.6]:

$$E : y^2 = x^3 + 1, \quad b = (0, 1), \quad A_0 = \ker(E^6 \xrightarrow{\sum} E), \quad A = A_0/\langle (b, \ldots, b) \rangle.$$

On $A_0/S_6 = |6O|$, use as coordinates the coefficients $(c_0, c_x, c_y, c_{x^2}, c_{xy}, c_{x^3})$ of the monomials $1, x, y, x^2, xy, x^3$. The sections of degree one are $\theta_0 = c_0 + 2c_{x^3}$, $\theta_1 = c_{x^2}$, with $\theta_0(0) \neq 0$. Translation by b is $(x, y) \mapsto (-2x/(y+1), (y-3)/(y+1))$ and multiplies both sections by four, so the action on the fibers, normalized by this scalar, fixes them. The scalar acts on these coordinates with weights $(0, 2, 0, 1, 2, 0)$ modulo three.

Here are explicit nonzero minors for the two new characters of degree three. Work over $\mathbb{F}_{1009}$, with $\omega = 374$, and use the following five points for each tuple; the sixth point is the negative of their sum in E.

a	P_1	P_2	P_3	P_4	P_5
1	$(43, 657)$	$(773, 520)$	$(318, 858)$	$(690, 511)$	$(417, 539)$
2	$(227, 130)$	$(228, 395)$	$(142, 879)$	$(864, 669)$	$(981, 805)$
3	$(67, 456)$	$(655, 43)$	$(62, 397)$	$(350, 949)$	$(224, 424)$
4	$(394, 779)$	$(165, 206)$	$(671, 577)$	$(653, 869)$	$(871, 634)$
5	$(695, 1007)$	$(243, 824)$	$(734, 374)$	$(792, 43)$	$(370, 183)$
6	$(461, 528)$	$(350, 60)$	$(773, 520)$	$(543, 63)$	$(492, 183)$
7	$(572, 539)$	$(231, 649)$	$(797, 632)$	$(313, 399)$	$(829, 358)$
8	$(297, 479)$	$(228, 395)$	$(166, 422)$	$(67, 456)$	$(133, 267)$
9	$(133, 742)$	$(864, 340)$	$(717, 489)$	$(224, 585)$	$(734, 374)$.

Every point lies in $E[316]$, where division by three means multiplication by 211. Thus the reflection ρ in $v = (1, 1, \omega, \omega, \omega^2, \omega^2)$ acts on each tuple by

$$\rho(P)_i = P_i - [\omega^{e_i}][211] \sum_k [\omega^{-e_k}] P_k, \qquad (e_1, \ldots, e_6) = (0, 0, 1, 1, 2, 2).$$

Here $[\omega](x, y) = (\omega x, y)$, and put $\tau(P)_i = P_i + b$. For any of these tuples or their transforms, let $c(P)$ be the kernel vector of the evaluation matrix on $(1, x, y, x^2, xy, x^3)$, normalized by

$c_0 + 2c_{x^3} = 1$. The kernels have dimension one and the normalizing denominators are nonzero. For $j = 0, 1, 2$, order lexicographically the exponent vectors

$$\mathcal{M}_j = \{u \in \mathbb{Z}^6_{\geq 0} : \textstyle\sum_i u_i = 3,\ (0,2,0,1,2,0) \cdot u \equiv j \pmod 3\}.$$

Define the matrix M_j by the rows

$$(M_j)_{2a-1,u} = c(P^{(a)})^u - c(\rho P^{(a)})^u, \qquad (M_j)_{2a,u} = c(P^{(a)})^u - c(\tau P^{(a)})^u.$$

The following minors use the first indicated rows and all columns except the listed ones; the last column gives their determinants modulo 1009.

j	$\lvert\mathcal{M}_j\rvert$	number of rows	omitted columns	det
0	21	18	$4, 19, 21$	775
1	17	15	$13, 17$	457
2	18	17	13	523.

The chord and tangent law on $y^2 = x^3 + 1$ and ordinary determinant expansion give these residues. The points and their transforms have torsion of order dividing 948, prime to 1009. Thus the minors lift to characteristic zero, since every denominator is a unit. Rains's section degrees give total dimension six, so the upper character bounds $(3, 2, 1)$ are equalities. Cubics in θ_0, θ_1 account for $(2, 1, 1)$, leaving new characters $1, \omega$. The polynomial degrees $(4, 6, 10, 12, 18)$ of the basic invariants then give the asserted characters in degrees four and six as well.

We compute the projections of the differentials, keeping the polynomial degree distinct from the degree of a section. Write $(c_0, \ldots, c_5) = (c_0, c_x, c_y, c_{x^2}, c_{xy}, c_{x^3})$. Let $\mathfrak{n}$ be the maximal ideal at the image of the origin in A/G_{33}. For $k \geq 1$, let $F_k \subset \mathfrak{n}/\mathfrak{n}^2$ be the span of the differentials at the origin of s/θ_0^m, where s is an invariant section of degree $m \leq k$. Rains's polynomial ring of invariant sections gives

$$(\dim F_1, \dim F_3, \dim F_4, \dim F_6) = (1, 3, 4, 5).$$

We verify that projection onto the classes of the differentials of the first $1, 3, 4, 5$ basic invariants, in increasing polynomial degree, has the corresponding rank.

We first specify finite inputs over $\mathbb{F}_{1009}$. List the 315 nonzero points of $[3]E(\mathbb{F}_{1009})$ as $P_0, \ldots, P_{314}$, in lexicographic order of their coordinate representatives in $\{0, \ldots, 1008\}^2$. Put

$$(m_1, \ldots, m_5) = (1, 7, 31, 73, 137), \qquad (b_1, \ldots, b_5) = (0, 17, 51, 83, 121).$$

For $0 \leq a < 315$, take the first five points of a tuple to be $P_{am_i + b_i \bmod 315}$, and take the sixth point to be the negative of their sum. Discard a tuple if the tuple or either of its transforms under ρ, τ has a repeated point, contains the origin, or has no normalized coefficient vector c. There are 285 remaining tuples, ordered by a. For $m = 1, 3, 4$ and $j = 0, 1, 2$, let $M_{m,j}$ be the matrix whose columns are indexed by the lexicographically ordered set

$$\mathcal{M}_{m,j} = \{u \in \mathbb{Z}^6_{\geq 0} : \textstyle\sum_i u_i = m, (0,2,0,1,2,0) \cdot u \equiv j \pmod 3\}.$$

The kth remaining tuple P, $1 \leq k \leq 285$, gives the rows $2k - 1$ and $2k$, with entries $c(P)^u - c(\rho P)^u$ and $c(P)^u - c(\tau P)^u$ in the column u, where $c^u = \prod_i c_i^{u_i}$. In each matrix, scan the columns in increasing order, choose the first available nonzero row after elimination at each pivot, and retain the original row indices in pivot order. The selected rows and pivot columns give the following nonzero determinants:

m	$\lvert\mathcal{M}_{m,j}\rvert$	$\operatorname{rank} M_{m,j}$	$\dim\ker M_{m,j}$	det
1	$(3, 1, 2)$	$(2, 0, 2)$	$(1, 1, 0)$	$(544, 1, 160)$
3	$(21, 17, 18)$	$(18, 15, 17)$	$(3, 2, 1)$	$(701, 552, 919)$
4	$(45, 39, 42)$	$(41, 35, 40)$	$(4, 4, 2)$	$(580, 438, 964)$.

The three entries in each tuple correspond to $j = 0, 1, 2$, and the empty determinant equals one. The total kernel dimensions are $2, 6, 10$, equal to the dimensions of the invariant sections given by Rains. Every invariant section lies in the corresponding kernel, so equality of these total dimensions identifies all the kernels with the spaces of invariant sections. Use the nullspace basis obtained by setting each free variable in turn to one and the others to zero in the reduced row echelon form. Order these basis vectors by their free columns and concatenate the three characters in the order $0, 1, 2$. With indices starting at zero, select the sections

$$(m, i) = (1, 1), (3, 1), (3, 3), (4, 1).$$

For the Taylor expansion at the origin of E, use the formal coordinate z satisfying $dz = dx/(2y)$. Write

$$x(z) = z^{-2} \sum_{r \geq 0} a_r z^{6r}, \qquad y(z) = x'(z)/2, \qquad a_0 = 1, \quad a_1 = -1/7.$$

For $r \geq 2$, the coefficients satisfy

$$((6r-2)(6r-3) - 12)a_r = 6 \sum_{v=1}^{r-1} a_v a_{r-v}.$$

Put $B(z) = (1, x(z), y(z), x(z)^2, x(z)y(z), x(z)^3)$. For each tuple $(\lambda_1, \ldots, \lambda_6) \in \mathbb{F}_{1009}^6$ listed below, let u be a formal parameter. Solve $\sum_{j=0}^{5} c_j B_j(\lambda_i u) = 0$, $1 \leq i \leq 6$, with $c_0 = 1$, and divide the resulting vector by $c_0 + 2c_5$. The three choices of $(\lambda_1, \ldots, \lambda_6)$ needed below are

$$(94, 181, 986, 534, 540, 692), \quad (980, 438, 34, 975, 252, 348),$$
$$(902, 849, 41, 952, 851, 441).$$

Their coordinate sums vanish modulo 1009. The full monomial supports of the selected numerators have minimum weights $4, 6, 10, 12$ for the weights $(0, 2, 3, 4, 5, 6)$ on $(c_0, \ldots, c_5)$. Let h_4, h_6, h_{10}, h_{12} be the corresponding first homogeneous terms. Evaluation at the first $1, 1, 2, 3$ directions, respectively, gives

d	rows of the evaluation matrix	det
4	h_4	895
6	h_6	722
10	$h_4h_6,\ h_{10}$	844
12	$h_4^3,\ h_6^2,\ h_{12}$	987.

(A.3)

Hence the classes of h_d are nonzero modulo products of invariants of smaller positive degrees. Complete h_4, h_6, h_{10}, h_{12} to homogeneous basic invariants by a generator of degree eighteen. Modulo products of invariants of positive degree, the four selected differentials project to

$$\begin{pmatrix} 1 & 0 & 144 & 0 \\ 0 & 1 & 0 & 217 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \qquad \text{in the directions } (4, 6, 10, 12). \tag{A.4}$$

Every leading principal minor is one.

We pass to characteristic zero through the completion of $\mathbb{Z}[\omega]$ at the prime defined by 1009 and $\omega = 374$. The curve has good reduction, the quotient isogeny has degree three prime to 1009, and the quotient remains an abelian scheme with its ample descended line bundle. Generic automorphisms and their inverses extend by the Néron mapping property of the abelian scheme. Normalize the isomorphisms of the descended line bundle on the identity fiber; their cocycle

equalities then extend, so the full linearization is retained. Since $1009 \nmid |G_{33}|$, the Reynolds operator commutes with reduction. It follows from the vanishing of higher cohomology for an ample line bundle on an abelian variety that the modules of invariant sections commute with reduction as well. The formal logarithm through polynomial degree twelve has integral coefficients at this prime. The polynomial invariant rings on both fibers have basic invariants of degrees $(4, 6, 10, 12, 18)$. Lift homogeneous generators by the Reynolds operator and apply Nakayama's lemma in each degree; the indecomposables in these degrees are therefore free modules of rank 1. Thus the nonzero minors in (A.3) and (A.4) are reductions of minors of the actual maps to $\mathfrak{n}/\mathfrak{n}^2$, and the projections over $\mathbb{C}$ have ranks $1, 3, 4$. The chart $\theta_0 \neq 0$ is an affine space of dimension five, with coordinates given by the other five generators divided by the corresponding powers of θ_0, so $F_6 = \mathfrak{n}/\mathfrak{n}^2$. A change of homogeneous basic invariants multiplies each class of degree d by a nonzero scalar, since the degrees of the five basic invariants are distinct. The projection ranks therefore hold for the basic invariants $p_4, p_6, p_{10}, p_{12}, p_{18}$ used above. The projections in question are isomorphisms; higher polynomial components of the differentials need not vanish. They give the required successive elimination of coefficients in the directions of degrees $6, 12, 18$, by division by powers of q_* with exponents $3, 4, 6$, and the analogous projections in the scalar character ω.

For completeness, the group of common fixed points of G_{33} on A is the diagonal image of $E[2]$. The lift $(\lambda/6, \dots, \lambda/6, -5\lambda/6)$ changes under the additional reflection by $(\omega\lambda/3)v$. This change vanishes modulo $\langle (b, \dots, b) \rangle$ exactly when $1 - \omega \mid \lambda$; the Weil pairing gives the same kernel of the polarization, of order four. At a point with trivial character on the fiber of $\mathcal{V}_*$, a lift of its translation to the theta group commutes with G_{33}, by the argument comparing the characters used above. Conjugate the scalar by that lift to transport its eigenspaces. The spaces of derivatives, with their components along the differentials of basic invariants of higher degree retained, therefore have projections of the same ranks at all relevant points. The derivatives in these directions first occur for generators of degrees $1, 3, 3, 4, 6$, which gives the stated powers of q_*. □

B Curves in projective space obtained from disks fixed by finite reflection groups

The comparisons in this appendix concern the class L of the original lattice, including its finite characters. They do not assume that K_W plus any multiple of L is nef on the quotient of ten points. Write $H = c_1(\mathcal{O}(1))$.

Lemma B.1. If $f : C \to \mathbb{P}^k$ is nonconstant, $u = \deg f^*\mathcal{O}(1) > 0$, and $f^*T_{\mathbb{P}^k} \to Q_h$ is nonzero, then $h = \deg Q_h \geq u$. The corresponding map $\kappa : C \to (\mathbb{P}^k)^*$ to the dual projective space, after common zeros are removed, has degree at most $h - u$. If κ is nonconstant and its derivative vanishes at R distinct points, then

$$R \leq 2(h - u) + 2g(C) - 2. \tag{B.1}$$

A constant κ is a hyperplane containing $f(C)$.

Proof. The Euler sequence represents the map by the sections of $Q_h \otimes f^*\mathcal{O}(-1)$ given by the $k + 1$ summands $\mathcal{O}(1)$ of its middle term. Their span, which has no common zero, defines κ and proves the degree statements. A general projection of its image to $\mathbb{P}^1$ preserves its degree and every prescribed zero of its derivative. Riemann–Hurwitz gives (B.1). □

Proposition B.2. If the Baily–Borel compactification of an arithmetic ball quotient of dimension six is $\mathbb{P}^6$ and $L = \lambda H$, then $\lambda \geq 1$. The group may be compact or nonuniform.

Proof. Assume $0 < \lambda < 1$. On every curve obtained from a disk fixed by a finite reflection group, $a = 7u$, $t = \lambda u$, and each nonzero row of the inverse differential has $h \ge u > t$. The classification of parabolic subgroups [20, Tables 10–12], together with Proposition 2.14, leaves parabolics of $A_6, B_6, D_6, G(3,3,6), E_6, G_{34}$ and the disks $G(m,p,5)$ with $m \ge 4$ or $(m,p) = (3,1)$. An irreducible factor of rank at most 4 satisfying the condition on full interior stabilizers in Lemma 2.12 contradicts the ample tangent bundle on a complete curve outside the boundary. For a compact disk with full generic stabilizer $G(m,p,5)$, the line bundle corresponding to P_0 and the Riemann–Hurwitz formula $2g - 2 + (1 - 1/m)R + \sigma = 2t$ for the disk give

$$(3m-5)j + (m-1)pu + 2mg + m\sigma \le 2m, \qquad j = mt \in \mathbb{Z}_{>0}.$$

For $m \ge 4$ this is impossible already with $j, u \ge 1$. The exceptional $(3,1)$ equality has $h = -1$. For a nonuniform disk, the line bundle wL gives $t \ge u/w$. The computation with P_0 for $m = 6$, with all polarizations and widths retained, gives

$$a \ge (23p/6 + 1)h_P + 53t + R + \sigma \ge (23p + 59)u/6 > 7u.$$

For the groups $G(3,1,5)$ and $G(4,1,5)$ the same computation gives

$$a \ge 23t + 14h/3 + R + \sigma \ge 17u/2, \qquad a \ge 33t + 19h/4 + R + \sigma \ge 13u.$$

Here respectively $6L$ and $4L$ are line bundles, and the relevant rows of the inverse differential, for P_0 or for the coordinate of greatest invariant degree, are nonzero. For the remaining groups over the Gaussian field, P_0 gives a shorter argument than a finite enumeration. After the group $G(4,1,5)$ is excluded, the $G(4,2,5)$ ends have width one and types $A = (1^3, 2^3)$, $B = (1^2, 2^4)$, $C = (1, 2^2, 4^3)$, $D = (1, 2, 4^4)$. Their sums B_c and the pole orders b_c in the direction of P_0 are $(9,1), (10,1), (17,2), (19,2)$. The torsor X has no point fixed by all of $G(4,2,5)$, so it is not an end of this disk. We justify the projection onto the direction of P_0 in the Gaussian model $Y^2 = X^4 - Z^4$. Write $\mathsf{X} = \prod_i X_i$, $\mathsf{Y} = \prod_i Y_i$, $\mathsf{Z} = \prod_i Z_i$, $z_i = Z_i/X_i$, $f_j = e_j(z_1^4, \dots, z_5^4)$, and $P = (\prod_i z_i)^{4/p}$, where $p = 2$ or 4. The symmetric theta ratios and the product ratio satisfy

$$\frac{\mathsf{Y}}{\mathsf{X}^2} = (1 - f_1 + f_2 - f_3 + f_4 - P^p)^{1/2}, \qquad \frac{\mathsf{Z}^{4/p}}{\mathsf{X}^{4/p}} = P.$$

Since $p > 1$, their linear parts separate dP from the direction of every elementary symmetric invariant. Thus the local lattice of the conormal bundle on the coarse curve, after removal of the radial direction, is $q^{b_c}\mathcal{O}\, dP \oplus \Lambda_{\mathrm{sym}}$. Changing to additive elliptic coordinates is triangular on the symmetric invariants and a nonzero scalar on P, so this is the original root representation. At an interior point with stabilizer $G(4,p,6)$, $P = u^{4-4/p}w$, $v = u^4$, and P_0 for $G(4,p,6)$ is $Z = vw$; the block of the elementary symmetric invariants in the normal directions is $vI + J_4$. The equation $g_1 = v + f_1$ and the generators of degree one that are nonzero at the ends make the curve immersed at the points with stabilizer $G(4,p,6)$ and at the ends. There is no first-order product mixing, and product inertia is trivial on the root space. The same rational projection onto the direction of dP therefore preserves every local lattice on the coarse curve. At a point with stabilizer $G(4,p,6)$, the line bundle corresponding to P_0 on the coarse quotient contributes $-(1-1/p)$, and the pullback of P_0 has one zero, so that together they contribute $1/p$. Consequently the nonzero quotient associated with P_0 has degree $h_P = -20t/p + R/p + \sum b_c$. For $p = 2$, put $\beta = \sum b_c$, so $h_P = -10t + R/2 + \beta$. The bounds $B_c \ge 8b_c + 1$ give

$$a \ge 8h_P + 28t + s + 5R/4 + \sigma > 7u,$$

because Lemma B.1 gives $h_P \ge u$. At the point over $Z = 0$ fixed by all of $G(4,2,5)$, the invariant nonzero section of degree one is X^2 for A, B, and X for C, D. The character on the fiber of $\mathcal{V}_*$ is trivial there, so Lemma 2.17 supplies the excluded full $G(4,2,5)$ disk. This

removes the entire A–D cusps, including the opposite side of C, D where the kernel is $G(4,4,5)$ and the width is two. Only $E = (1^2, 2, 4^3)$ and the torsor $X = (1^2, 4^4)$ remain for $G(4,4,5)$, both with width one. At every relevant point their invariant section X is nonzero of degree one, giving trivial character on the fiber of $\mathcal{V}_*$. Here $b_c = 1$, $B_c \geq 16$, and the same quotient associated with P_0 has $h_P = -5t + R/4 + s$ and

$$a \geq 16h_P + 33t + R + \sigma > 7u.$$

These local computations with the quotient associated with P_0 use neither descent of $2L$ nor nefness of K_W plus a multiple of L. They retain the original actions of the groups F_P, the scalar characters, and the central translations at the cusps. In particular the preliminary finite tests for the larger values of m are unnecessary.

All remaining reflections have order two, and the orders of stabilizers of rank 2 divide 24. Thus $14L = bH$ is integral, and Proposition 3.12 gives $24c_2^{\mathrm{orb}} = 504L^2 = (18b^2/7)H^2$ integral. It follows that $7 \mid b$ and $\lambda = 1/2$. Consequently $u = 2t$, $a = 14t$, and $t \in \frac{1}{2}\mathbb{Z}_{>0}$. For G_{33}, lifting the entire group at a cusp of width one gives

$$51s + 38R = 192t - 6, \quad s + 5R/6 \leq 2 + 2t, \quad t = 1/2, 1.$$

Neither value has an integral solution. For B_5, the four possible pairs at an end are

$$(B_c, \nu_c) = (6,1), (9,2), (10,2), (11,2), \qquad B_c \geq 9\nu_c/2.$$

Equation (3.4) and $h = -10t + R + \sum \nu_c$ therefore give

$$a \geq 13t + 9h/2 + R + \sigma \geq 22t > 14t.$$

If an E_6 point exists, take a D_5 curve through it. Let R, D count its E_6, D_6 points. The row of the inverse differential associated with the invariant of greatest degree, divided over $\mathbb{C}\{\xi\}$ by the common order of the zeros and poles of its entries, has direction

$$\nu = dT_{12} + \xi^3 dT_9/2 + \xi^6 dT_6/4.$$

Since the ratio $\xi^3/2$ does not lie in the image $A = \mathbb{C}\{\xi^2, \xi^5\}$ of the ambient local ring, there is no constant map κ as in Lemma B.1; this remains true under $\mathrm{GL}(6, A)$ coordinate changes. Both the image and the row have zero derivative at $\xi = 0$, so (B.1) applies at all R points. The degree equations give

$$16R + 8s + 5D + \sigma = 41t, \quad h - u = (t - D - \sigma)/4,$$

and their excess over (B.1) is $(t + 8s + 11D + 23\sigma)/16 > 0$. After these exclusions, the full $G(3,3,5)$ ends have width one and degrees $(1^3, 3^3)$. The nonzero target line bundle corresponding to P_0 has degree $z = -5t + R/3 + s$. Equation (3.4) and the Riemann–Hurwitz formula for the disk give

$$a = 12z + 23t + R + \sigma \geq 47t > 14t,$$

using $z \geq u = 2t$ from Lemma B.1. For D_5 there are now no points with stabilizer E_6; its cusps have width one and, for the full group, generators of degrees $(1^4, 2^2)$. The quotient associated with P_0, obtained by the direction of the odd theta function, taken separately, and the frame P/u on the coarse quotient at D_6, has degree $p = -5t + D/2 + s$. Here D counts D_6 points. Equation (3.4) and the Riemann–Hurwitz formula for the disk give

$$a = 8p + 13t + D + \sigma \geq 29t > 14t,$$

again with $p \geq u = 2t$ from Lemma B.1. Finally A_5 gives $6h = a - 14t - \sigma \leq 0$, contrary to $h \geq u$. □

Lemma B.3. Let $Z = \mathbb{P}^k$, $k \geq 2$, be a closed locus of coincident labels in a smooth GIT model, with $N_{Z/W} = \mathcal{O}(-1)^c$. Assume $\pi|_Z$ is an isomorphism and its image meets the boundary in a finite set. If $L|_Z = \lambda H$ and $\lambda < 1$, then Z is the compactification of the induced arithmetic quotient of the ball fixed by the corresponding finite reflection group, whose automorphic line bundle has class $L|_Z$.

Proof. Since $\pi|_Z$ is an isomorphism, the image of Z has the same normal bundle away from its finitely many cusps. Lemma 3.10 applies with the restriction of the class L of the original lattice. The inequality $\lambda < 1$ forces $s = c$, so the Baily–Borel compactification of the quotient of the fixed ball over Z by the full normalizer is Z. □

Proposition B.4. For a compact arithmetic ball quotient with coarse space $\mathbb{P}^k$ and $L = \lambda H$, one has $\lambda \geq 1/3$ if $k = 4$ and $\lambda \geq 2/3$ if $k = 5$.

Proof. For every complete curve obtained from a disk fixed by a finite reflection group, use $a = (k+1)u$ and $h \geq u$. In dimension five the direction of P_0 for $G(m, p, 4)$ yields

$$(4m-6)j + (m-1)pu + 2mg + m\sigma \leq 2m.$$

It excludes $m \geq 4$ and $(m, p) = (3, 1)$. The remaining groups are $A_5, B_5, D_5, G(3, 3, 5), G_{33}$. In these groups, the orders of stabilizers of rank 2 divide 24. The integral classes $12L$ and $24c_2^{\mathrm{orb}} = 360L^2$ imply $\lambda \in \frac{1}{6}\mathbb{Z}_{>0}$. For $\lambda = 1/6, 1/3, 1/2$, a D_4 curve, with R, M counting G_{33}, D_5 points, gives

$$6u + 18t = 11R/2 + 4M + \sigma, \qquad 5R/6 + M/2 + \sigma = 2t + 2 - 2g.$$

Thus $g = 0, u \leq 2$ and $28R + 21M = (36 + 96\lambda)u - 12$. Reduction modulo seven excludes $\lambda = 1/6$ and $\lambda = 1/2$. The only remaining case is

$$\lambda = 1/3, \quad u = 1, \quad R = 2, \quad M = 0, \quad \sigma = 1. \tag{B.2}$$

We exclude it below, retaining the characters of the triality automorphisms of D_4. Then a B_4 curve obeys $(1 + 4\lambda)u + 4g \leq 4$; integrality of $2t$ excludes $\lambda = 1/6, 1/3$, and $\lambda = 1/2$ requires $4R = 16u - 2$. With G_{33} absent, the inequality in dimension five for $G(m, p, 4)$, applied to $G(3, 3, 4)$, gives $u + j \leq 1$. Finally A_4 gives $5h = 6u - 9t - \sigma < 5u$.

For $k = 4$ the analogous inequality $(3m-5)j + (m-1)pu + 2mg + m\sigma \leq 2m$ excludes $m \geq 5$ and $G(4, 1, 4)$. The remaining full groups are

$$A_4, B_4, D_4, G(3, 1, 4), G(3, 3, 4), G(4, 2, 4), G(4, 4, 4),$$
$$G_{28}, G_{29}, G_{30}, G_{31}, G_{32}.$$

Now $30L = bH$ and $25920c_3^{\mathrm{orb}} = -259200L^3$ give $5 \mid b$, hence $\lambda \in \frac{1}{6}\mathbb{Z}_{>0}$. At $\lambda = 1/6$, the G_{25} curve through a G_{32} point satisfies

$$5u + 29t = 28R/3 + \sigma, \quad 5R/6 + \sigma = 2t + 2 - 2g,$$

leaving $u = R = 2, \sigma = 1$. Its image is a smooth conic. The effective action on the line fixed by $G_{25} \subset G_{32}$ has order six, so for a coordinate x on this line, $v = x^6$ is a coordinate on the coarse curve. The restrictions of the basic invariants of degrees $12, 18, 24, 30$, when nonzero, have orders $2, 3, 4, 5$ in v; hence the differential of the curve vanishes at $v = 0$, a contradiction. For $G(3, 1, 3)$ the equations

$$5u + 20t = 11R/3 + \sigma, \quad 2R/3 + \sigma = 2t + 2 - 2g, \quad -9t + R \geq u$$

leave $u = 1, R = 2 < 5u/2$. No reflections of order three remain, and the integral bundle $10L = 2K + B$ excludes $10\lambda = 5/3$. □

Exclusion of (B.2) *by triality.* The curve is a line, with normal bundle $\mathcal{O}(1)^4$. Its two G_{33} points are p, q, and its other two inertia points have order two and trivial action on the root space. For D_4 put $e_j = e_j(x_1^2, \dots, x_4^2)$ and

$$h_4 = e_2 - e_1^2/4, \quad P = x_1x_2x_3x_4, \quad h_6 = e_3 - e_1e_2/6 + e_1^3/48.$$

The outer S_3 preserves h_6 and $h_4^2 + 12P^2$. Its eigenlines for the action of order three are $h_4 \pm 2\sqrt{-3}P$. The product relation in the fundamental group says that the two nontrivial triality automorphisms given by the local monodromy at p, q are inverse to each other. Thus these two eigenlines, though not preserved by the outer S_3, are globally defined here. In terms of the line bundles for the invariants indexed by $2, 4a, 4b, 6$, one has $E_{4a}E_{4b} = E_2E_6$, including the scalar and automorphic characters. Their degrees on the coarse curve are all -2. The columns of the differential of the quotient map associated with E_2, E_{4a}, E_{4b}, E_6 vanish at p and q to orders $(1, 0, 2, 1)$ and $(1, 2, 0, 1)$, respectively. Consequently

$$S_2 = E_2(p+q), \quad S_6 = E_6(p+q), \quad S_{4a} = E_{4a}(2q), \quad S_{4b} = E_{4b}(2p)$$

are line bundles of degree zero with $S_{4a}S_{4b} = S_2S_6$. After the tangent direction is removed, the local blocks of size two by two have leading forms

$$\begin{pmatrix} a_2v & a_4 \\ a_2'v^2 & a_4'v \end{pmatrix}, \qquad \begin{pmatrix} b_4v^2 & b_6v \\ b_4'v^3 & b_6'v^2 \end{pmatrix}, \qquad \frac{a_4b_4}{a_2b_6} = \frac{1}{12}. \tag{B.3}$$

The ratio follows from the following polynomial identities. Use $\sum z_i = 0$, the permutation action of S_6, and the reflection in $(1, 1, \omega, \omega, \omega^2, \omega^2)$, where $\omega^2 + \omega + 1 = 0$. Put $p_j = \sum_i z_i^j$ and

$$\begin{aligned} F_4 &= p_4 - p_2^2/2, \qquad F_6 = 5p_2^3 - 30p_2p_4 - 20p_3^2 + 48p_6, \\ F_{12} &= 300p_2^3p_3^2 + 315p_2^3p_6 - 990p_2^2p_3p_5 - 450p_2^2p_4^2 - 300p_2p_3^2p_4 - 90p_2p_4p_6 \\ &\quad + 864p_2p_5^2 - 250p_3^4 + 840p_3^2p_6 - 540p_3p_4p_5 + 300p_4^3 + 72p_6^2. \end{aligned}$$

These three polynomials are invariant under the permutation action of S_6 and the reflection in $(1, 1, \omega, \omega, \omega^2, \omega^2)$. Set $\vartheta = \sqrt{-3}$, $\omega = (\vartheta - 1)/2$, $s = \sum_{i=1}^4 x_i$, and substitute

$$z_1 = u + \vartheta s/6, \quad z_2 = -u + \vartheta s/6, \quad z_{i+2} = x_i - (3+\vartheta)s/12 \quad (1 \le i \le 4).$$

Write $H_A = h_4 + 2\vartheta P$, $H_B = h_4 - 2\vartheta P$, and $T_{12} = F_{12} + 3F_6^2/4$. Up to terms of degree at least seven in $x_1, \dots, x_4$, direct expansion gives

$$\begin{aligned} F_4 &= -2u^2e_1 + \omega H_A, \\ F_6 &= 16u^6 + 30(1+\vartheta)u^2H_B - 36h_6 + 5e_1^3/4, \\ T_{12} &= 2700(1+\vartheta)u^8H_B + 32400u^6h_6 - 1725u^6e_1^3. \end{aligned}$$

The terms in e_1^3 disappear in the indecomposable normal directions. Along the tangent direction the invariant F_6 restricts to $16u^6$, so use $v = u^6$ and the frames $(e_1/u^4, H_A, H_B/u^4, h_6)$ on the coarse quotient. Removing the tangent direction therefore gives the two blocks in (B.3) and the required product

$$\frac{a_4b_4}{a_2b_6} = \Big(-\frac{\omega}{2}\Big)\frac{1+\vartheta}{12} = \frac{1}{12}.$$

Choose global frames compatible with $S_{4a}S_{4b} = S_2S_6$. The four columns of the differential of the quotient map, with entries in $\mathcal{O}(1)^4$, then satisfy

$$\begin{aligned} v_{4a}(p) &= A_pv_2(p), \qquad v_{4b}(p) = B_pv_6(p), \\ v_{4b}(q) &= A_qv_2(q), \qquad v_{4a}(q) = B_qv_6(q), \qquad A_pB_p = A_qB_q. \end{aligned}$$

For $p = (1:0), q = (0:1)$ their determinant is a constant multiple of $(A_pB_p - A_qB_q)x^2y^2$, hence zero. The differential of the full quotient map has rank 4 on the normal directions at a general point, a contradiction. □

Proposition B.5. A nonuniform arithmetic ball quotient with compactification $\mathbb{P}^5$ cannot have $L = H/6$.

Proof. The line bundle wL forces $K = \mathbb{Q}(\sqrt{-3})$. For $m = 6$ and rank 4, the comparison using P_0 gives $a \geq (17p/6 + 1)h_P + 30t + R + \sigma$. Using $a = 6u$, $h_P \geq u$, and $t \geq u/6$ gives a contradiction. For G_{32}, equation (3.1), the Riemann–Hurwitz formula for the disk, and Lemma B.1 leave only $(g, u, t, s, \sigma) = (0, 2, 1/3, 2, 2/3)$. The image is a smooth conic, but the degrees $(2, 3, 4, 5, 6)$ of the generators at cusps of width one make the differential zero. For F_4, the types with width one are $A = (1, 1, 2, 2, 3)$ and $B = (1, 2, 2, 3, 4)$; the only remaining case is $(g, t, s_A, s_B, \sigma, u) = (0, 1/2, 1, 2, 0, 3)$. The flags defined by the theta functions associated with the root lattice project onto $\langle dp_2 \rangle$, $\langle dp_2, dp_6 \rangle$, and $\langle dp_2, dp_6, dp_8 \rangle$. These projections are nonzero since the matrices formed by the values of the theta functions at the fixed points and the Taylor coefficients of the lattice sums in degree 2, in degrees $2, 6$, and in degrees $2, 6, 8$ have determinants $3\vartheta_2^4\vartheta_3^4\vartheta_4^4/8$, $-5760A\eta(\tau)^{12}\eta(2\tau)^{12}$, and $-696729600A^4\eta(\tau)^{16}\eta(2\tau)^{16}$, where $A = (\vartheta_3^4 + \vartheta_4^4)/2 \neq 0$ at a modulus τ for which $\mathbb{Z} + \tau\mathbb{Z}$ is homothetic to the ring of Eisenstein integers. The final identity is an identity of weight 24 on $\Gamma_0(2)$; its Sturm bound is six, so the expansion of the lattice sums through q^6 is an exact finite verification. Successive divisions by powers of q_* subtract $5, 6, 8, 10$ from the degrees $3 + d/2$ of the coefficients. Thus $\dim \operatorname{Hom}(f^*T_{\mathbb{P}^5}, \mathcal{O}(3)) \leq 1$, but the Euler sequence gives dimension at least two. For $G(3, 1, 4)$ only $u = 2, t = 1/3, R = 0, s = 2, \sigma = 2/3$ remains, with both ends $(1^2, 3^3)$. Let B_3 be the total reduced branch divisor whose generic inertia has order three, and define $d_3 \in \mathbb{Z}_{\geq 0}$ by $\mathcal{O}_{\mathbb{P}^5}(B_3) = \mathcal{O}_{\mathbb{P}^5}(d_3)$. Its differential maps the line bundle corresponding to P_0, of degree $-12t = -4$, to $f^*\mathcal{O}_{\mathbb{P}^5}(B_3)$, with one interior zero and three at each end. Since $\deg f^*\mathcal{O}_{\mathbb{P}^5}(1) = u = 2$, its degree equation is $2d_3 + 4 = 7$, impossible. The B_4 equations give $10t \leq 4(1 - g) - 2\sigma$ with $2t \in \mathbb{Z}_{>0}$, impossible. For D_4 retain its points with stabilizers G_{33} and D_5: $g = 0, u, s \leq 2$ contradict $2s \equiv 3u + 2 \pmod 7$. Only after excluding them use $G(3, 3, 4)$: $u = 2, t = 1/3$ requires $12s + 5R = 26$, $s \leq 2$. Irreducible factors of lower rank satisfying the condition on full interior stabilizers in Lemma 2.12 contradict Lemma B.1. Finally A_4 gives $5h = 27t - \sigma < 5u$. □

Proposition B.6. A nonuniform arithmetic ball quotient with compactification $\mathbb{P}^5$ cannot have $L = H/4$.

Proof. The field is Gaussian. For G_{31} with a common fixed point the only possible case is $(g, u, t, s, \sigma) = (0, 2, 1/2, 3, 0)$. Let e_j denote the elementary symmetric polynomial of degree j in six variables, so $e_6(x) = x_1 \cdots x_6$, and set $i_{2j} = e_j(x_1^2, \ldots, x_6^2)$ for $j = 1, 2, 3$. Near $(1 : \cdots : 1)$ in the sextic model $i_2^3 - 4i_2i_4 + 8i_6 - 16e_6(x) = 0$, normalize $e_1(x) = 6$, write $x_i = 1 + \xi_i$, and set $q_j = e_j(\xi_1, \ldots, \xi_6)$ for $2 \leq j \leq 6$. The cotangent coordinates are ordered as (q_2, q_3, q_5, q_6). Up to a constant factor in each row, the differentials of the ratios $e_j(x)/e_1(x)^j$ for the generators e_2, e_3, e_4 of degrees $2, 3, 4$ have the following coefficients in the directions of the basic invariants of degrees $8, 12, 20, 24$ on the normal space:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 4 & 1 & 0 & 0 \\ 6 & 3 & 1/2 & -1/4 \end{pmatrix}.$$

Linearizing the sextic equation gives $dq_4 = dq_5/2 - dq_6/4$. After k successive divisions by powers of q_* at the three cusps, the coefficient for the basic invariant of degree d has degree $2 + d/2 - 3k$; this degree is zero for $d = 8$, $k = 2$ and for $d = 20$, $k = 4$, so these two coefficients are constant. Setting them to zero in turn bounds $\dim \operatorname{Hom}(f^*T_{\mathbb{P}^5}, \mathcal{O}(2))$ by two; the Euler sequence gives at least three. The G_{31} torsor has sixteen fixed points for $J = 2^{2+4}$, in residual orbits of sizes six and ten. Its stabilizer orders $120, 72$ are not divisible by $|G(4, 2, 4)/J| = 48$. It therefore contains no point fixed by all of $G(4, 2, 4)$. For $G(4, 2, 4)$ with a fixed point, the four types with width one are $A = (1^3, 2^2)$, $B = (1^2, 2^3)$, $C = (1, 2^2, 4^2)$, $D = (1, 2, 4^3)$. In

each of these types, the degree of the generator associated with P_0 is half the highest degree. The seven remaining combinations of types at the ends are C, AA, CD, AAD, BBC, CDD, $CDDD$, with $u = 1, 1, 2, 2, 2, 3, 4$ and $\nu = 4u$. Every row has $R = 0$ and a nonzero target line bundle corresponding to P_0, of degree $-8t+\nu/2 = 0 < u$. The remaining factors are parabolics of $A_5, B_5, D_5, G(4, 4, 5), F_4, G_{29}$, but irreducible factors of smaller rank satisfying the condition on full interior stabilizers in Lemma 2.12 contradict Lemma B.1. Here the orders of stabilizers of rank 2 divide 24, so $24c_2^{\mathrm{orb}} = 360L^2$ extends across the finite cusp set. Its value on a plane is $45/2$, contradicting integrality. □

B.1 Projective spaces of dimensions seven through nine

Proposition B.7. For $7 \le k \le 9$, projective space $\mathbb{P}^k$ is not the Baily–Borel compactification of an arithmetic quotient of $\mathbb{B}^k$.

Proof. The argument for generation by reflections in the proof of Proposition 2.11 applies to this smooth simply connected rational compactification. Retain the group over K generated by the normalized reflections, and put $w = |\mu(K)|$. On $\mathbb{P}^k$, put $a = (k+1)u$, $u > 0$. The Euler sequence gives $h \ge u$ for every nonzero quotient $f^*T_{\mathbb{P}^k} \to Q_h$. An irreducible factor of rank at most $k-2$ satisfying the condition on full interior stabilizers in Lemma 2.12 contradicts this on a complete curve avoiding the finitely many cusps. The classification of parabolic subgroups thus leaves the curves fixed by subgroups of corank 1 listed below. At a smooth cusp the link is a sphere, which makes the linear action of F_P on $\mathbb{C}^r$ irreducible.

For $k = 7$, first remove the primitive group G_{34}, before applying Proposition 3.4 at its subgroups that can occur as F_P. Its polynomial degrees are $(6, 12, 18, 24, 30, 42)$, the generators of its ring of invariant sections have degrees $(1, 3, 4, 6, 7, 9, 12)$, and

$$a = 41s + 2 - 2g - 132t, \quad h = -42t + 12s, \quad 6t \in \mathbb{Z}_{>0}.$$

There is no torsor, and the reflection divisors give a lift of the entire group with trivial character at the origin. For disks without ends, $h < 0$, which excludes compact disks immediately. For disks with ends, the Riemann–Hurwitz formula for the disk reads $2g - 2 + s + \sigma = 2t$. Its combination with $h \ge a/8 > 0$ gives $94t + 55\sigma \le 108(1 - g)$, hence $g = 0$ and in particular $6t \le 9$. In that complete finite range, $a \in 8\mathbb{Z}_{>0}$, $a \le 48t$, and $h \ge a/8$ leave no possibility. The lift of the entire group then excludes G_{34} cusps as well.

Now consider the monomial groups. For both compact and nonuniform lattices, defer D_6 and D_7 until their possible points with stabilizers E_7 and E_8 have been removed below. For disks without ends and every $m \ge 2$, the Riemann–Hurwitz formula for the disk gives

$$h \le -(r-1)j + \frac{2j + 2m}{m - 1} \le 7 - r, \qquad r = k - 1 \ge 7.$$

For $r = 6$, use $h \le 4 - 4j$ for B_6 and $h \le -5j + (2j + 2m)/(m - 1)$ for $m \ge 3$; these again contradict $h > 0$. For nonuniform disks, use all normalized entries in Appendix C.1, in descending order of m. For $r = 6, 7$, omit D_r here and treat it below with the other ADE groups. The bound $A \le rm + 1$ of (C.2) and the Riemann–Hurwitz formula for the disk give

$$a \le (2rm + 2)(1 - g) - (D - 2rm)t,$$

since the omitted coefficient of R is $\eta - r(m-1) = 2r - 1 + 1/p - rm \le 0$. As $a = (r+2)u \ge r+2$ and $D > 2rm$ for $r = 6, 7$, this proves $g = 0$ and

$$1 \le j \le \left\lfloor \frac{mr(2m - 1)}{D - 2rm} \right\rfloor.$$

These are the following complete j-ranges:

(m,p)	$(2,1)$	$(3,1)$	$(3,3)$	$(4,1)$	$(4,2)$	$(4,4)$	$(6,1)$	$(6,2)$	$(6,3)$	$(6,6)$
$r=6$	2	3	6	4	7	9	7	11	13	16
$r=7$	1	2	3	3	4	5	5	7	8	9.

In (C.5), impose $a \in (k+1)\mathbb{Z}_{>0}$, $a \le (k+1)wt$, and $h \ge a/(k+1)$. For every listed pair there are zero survivors. For these non-ADE monomial groups, comparison of group orders excludes ends whose groups are exceptional Weyl groups. In rank 7, for example, $m^6\,7! > 576\,7!$ for $m \ge 3$, but $|W(E_7)|/|G(2,1,7)| = 9/2$ is not integral.

For $r = 8$, there is no point with a primitive stabilizer of rank 9 and

$$h \le 2m(1-g) - (r-3)j + (2-m)R - m\sigma.$$

The estimate is nonpositive unless $g = 0$ and $(m,j) = (3,1),(4,1),(6,1),(6,2)$. For $(m,j) = (3,1)$, $s = 2$ gives $R \le 1, h \le 0$, and $s = 1$ gives $R \le 2, h \le -2$. For $(m,j) = (4,1), s = 1$, one has $h \le -1$. For $(6,1), s = 1$, one has $h \le 0$, but $s = 2$ is impossible because it requires $5R/6 + \sigma = 1/3$. For $(6,2)$, the two cases have $h \le -6, -2$. The sole remaining case $m = 4, j = 1, s = 2$ has $h > 0$ only when $p > 1, R = 0, \nu_1 = \nu_2 = 4$, both widths one, and $B_c \ge 28$. Thus $a \ge 24$, contrary to $a \le 40t = 10$. This excludes all these disks with monomial full generic stabilizer and then their full cusps.

For $k = 7$, after these non-ADE groups have been removed, the orders of stabilizers of rank 4 in the remaining ADE factors divide 2880. For $16L = bH$, Proposition 3.12 gives $1575b^4/512 \in \mathbb{Z}$, hence $8 \mid b$ and $a \le 16t$. An E_7 point or E_6 cusp gives an E_6 disk with

$$2g - 2 + s + R/2 + \sigma = 2t, \quad a = 11s + 13R + 2 - 2g - 42t, \quad h = -12t + 3R + 3s, \quad 2t \in \mathbb{Z}_{>0}.$$

The coefficient $13R$ includes the sign character of the odd invariant degrees. Eliminating s in the inequality $8h \ge a$ from Lemma B.1 gives $10t + 22\sigma \le 42(1-g)$, hence $g = 0, 2t \le 8$. The conditions $a \in 8\mathbb{Z}_{>0}$, $a \le 16t$ leave only $(t,s,R,\sigma,a,h) = (3/2,3,4,0,24,3)$, for which the line bundle associated with an invariant of odd degree would have degree $-19/2$. For D_6, projection of the inverse differential to the directions of the elementary symmetric invariants gives $5h = a - 12t - R - \sigma$. Together with $h \ge a/8$, this requires $a \ge 32t$, a contradiction. The last A_6 disk has $7h = a - 20t - \sigma < 0$.

For $k = 8$, the orders of stabilizers of rank 3 in the ADE groups divide 24. The integral class $18L = bH$ and $24c_3^{\mathrm{orb}} = -2016L^3$ give $28b^3/81 \in \mathbb{Z}$, hence $9 \mid b$ and $a \le 18t$. An E_8 point or E_7 cusp gives an E_7 disk. At E_8, the local coordinate of the coarse curve is $v = u^2$. Homogeneity gives pole order six for the row of the inverse differential associated with the invariant of degree eighteen, and order 28 for the determinant; on the coarse quotient, the stabilizers act trivially on the fibers of the line bundles associated with the invariant directions. Consequently

$$2g - 2 + s + R/2 + \sigma = 2t, \quad a = 17s + 28R - 70t + 2 - 2g, \quad h = -18t + 6R + 4s, \quad 2t \in \mathbb{Z}_{>0}.$$

Elimination gives $9h = 2a - 18t - 3R - 2\sigma$. The inequality $9h \ge a$ from Lemma B.1 and $a \le 18t$ force $R = \sigma = 0, a = 18t, s = 5t, 3t = 2 - 2g$. Thus $g = 0, t = 2/3$, contradicting $2t \in \mathbb{Z}$. For the remaining D_7 disk, $6h = a - 21t - R - \sigma$ gives $a \ge 63t + 3R + 3\sigma$, again impossible. The last A_7 disk has $8h = a - 27t - \sigma < 0$.

For $k = 9$, the only remaining primitive group is E_8. Its tuple in (C.4), with width one and the full group, gives

$$2g - 2 + s + \sigma = 2t, \quad a = 29s + 2 - 2g - 128t, \quad 2t \in \mathbb{Z}_{>0}, \quad a \in 10\mathbb{Z}_{>0}.$$

The upper bound $a \le 60(1-g) - 70t$ forces $g = 0, t = 1/2, s = 3$, and then $a = 25$, a contradiction. Only type A groups remain in $\mathbb{P}^9$. Write $20L = bH$. Proposition 3.12 gives $27b^2/20, 9b^3/25 \in \mathbb{Z}$, hence $10 \mid b$ and $a/t \le 20$. The last A_8 disk satisfies $9h = a - 35t - \sigma < 0$, contrary to Lemma B.1. This completes the independent exclusions of $\mathbb{P}^7, \mathbb{P}^8, \mathbb{P}^9$. □

C Finite ranges and the remaining cases of monomial groups

This appendix specifies the finite ranges of parameters and every rejection test used in Sections 5 and 6. All operations are on integers and rational numbers. The bounds below define finite sets of multisets of normalized entries and widths at the ends, and (C.8) treats every value of e, including unbounded intervals. Repeated affine actions remain distinct geometric possibilities; actions with the same group, section degrees, width, and condition on the existence of a common fixed point are counted as one row of the enumeration. Every action giving the same row is excluded by the test specified for that row in Appendices C.3–C.5. The three finite ranges of parameters used in the induction are organized as follows; the counts refer to these rows, with all permitted values of e retained.

Labels	Rows	Tests	Proof	Survivors
$n \geq 13$	170	Proposition 3.7, widths and torsors	C.3	0
$n = 12$	117	The same tests, with the injection (3.8)	C.4	0
$n = 11$	413	Tests using only $a \leq 7wt$, then the nefness of K_W+10L and the tests that use it	C.5	0

The last row is used in two stages, before and after (6.5) and the exclusion of E_8; the stronger bound is not assumed in the first stage.

C.1 Cusps and their widths

For F_P of rank $r \geq 6$, append m to each prefix until its length is $r + 1$. Rains's strongly crystallographic classification gives the following degrees of the generators before normalization [26, Section 4.4].

(m, q)	prefixes	(m, q)	prefixes
$(2, 1)$	$(1, 1, 1), (1, 1), (1), ()$	$(6, 1)$	$(1), (2, 3), (2), (3), ()$
$(2, 2)$	$(1, 1, 1, 1)$	$(6, 2)$	$(1, 3), (2, 3, 3), (3, 3)$
$(3, 1)$	$(1, 1), (1), ()$	$(6, 3)$	$(1, 2), (2, 2, 3), (2, 2)$
$(3, 3)$	$(1, 1, 1)$	$(6, 6)$	$(1, 2, 3)$
$(4, 1)$	$(1, 2), (2, 2), (1), (2), ()$	$(4, 4)$	$(1, 1, 2)$
$(4, 2)$	$(1, 2, 2), (2, 2, 2), (1, 1)^*, (1, 2), (2, 2)$		

The $(6, 1)$ prefixes $(2, 3), (2), (3)$ occur three, two, and two times, respectively; there are 36 entries before normalization. The asterisk denotes the nontrivial torsor, which has no point fixed by all of $G(4, 2, r)$. It does not exclude smaller stabilizers or nontrivial characters on the fibers of $\mathcal{V}_*$.

Here is the complete normalization rule. First divide the tuple of degrees given by the entry by its greatest common divisor, obtaining $a = (a_0, \dots, a_r)$. Put

$$h_i = \gcd(a_j : j \neq i), \qquad H_0 = \operatorname{lcm}_i h_i.$$

For every $v \mid H_0$, the Veronese subring of degree v is a polynomial ring whose generators have degrees

$$\alpha_i = \gcd(v, h_i)a_i/v. \tag{C.1}$$

These are precisely the Veronese subrings that are polynomial rings. The coordinate reflection subgroups have pairwise coprime orders $\gcd(v, h_i)$, and Chevalley–Shephard–Todd gives the criterion. Removing first a group of scalars of order g acting only on the fibers of $\mathcal{V}_*$ replaces

q_{old} by q_{old}^g and divides c_* by g. This gives 64 entries after normalization and 32 distinct tuples with their group parameters retained. Every normalized entry satisfies

$$(r-1)b \le A = \sum_i \alpha_i \le rm+1, \qquad b = \max_i \alpha_i \le m. \tag{C.2}$$

Before normalization, the greatest degree occurs at least twice and $\sum_i (b-\alpha_i) \le 2b$; applying (C.1) proves the same bounds after normalization.

The lifts of the entire groups at the cusps follow directly from the reflection roots. For $G(m,q,r)$, the reflections exchanging adjacent coordinates up to a factor that is one or a primitive mth root of unity lift with their positive Gram matrix on $\mathbb{C}^r$. Their rational span fixes a Hermitian plane through the isotropic line of the cusp, and the full generic stabilizer of the corresponding disk is $G(m,p,r)$, $q \mid p \mid m$. For $q = 1, m = 2,3,4$, a diagonal image of order m supplies a lift of the entire group. For $m = 6$, if the kernel of the character on the fiber of $\mathcal{V}_*$ is $G(6,p,r) \lhd G(6,q,r)$, this character has order exactly

$$\kappa = p/q. \tag{C.3}$$

Put $\zeta_6 = \exp(\pi i/3)$. Let T_{ij} be the diagonal operator on $\mathbb{C}^r$ that multiplies coordinates i, j by ζ_6, ζ_6^{-1}, respectively, and fixes the other coordinates. Let E_i be the projection onto coordinate i. For distinct i, j, k, the identities $2I - T_{ij} - T_{ij}^{-1} = E_i + E_j$ and $(E_i + E_j)(E_i + E_k) = E_i$ split the lattice Λ_P by coordinates. Since $1 - \zeta_6$ is a unit, the kernel has a unique fixed point; the full group therefore fixes it, and the character image is the full quotient.

For later use, the Weyl groups E_6, D_7, E_7, E_8 lift in their entirety, have no torsor, and have width one, and the generators of their rings of invariant sections have degrees

$$(1^3, 2^3, 3), \quad (1^4, 2^4), \quad (1,1,2,2,2,3,3,4), \quad (1,2,2,3,3,4,4,5,6). \tag{C.4}$$

For these groups $(s_\alpha - 1)\beta = \alpha$ gives zero coinvariants and no invariant Pic^0. The polarization is an integral power of the minimal one even at CM moduli. Connected reflection divisors through the origin force trivial linearization, and the criterion using greatest common divisors after deleting one entry in (C.1) forces that power to be one. The positive rational root span supplies the Hermitian plane, and the trivial character of the full stabilizer on the fiber of $\mathcal{V}_*$ gives width one.

C.2 The finite range and the unbounded values of e

The proof of Lemma A.1 specifies the computation over $\mathbb{F}_{1009}$ and the projections of the Taylor expansions for G_{33}. For the rows in the table, (C.5)–(C.8) and the following subsections supply the search ranges and rejection criteria.

For fixed r, m, p, j, after $g = 0$ has been proved, enumerate

$$1 \le s \le \lfloor 2 + 2j/m \rfloor, \quad 0 \le R \le \left\lfloor \frac{2 + 2j/m - s}{1 - 1/m} \right\rfloor, \quad \sigma = 2 + 2j/m - s - (1-1/m)R. \tag{C.5}$$

The allowed values of σ are sums of $1 - 1/e$, $1 < e \mid w$, with at most $\lfloor 2\sigma \rfloor$ summands. Use $w = 4$ for $m = 4$, $w = 6$ for $m = 3, 6$, and the union of the three possibilities $w = 2, 4, 6$ for $m = 2$. At each end independently use every normalized entry above with $q_c \mid p$ and every $\kappa_c \mid p/q_c$, before applying the stricter exclusions using the widths κ_c. The quantities B_c, ν_c, a, h are those in (6.1)–(6.3). When all widths are one, the line bundle corresponding to P_0 imposes

$$rj/p + (1 - 1/p)R \in \mathbb{Z}.$$

Take only integral a, impose $a \le rwt$, and impose even a for twelve labels.

For completeness, the coefficient bound used in the enumeration is as follows. Let v_c be the zero forced at the c-th end after the map kills df, and set $V = \sum_c v_c$, $\nu = \sum_c \nu_c$. If $\kappa_c \neq 1$, use $v_c = 1$; if $\kappa_c = 1$, use $v_c = \min_i \alpha_{c,i}$. In the latter case, if the tuple has a unique generator of degree one and the other degrees have a common divisor greater than one, the equation in the radial direction instead forces the least of those other degrees. Nonvanishing follows from the greatest common divisor test using the width in Lemma 3.8. The coefficients for the elementary symmetric invariants have degrees

$$R + \nu - V - (r - 1 - i)j \quad (p > 1), \qquad R + \nu - V - (r - i)j \quad (p = 1).$$

The triangular differential at the points with a larger irreducible stabilizer imposes R zeros. There is at most one extra coefficient for P_0. Thus the bounds

$$\ell \leq \left\lfloor \frac{\nu - s}{j} \right\rfloor + 3, \qquad \ell \leq \left\lfloor \frac{\nu - V}{j} \right\rfloor + 2 \quad (p = 1) \tag{C.6}$$

are valid. The sharper count used in Appendices C.3–C.5 is

$$\ell \leq 1 + \max\left(0, \min\left(q_{\mathrm{sym}}, \left\lfloor \frac{\nu - V}{j} \right\rfloor + 1\right)\right) + \epsilon_P, \tag{C.7}$$

where $q_{\mathrm{sym}} = r$ for $p = 1$, and $r - 1$ otherwise. Here $\epsilon_P = 0$ for $p = 1$; for $p > 1$, take $\epsilon_P = 1$ unless the following test forces $\epsilon_P = 0$. When all widths are one and $p > 1$, the line bundle corresponding to P_0 has degree $E_P = -rj/p - (1 - 1/p)R$, and $h - E_P - V < R$ forces $\epsilon_P = 0$.

There is no search cutoff on e. For each permitted ℓ, solve exactly

$$e \geq \max(1, -h), \qquad a - (n - 2\ell)e \in 2\mathbb{Z}_{\geq 0}. \tag{C.8}$$

If $n - 2\ell > 0$, this is a finite interval with its parity condition. If $n - 2\ell = 0$, it is either empty or the entire lower half-line, depending on $a \in 2\mathbb{Z}_{\geq 0}$. If $n - 2\ell < 0$, it is an unbounded half-line with the indicated congruence. When the coefficient is even and a is even, both parities of e must be retained. Condition (3.9) is monotone in e for both injections of Proposition 3.7, so failure at the smallest permitted integer excludes its whole interval or half-line. In particular, no positivity condition on a is added to the enumeration.

C.3 Ranks at least 9: the 170 rows

The analytic range proved in Proposition 6.1 is

$$9 \leq r \leq 28, \qquad 1 \leq j \leq \begin{cases} 5 & r = 9, \\ 4 & r \geq 10. \end{cases}$$

With (C.7), the domain (C.5)–(C.8) gives

r	9	10	11	14
rows	127	41	1	1

and zero rows in every other rank. At rank 9 the nonzero counts by (m, p) are $3, 4, 4, 30, 3, 4, 79$ at $(3,1), (3,3), (4,2), (4,4), (6,1), (6,2), (6,6)$, respectively. The remaining comparisons using the tangent bundle have the following inputs. Here ρ is the cover degree and β its total allowed

pole order.

r	(m,p)	j	s	ρ	β	a	ℓ	$e_{\min}$
9	$(4,2)$	2	3	1	12	17	5	5
9	$(4,4)$	2	3	2	20	$6,7,8,9$	6	6
9	$(6,1)$	$1,2,3$	j	1	$6j$	$7j+2$	≤ 6	$3j$
9	$(6,6)$	3	3	2	30	$8,7$	7	$10,9$
9	$(3,1)$	1	2	1	6	7	6	3
9	$(3,3)$	1	2	1	6	9	$5,6$	3
10	$(6,2)$	2	2	1	12	12	$6,7$	6
10	$(4,4)$	1	2	2	12	$8,10$	$6,7$	3
10	$(4,4)$	2	3	1	12	$12,14,16,18$	6	6

For the comparison using the tangent bundle, sort

$$2+\beta, \quad \beta-\rho ij \ (1\leq i<r), \quad \beta-\rho rj/p,$$

with the last entry interpreted as the $i=r$ symmetric entry when $p=1$. The required positive rank and degree are $n-\ell-2$ and $\rho a+(\ell-1)\rho e$. Rows 1,2,3,5,7 have available/required ranks $(4,6),(3,5),(4,5),(4,5),(4,5)$. Row 6 gives $5<6$, or $9+5e\geq 24>23$. Row 4 requires degree $122>95$ in rank 4. The last two rows give $5<6$, or $2a+6\rho e\geq 52>45$. Every comparison improves as e increases. The other rows of rank 10 with $m=6$ violate (C.3). The unique rows of ranks 11 and 14 are

$$(m,p,j,s,R,\sigma,a,h)=(4,2,1,2,0,1/2,4,-4), \quad (4,2,1,2,0,1/2,10,-5).$$

They require respectively one and two points of the Gaussian torsor fixed by all of $G(4,2,r)$, which do not exist. Only the two rows of rank 9

$$(m,p,R,a,t,h)=(3,1,1,16 \text{ or } 18,1/3,-2)$$

remain. They have two cusps of width one, with section degrees one and three. The comparison using the normal bundle has target line bundles whose degrees, divided by ρ, are $7-i$, $1\leq i\leq 9$. For $\ell=5,\dots,10$, required degrees $22,24,26,28,30,32$ exceed available sums $20,18,15,11,6,0$. For $\ell<5$, too few target summands have degree at least $e\geq 2$. This completes all 170 exclusions.

C.4 Rank 8: the 117 rows

For twelve labels, a is even and $a\leq 8wt$. The bounds

$$h\leq -5j+2m(1-g)+(2-m)R-m\sigma, \qquad a\leq (16m+2)(1-g)-(D-16m)t$$

show that $g\geq 1$ is impossible: counting coefficients as in Appendix C.2 gives $\ell\leq 4$ and $a>0$, but the second bound gives $a<0$. If $h\geq 0$, then $j\leq 2$; all normalized entries and widths in that finite range contradict integral $a\leq 8wt$. If $j\geq 8$, then $\ell\leq 6$ and $a\geq 0$, contrary to $a\leq 16m+2-j(12+8/p)<0$. Thus $g=0, h<0, 1\leq j\leq 7$. For this rank, replace the condition on the allowed values of σ by the weaker necessary condition $\sigma=0$ or $\sigma\geq 1/2$, which every allowed value satisfies. Using (C.7) gives

(m,p)	$(2,1)$	$(2,2)$	$(3,1)$	$(3,3)$	$(4,2)$	$(4,4)$	$(6,2)$	$(6,3)$	$(6,6)$
rows	9	20	2	2	12	40	1	8	23.

The pairs $(4,1),(6,1)$ leave no row. Two rows have $a=0$, and 27 domains for e are unbounded; these are retained.

Apply Proposition 3.7 with the widths κ_c and the sharper comparison using the normal bundle whenever the end is immersed. Lifts of the entire groups exclude $G(4,1,8)$ and $G(3,1,8)$ cusps after the comparisons on the curves obtained from the disks ending at these cusps. The tuple $(1,1,4^7)$ is excluded only when its width requires a point fixed by all of $G(4,2,8)$, which does not exist. Equation (C.3) removes 29 rows with $m = 6$. For the three remaining rows with $m = 6$,

$$T = G(6,3,8), \quad t = 1/2, \quad s = 3, \quad R = \sigma = 0, \quad h = -6, \quad a = 20, 18, 16,$$

the end types are $A = (1,1,3^7)$, $B = (1,2,6^7)$, and $C = (2,2,3,6^6)$, with combinations ABB, ABC, ACC. Width one forces a nonzero generator of degree one at A, B, and nonzero generators of degrees two and three at C. Thus the critical divisor of the original map has degree $c_f = 0, 1, 2$, respectively. The normal targets in Proposition 3.7 have normalized degrees

$$(12, 9, 7, 6, 3, 0, -3, -6).$$

Each row has $\ell = 5, e \geq 6$, so the bundle $\mathcal{K}$ of Proposition 2.3 has rank 4 and degree at least $42, 39, 36$, respectively. All exceed the sum $12 + 9 + 7 + 6 = 34$. No assertion of immersion at those ends is required.

The two remaining comparisons over the Gaussian field using the normal bundles have

T	a	j	s	widths
$G(4,4,8)$	6	1	2	$(2,1)$
$G(4,4,8)$	10	2	3	$(1,1,1)$.

On the double and identity covers the upper bounds on the degrees of the normal targets are $(8,6,6,4,2,0,-2,-4)$. Only $\ell = 5, e' = 4$ remains, and the bundle $\mathcal{K}$ of Proposition 2.3 has rank 4 and degree $24 = 8+6+6+4$. It would have to equal the four largest target line bundles. The direction of P_0 instead has pole at most one at every point fixed by all of $G(4,4,8)$ with trivial character on the fiber of $\mathcal{V}_*$. Write

$$\Omega = \{z \in \mathbb{Z}[i]^8 : \textstyle\sum z_i \in (1-i)\mathbb{Z}[i]\}, \quad A = \mathbb{C}^8/\Omega, \quad M = p^*\mathcal{O}([0])^{\boxtimes 8},$$

where $p : A \to E_i^8$ has degree two. The character summands of $p_*\mathcal{O}_A$ give odd and even theta products. The even theta product centered at $(1+i)/2$ is nonzero at the origin; their ratio has initial indecomposable term $c\prod z_i$, $c \neq 0$. Changing one coordinate sign separates this term from symmetric invariants. The four fixed points are $\Omega^\#/\Omega = K(M)$, with $\Omega^\# = \mathbb{Z}[i]^8 + \mathbb{Z}(1+i)(1,\dots,1)/2$. Normalized translations by fixed points at which the group acts trivially on the fiber of M preserve M and transport the calculation. Thus the contraction contains $q\, d(\prod z_i)$ at each required fixed point. This lowers the degree of the line bundle corresponding to P_0 from six to zero on the double cover, or to at most three on the identity cover, contradicting its required minimum four.

The remaining B_8 row is

$$a = 8, \quad t = 1/2, \quad s = 3, \quad R = \sigma = 0, \quad h = -2, \qquad \alpha_c = (1^3, 2^6).$$

For the minimal D_8 polarization, three even theta functions and one odd theta function of degree one give the first two independent projections of first derivatives of invariant sections. For a map to $\mathcal{O}(-2)$, the first division by powers of q_* at the cusps leaves coefficients of degrees $i-5$, so those first two projections vanish. The equation in the radial direction forces a second division by powers of q_* in all coefficients of degree one. The resulting degrees $i-8$ leave only the projection of the inverse differential to the coordinate of greatest invariant degree. A Hom

space of dimension one now gives $a \geq 8(-h) = 16$, a contradiction. All twenty D_8 rows require a B_8 end. The complete rejection counts are

Proposition 3.7	16	B_8 theta	1
B_8 ends	20	$G(3,1,8)$ tests/ends	$2+2$
$G(4,1,8)$ ends	33	Gaussian torsors/product tests	$9+2$
equation (C.3)	29	normal bundles over the field $\mathbb{Q}(\sqrt{-3})$	3.

These sum to 117.

C.5 Rank 7: the 413 rows

Use only the bound $a \leq 7wt$, deferring D_7 when a point with stabilizer E_8 is possible. The inequalities

$$h \leq -4j + 2m(1-g) + (2-m)R - m\sigma, \qquad a \leq (14m+2)(1-g) - (D-14m)t$$

exclude $g \geq 1$ by $\ell \leq 4$, and leave no row with $h \geq 0$ and $j \leq 3$. For $j \geq 8$, the inequalities $a \geq -h \geq 4j - 12$, $a \leq 86 - 49j/6$, and $\ell \leq 5$ leave only $m = p = 6, j = 8$, where $-h \geq 24 > a$. Hence $g = 0, h < 0, 1 \leq j \leq 7$. Use the weaker necessary condition $\sigma = 0$ or $\sigma \geq 1/2$, which every allowed value of σ satisfies. With (C.7), the finite range gives

(m,p)	$(2,1)$	$(2,2)$	$(3,1)$	$(3,3)$	$(4,1)$	$(4,2)$	$(4,4)$	$(6,2)$	$(6,3)$	$(6,6)$
rows	21	43	2	5	1	51	132	18	16	124.

All 25 unbounded domains for e are retained. The exclusions have two stages, with different available upper bounds on a. Before (6.5), use only $a \leq 7wt$; the following tests exclude 166 rows.

groups	number	test
$G(6,p,7)$	148	$\kappa_c = p/q_c$
$G(6,p,7)$	10	cyclic covers
$G(3,1,7)$	2	normal bundle
$G(3,3,7)$	5	excluded $G(3,1,7)$ ends
$G(4,1,7)$	1	normal bundle.

These tests suffice for the list of possible factors used to prove (6.5); no row is excluded here using $a \leq 10t$. Keep the other 247 rows, including D_7, until (6.5) and the exclusion of E_8 have been proved. The new bound excludes 205 of these rows by $a > 10t$. The remaining 42 rows are excluded by a B_7 end in one case, a $G(4,1,7)$ end in 25 cases, absence of a point of the Gaussian torsor fixed by all of $G(4,2,7)$ in eight cases, and the comparison using the normal bundle in eight cases. The total is $166 + 205 + 42 = 413$. The ten rows tested using cyclic covers have $R = 0$ and at most two nontrivial widths. At the least permitted e, one of the two monotone inequalities in Proposition 3.7 fails. The same comparison using the normal bundle excludes the two $G(3,1,7)$ rows and the $G(4,1,7)$ row, and lifts of the entire groups exclude the corresponding cusps. The absence of a common fixed point again excludes only $(1,1,4^6)$ when a point of its torsor fixed by all of $G(4,2,7)$ is required. In the eight cases of this comparison over the Gaussian field, $R = \sigma = 0$ and the section degrees are one, two, and four. The width κ is the least nonzero degree, so the end is immersed in $Q = q^\kappa$. Seven cases have $j = 2, s = 3$ and one cusp of width two. After the double cover, the upper bounds on the degrees of the normal targets $(12, 9, 8, 4, 0, -4, -8)$ have sum at most 29 in rank 3, but the bundle $\mathcal{K}$ of Proposition 2.3 requires 36. The last case $j = 4, s = 4, a = 9, \ell = 5, e \geq 9$ has width one and requires 43 against the same bound. The arguments for D_7 and for E_7 without points with a larger irreducible stabilizer are given explicitly in Proposition 6.2, after the bound $a \leq 10t$ has been established.

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